PEARSON EDEXCEL INTERNATIONAL A LEVEL CHEMISTRY · WCH12

Unit 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Revision notes, spec point by spec point (Topics 6 to 10)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

Contents

Topic 6 – Energetics

6.1 Enthalpy change and standard conditions

6.2 Sign convention: exothermic and endothermic

6.3 Enthalpy level diagrams

6.4 Standard enthalpy change definitions

6.5 Calculating enthalpy changes from experiments

6.6 Hess's law and enthalpy cycles

6.7 Core Practical 2: an enthalpy change by Hess's law

6.8 Evaluating calorimetry results

6.9 Bond enthalpy and mean bond enthalpy

6.10 Calculating mean bond enthalpies from enthalpy changes

6.11 Bond enthalpy, which bond breaks and how fast

Topic 7 – Intermolecular Forces

7.1 The three types of intermolecular force

7.2 Hydrogen bonding in water, ammonia and hydrogen fluoride

7.3 The anomalous properties of water

7.4 Predicting hydrogen bonding in other molecules

7.5 Intermolecular forces and physical properties

7.6 Choosing a solvent

Topic 8 – Redox Chemistry and Groups 1, 2 and 7

8.1 Oxidation numbers and the rules

8.2 Calculating oxidation numbers, including peroxides and hydrides

8.3 Oxidation numbers as Roman numerals

8.4 Writing formulae from oxidation numbers

8.5 Oxidation and reduction in s- and p-block reactions

8.6 Oxidising agents and reducing agents

8.7 Disproportionation

8.8 Using oxidation numbers to classify reactions

8.9 Metals form positive ions; non-metals form negative ions

8.10 Ionic half-equations and full ionic equations

8.11 Ionisation energy down Groups 1 and 2

8.12 Reactivity trends in Groups 1 and 2

8.13 Reactions of Group 1 and 2 elements with oxygen, chlorine and water

8.14 Reactions of oxides and hydroxides with water and acid

8.15 Solubility of Group 2 hydroxides and sulfates

8.16 Thermal stability of nitrates and carbonates

8.17 Flame colours

8.18 Practical procedures: thermal decomposition and flame tests

8.19 Tests for carbonate, hydrogencarbonate, sulfate and ammonium ions

8.20 Solution concentrations and acid–base titrations

8.21 Core Practical 3: finding the concentration of hydrochloric acid

8.22 Minimising uncertainty in volumetric analysis

8.23 Core Practical 4: a standard solution from a solid acid

8.24 Group 7 trends: physical properties, electronegativity and reactivity

8.25 Halogen displacement reactions and colours

8.26 Redox reactions of the halogens

8.27 Reactions of halides and hydrogen halides

8.28 Predicting the properties of fluorine and astatine

Topic 9 – Introduction to Kinetics and Equilibria

9.1 Collision theory: concentration, temperature, pressure and surface area

9.2 Activation energy

9.3 Calculating rate: 1/time and gradients

9.4 Maxwell–Boltzmann distribution and temperature

9.5 Catalysts: an alternative route with lower activation energy

9.6 Reaction profiles: uncatalysed and catalysed

9.7 Catalysts in industry: sustainability

9.8 Catalysts and the Maxwell–Boltzmann distribution

9.9 Reversible reactions and dynamic equilibrium

9.10 Position of equilibrium: temperature, pressure and concentration

9.11 Industrial compromise: yield v rate

Topic 10 – Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

10.1 Classifying reactions

10.2 Reaction mechanisms

10.3 Heterolytic bond breaking: electrophiles and nucleophiles

10.4 Nucleophiles

10.5 Bond polarity and the type of mechanism

10.6 Naming and drawing halogenoalkanes

10.7 Primary, secondary and tertiary halogenoalkanes

10.8 Reactions of halogenoalkanes

10.9 Nucleophilic substitution mechanisms (OH^- and NH_3)

10.10 Comparing rates of hydrolysis from experimental data

10.11 Core Practical 5: rates of hydrolysis of halogenoalkanes

10.12 Reactivity trend: primary, secondary and tertiary

10.13 Reactivity trend: chloro-, bromo- and iodoalkanes (bond enthalpy)

10.14 Core Practical 6: chlorination of 2-methylpropan-2-ol

10.15 Naming and drawing alcohols

10.16 Primary, secondary and tertiary alcohols

10.17 Reactions of alcohols: combustion, halogenation, dehydration

10.18 Oxidation of alcohols with acidified potassium dichromate(VI)

10.19 Techniques for preparing and purifying a liquid organic product

10.20 Core Practical 7: oxidation of propan-1-ol to propanal and propanoic acid

10.21 Mass spectra of organic compounds

10.22 Infrared spectra

10.23 Core Practical 8: analysis of some inorganic and organic unknowns

Topic 6 – Energetics

SPEC 6.1

Enthalpy change and standard conditions

WHAT THIS SPEC POINT ASKS

Know that ΔH is the heat energy change measured at constant pressure, and that standard conditions are 100 kPa and a stated temperature, usually 298 K.

6.1 What ΔH means

- **Enthalpy (H)** is the heat energy stored in a substance. You can't measure it directly, but you can measure the **change**, ΔH, when a reaction happens.
- **ΔH is the heat energy change measured at constant pressure.** A reaction in an open beaker or a polystyrene cup is at constant (atmospheric) pressure, so the heat it gives out or takes in is ΔH.
- Units: **kJ mol^{-1}**, i.e. per mole of the reaction as written in the equation (or per mole of the named substance in a definition).
- **System** = the reacting chemicals. **Surroundings** = everything else: the water or solution, the container, the air. A thermometer measures the surroundings.

6.1 Standard conditions

Condition	Value
Pressure	**100 kPa** (10^5 Pa)
Temperature	a stated temperature, **usually 298 K** (25 °C)
Solutions	1 mol dm^{-3}
Substances	in their **standard states**: their normal physical state at 100 kPa and 298 K (e.g. $H_2O(l)$, C(s, graphite), $Br_2(l)$, $O_2(g)$)

- A standard enthalpy change is written with the symbol $^{\ominus}$, e.g. $\Delta H^{\ominus}_{298}$. The temperature is "usually" 298 K because data can be quoted at other temperatures as long as the temperature is stated.
- **State symbols matter.** $H_2O(l)$ and $H_2O(g)$ have different enthalpies: condensing steam releases heat. So ΔH for a reaction making liquid water is more negative than one making steam. Always write state symbols in enthalpy equations.

MARK-SCHEME LANGUAGE

- Standard conditions: "298 K / 25 °C" and "100 kPa / 1 atm / 101 kPa / 1 × 10^5 Pa". ACCEPT "a specified temperature".
- DO NOT ACCEPT "298 °K", or a wrong unit such as "101 Pa". "Room temperature" alone is ignored.
- Final answers to enthalpy calculations need a **sign and units** (kJ mol^{-1}) for the last mark.

Often asked as: part of a definition ("...under standard conditions, which are..."), or the sign-and-units mark at the end of a calculation.

COMMON MISTAKE

Writing "standard conditions" when you mean "standard **states**". Conditions are pressure and temperature; states are the physical states of the substances (WCH12 Jan 2024 examiner report). Leaving out state symbols in an enthalpy equation.

SPEC 6.2 Sign convention: exothermic and endothermic

WHAT THIS SPEC POINT ASKS

Know that exothermic reactions have a negative ΔH and endothermic reactions a positive ΔH.

	Exothermic	Endothermic
Heat	given **out** to the surroundings	taken **in** from the surroundings
Temperature of surroundings	**rises**	**falls**
Products compared with reactants	**lower** enthalpy	**higher** enthalpy
Sign of ΔH	**negative** (ΔH < 0)	**positive** (ΔH > 0)
Examples	combustion, neutralisation, most oxidation, respiration, displacement	thermal decomposition (e.g. $CaCO_3$), photosynthesis, dissolving NH_4NO_3, $KHCO_3$ + HCl

- The sign describes the **system**. When the system loses heat, its enthalpy falls, so ΔH is negative, even though the thermometer reading goes up.
- Bond breaking is endothermic; bond making is exothermic (see 6.9). Whether the overall reaction is exo- or endothermic depends on which is bigger.

EXAM TIP

Use the experiment to check your sign before you write the answer. **Temperature went up → ΔH negative.** Temperature went down → ΔH positive. A thermal decomposition must come out positive; if yours is negative, recheck the cycle.

MARK-SCHEME LANGUAGE

- "Heat energy is **released** (to the surroundings)" for exothermic; "heat energy is **absorbed** / taken in" for endothermic.
- The final calculation mark is usually "answer with **correct sign** and units".

Often asked as: the sign of a calculated ΔH, or "explain why this reaction is exothermic" from a temperature change.

BEYOND THE SPEC, BUT EXAMINED

A temperature rise means the reaction is exothermic, so ΔrH must be given as negative even when the question doesn't remind you (WCH12 Jan 2024).
A thermal decomposition must be endothermic: examiners expect you to notice a negative answer is wrong (WCH12 Jun 2019).

COMMON MISTAKE

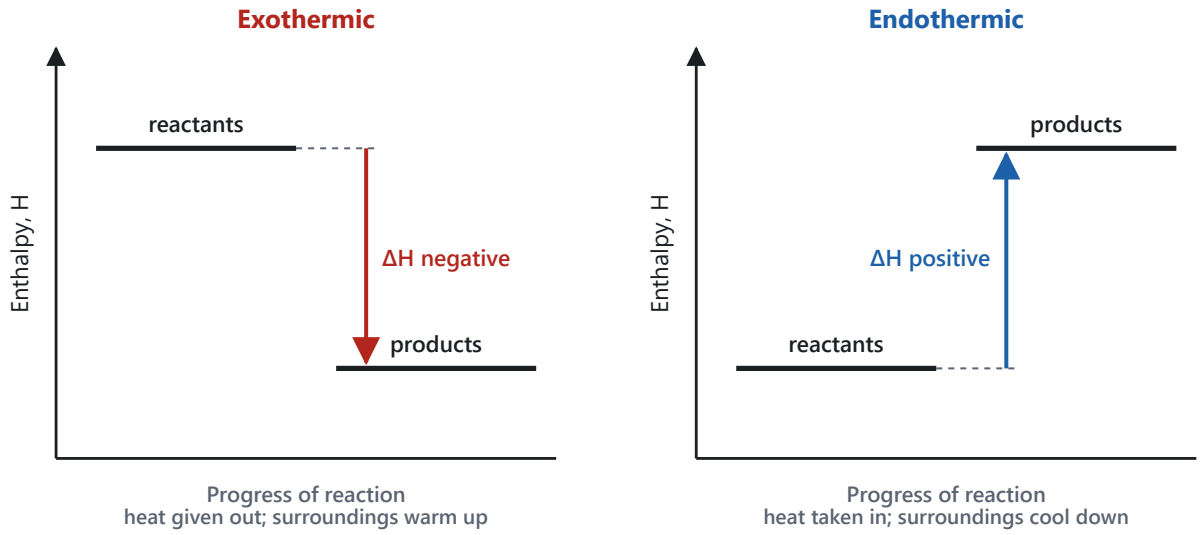
A "negative temperature change" is never credited: describe a temperature **fall** and a positive ΔH (WCH12 Jan 2022).
Saying neutralisation "requires energy". That implies it is endothermic; it releases energy (WCH12 Oct 2023 examiner report).
Dropping the minus sign on combustion and neutralisation enthalpies.

SPEC 6.3

Enthalpy level diagrams

WHAT THIS SPEC POINT ASKS

Draw and interpret enthalpy level diagrams for exothermic and endothermic changes.



Exothermic: products below reactants, arrow points down. Endothermic: products above reactants, arrow points up.

6.3 How to draw one for full marks

1. Vertical axis labelled **enthalpy** (or H, or energy). Not "ΔH" or "enthalpy change".
2. Two horizontal lines: **reactants** and **products**. Label each, ideally with formulae and state symbols (e.g. $CH_4(g) + 2O_2(g)$).
3. Exothermic: products **lower**. Endothermic: products **higher**.
4. One **single-headed arrow** from the reactant level to the product level, labelled ΔH (with its value if you know it, e.g. $\Delta cH = -890$ kJ mol^{-1}).

6.3 Level diagram or reaction profile?

- An **enthalpy level diagram** shows only the start and finish levels.
- A **reaction profile** also shows the hump: the **activation energy (Ea)** is measured from the reactants up to the top of the hump (the transition state). Catalysed profiles come in Topic 9 (9.6).
- For an endothermic reaction, Ea must be **larger** than ΔH, because the hump has to be above the products.

MARK-SCHEME LANGUAGE

- "Y axis labelled **enthalpy**" (ACCEPT energy / H). DO NOT ACCEPT "ΔH", "enthalpy change" or "energy change" as the axis label.
- "Products line below reactants" (exothermic), both levels labelled, arrow **downward** labelled with the ΔH.
- DO NOT ACCEPT a double-headed arrow, a line with no arrowhead, or an arrow labelled −ΔH. A hump is ignored on a level diagram, and so is a missing x-axis label.
- Endothermic profile: "products higher than reactants", with ΔH and Ea both labelled (ΔH may be written as its value, e.g. +133).

Often asked as: draw the diagram for a combustion with ΔcH labelled (WCH12 Jun 2022); draw an endothermic profile with ΔH and Ea (WCH12 Jan 2022).

COMMON MISTAKE

Drawing the familiar exothermic shape when the question gives a **positive** ΔH. Read the sign first (WCH12 Jan 2022 examiner report).
Arrows pointing the wrong way: each wrong arrow loses a mark. Ea arrows start at the **reactants** level, not the products.

SPEC 6.4 Standard enthalpy change definitions

WHAT THIS SPEC POINT ASKS

Know the definitions of the standard enthalpy changes of reaction, formation, combustion, neutralisation and atomisation.

6.4 The five definitions

All are measured under **standard conditions** (100 kPa, stated temperature, usually 298 K) with all substances in their **standard states**.

Change	Definition (learn the key words)	Example equation
Reaction, $\Delta rH^{\ominus}$	enthalpy change when the **molar quantities shown in the equation** react	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta rH = -92$ kJ mol^{-1}
Formation, $\Delta fH^{\ominus}$	enthalpy change when **one mole of a compound** is formed **from its elements** in their standard states	$2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
Combustion, $\Delta cH^{\ominus}$	enthalpy change when **one mole of a substance** is **completely burned in oxygen**	$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
Neutralisation, $\Delta neutH^{\ominus}$	enthalpy change when an acid reacts with an alkali to form **one mole of water**	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
Atomisation, $\Delta atH^{\ominus}$	enthalpy change when **one mole of gaseous atoms** is formed from the element in its standard state	$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$; $C(s) \rightarrow C(g)$

6.4 Points that catch people out

- **Elements have ΔfH = 0.** Forming an element in its standard state from itself is no change. So ΔfH of $O_2(g)$, C(graphite), Na(s) are all zero. (But ΔfH of O_3 or of $Br_2(g)$ is not zero: they aren't standard states.)

- **Formation equations can need fractions** to make exactly one mole of compound: $\frac{1}{2}O_2$, $\frac{3}{2}H_2$. Remember H_2, O_2, N_2 and the halogens are **diatomic**, and include the oxygen already inside an alcohol when you balance the elements.
- **Combustion and neutralisation are always exothermic.** Atomisation is always endothermic (bonds or metallic bonding must be broken).
- **Atomisation of a diatomic element** makes one mole of atoms, so the equation uses **$\frac{1}{2}X_2$**. ΔatH of chlorine = ½ × E(Cl–Cl) = ½ × 243 ≈ +122 kJ mol^{-1}.
- **One equation can fit two definitions.** $C(s) + O_2(g) \rightarrow CO_2(g)$ is both ΔcH of carbon and ΔfH of CO_2. $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ is both ΔcH of hydrogen and ΔfH of water.
- **Scaling.** If the equation is doubled, ΔrH doubles: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ has ΔrH = 2 × (−286) = −572 kJ mol^{-1}. But ΔfH and ΔcH are always **per mole** of the named substance.
- **Neutralisation** of any strong acid with any strong alkali gives about **−57 kJ mol^{-1}**, because the reaction is always $H^+ + OH^- \rightarrow H_2O$. Sulfuric acid + NaOH: the equation that matches the definition is $\frac{1}{2}H_2SO_4 + NaOH \rightarrow \frac{1}{2}Na_2SO_4 + H_2O$.

MARK-SCHEME LANGUAGE

- Why ΔfH of oxygen is zero: "oxygen is an **element** and (it is) in its **standard state**" (WCH12 Jan 2022).
- Formation: "**one mole** of compound", "from its **elements**", "in their **standard states**".
- Formation cycles: "balanced elements **with state symbols**", "inclusion of multiples".

Often asked as: write the equation for a named enthalpy change; explain why a value is zero; explain why a given equation is (or isn't) a formation reaction.

BEYOND THE SPEC, BUT EXAMINED

Writing the **ionic** equation for a named enthalpy change, with spectator ions cancelled, e.g. $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ for neutralisation (WCH12 Oct 2023).
Changes of state can appear as a step in a cycle: sublimation of iodine, $I_2(s) \rightarrow I_2(g)$, ΔH = +62 kJ mol^{-1} (WCH12 Jan 2026).

COMMON MISTAKE

Writing "under standard conditions" when the mark is for "elements in their **standard states**" (WCH12 Jan 2024 examiner report).
Writing H and O instead of H_2 and O_2 on the elements line, leaving out state symbols, or forgetting the O already in an alcohol (WCH12 Jan 2022 examiner report).
Defining combustion "in air" or without "complete".

SPEC 6.5 Calculating enthalpy changes from experiments

WHAT THIS SPEC POINT ASKS

(i) Use energy transferred = mass × specific heat capacity × temperature change. (ii) Turn that into ΔH in kJ mol^{-1}. Limited to mixing in an insulated container and to combustion calorimetry.

6.5 The two equations

q = m × c × ΔT

Symbol	Meaning	Unit
q	energy transferred to (or from) the surroundings	J
m	mass of the **water or solution being heated**, not the reagent	g
c	specific heat capacity of water, **4.18**	J g^{-1} $°C^{-1}$ (= J g^{-1} K^{-1})
ΔT	temperature change	°C (or K: a change of 1 °C = 1 K)

ΔH = −q ÷ n (with q in kJ), where n is the number of **moles of the substance the ΔH refers to**, usually the limiting reagent, or the water formed for neutralisation.

- Put the **sign** in by thinking: temperature up → negative; temperature down → positive.
- For solutions, assume the **density is 1.00 g cm^{-3}**, so 50.0 cm^3 of solution has m = 50.0 g. Add the volumes of two solutions that are mixed.
- A solid added to a solution: m is still just the solution. Don't add the mass of the solid.

6.5 Method 1: mixing in an insulated container

- Use a **polystyrene cup** (a good insulator with a low heat capacity), with a lid, standing in a beaker for stability.
- Measure a known volume of one solution, record its temperature for a few minutes, add the second reagent (in **excess** if the other is the one you'll calculate with), stir, and record the temperature regularly.
- Used for neutralisation, displacement (e.g. Zn + $CuSO_4$), dissolving, and the reactions in Core Practical 2.

WORKED EXAMPLE: ENTHALPY OF NEUTRALISATION

25.0 cm^3 of 2.00 mol dm^{-3} HCl is mixed with 25.0 cm^3 of 2.00 mol dm^{-3} NaOH. The temperature rises by 13.5 °C.

q = m c ΔT = 50.0 × 4.18 × 13.5 = 2821.5 J = 2.8215 kJ

n(H_2O formed) = n(HCl) = 2.00 × 25.0 ÷ 1000 = 0.0500 mol

ΔneutH = −2.8215 ÷ 0.0500 = **−56.4 kJ mol^{-1}** (3 s.f., negative because the temperature rose)

6.5 Method 2: combustion calorimetry

- A **spirit burner** of the fuel heats a known mass of water in a **copper can** (copper conducts heat well) clamped above it.
- Weigh the burner (with its cap on, to stop evaporation) before and after; the loss in mass is the fuel burned.
- Stir the water and record its starting and highest temperatures. Use a draught shield, keep the flame close to the can, and use a lid on the can.

WORKED EXAMPLE: ENTHALPY OF COMBUSTION OF ETHANOL

Burning 0.460 g of ethanol (Mr 46.0) heats 100.0 g of water by 13.2 °C.

q = 100.0 × 4.18 × 13.2 = 5517.6 J = 5.5176 kJ ← mass of the **water**, not the ethanol

n(ethanol) = 0.460 ÷ 46.0 = 0.0100 mol

ΔcH = −5.5176 ÷ 0.0100 = **−552 kJ mol^{-1}**

The data-book value is −1367 kJ mol^{-1}. Why the experiment is so far off is 6.8.

WORKED EXAMPLE: WORKING BACKWARDS TO A TEMPERATURE

A hand warmer reaction has ΔrH = −90.0 kJ mol^{-1}. 0.0150 mol reacts in 50.0 g of water at 20.0 °C. What is the final temperature?
q = 0.0150 × 90.0 = 1.35 kJ = 1350 J
ΔT = q ÷ (m c) = 1350 ÷ (50.0 × 4.18) = 6.46 °C
Final temperature = 20.0 + 6.5 = **26.5 °C**

MARK-SCHEME LANGUAGE

- Marks go: energy transferred (mcΔT) → moles → energy ÷ moles → "**sign and units and 2 or 3 s.f.**". Intermediate rounding is ignored.
- m is the mass of the **solution / water**: "50.0 × 4.18 × 5.2", not the salt (WCH12 Oct 2020). DO NOT AWARD 100.305 × 4.18 × 13.2 when 0.305 g of fuel was burned (WCH12 Jun 2022).
- c = 4.18 J g^{-1} °C^{-1} almost every time; 4.2 has been accepted where a paper used it.
- "Correct answer with no working scores full marks" is common, but show working so you can get transferred-error marks.

Often asked as: neutralisation, displacement or dissolving data in a cup; combustion of an alcohol; a thermometric titration; working back from ΔH to a temperature.

BEYOND THE SPEC, BUT EXAMINED

Thermometric titration: add acid in portions and plot temperature against volume. Two straight lines of best fit cross at the end point; the total mass heated includes the acid added (WCH12 Jun 2024).
Energy per gram or per cm^3 of a fuel from ΔcH, Mr and density (WCH12 Jun 2022).
Percentage **efficiency** of a heater or engine = useful energy ÷ energy from combustion × 100 (WCH12 Jun 2023).

COMMON MISTAKE

Using the mass of the salt or fuel in mcΔT, or adding it to the mass of water (WCH12 Oct 2020, Oct 2022, Jan 2022 examiner reports).
Adding 273 to ΔT. A temperature **change** is the same in °C and K (WCH12 Oct 2021).
Forgetting to convert J to kJ, or to find the limiting reagent before working out n.
Using the Mr of a hydrated salt without its water of crystallisation (WCH12 Jan 2021).

SPEC 6.6

Hess's law and enthalpy cycles

WHAT THIS SPEC POINT ASKS

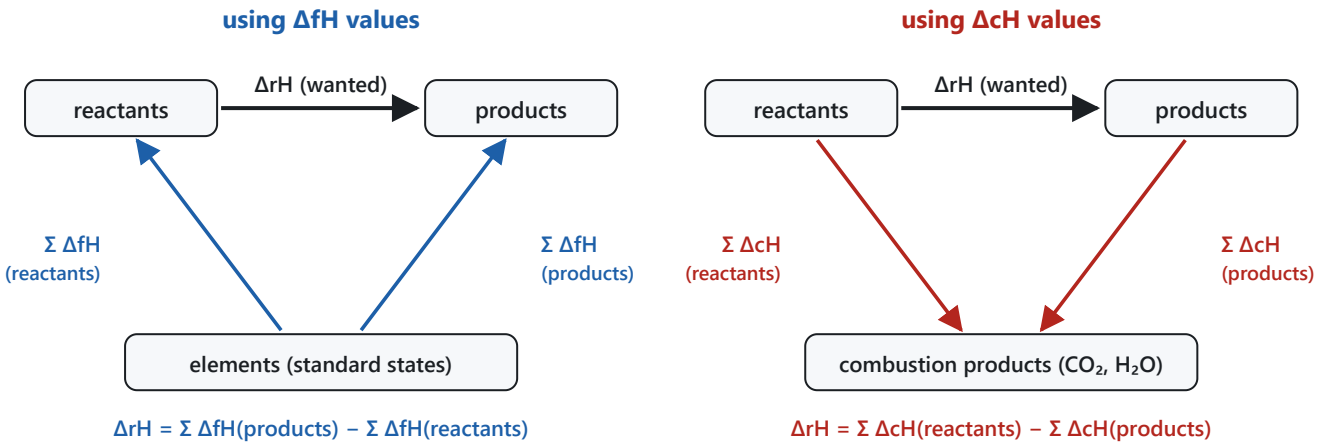
Know Hess's law and use it to (i) build enthalpy cycles and (ii) calculate enthalpy changes from given data, data from a table, or experimental results.

6.6 The law

Hess's law: the total enthalpy change for a reaction is independent of the route taken, as long as the starting and finishing conditions are the same. It follows from conservation of energy.

It lets you find ΔH for reactions you can't measure directly: reactions that are too slow, give side products, can't be started easily, or happen with a solid whose temperature you can't measure (e.g. $CaCO_3$ → CaO + CO_2, or anhydrous $CuSO_4$ + water → hydrated $CuSO_4$).

6.6 The two standard cycles



With ΔfH data the elements go at the bottom and the arrows point up. With ΔcH data the combustion products go at the bottom and the arrows point down.

Route rule: going round the cycle the other way, follow the arrows; when you go against an arrow, **change its sign**. Multiply every value by the number of moles in the equation.

WORKED EXAMPLE: ΔRH FROM ENTHALPIES OF FORMATION

Find ΔrH for $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$.
ΔfH / kJ mol^{-1}: $NaHCO_3(s)$ −950.8; $Na_2CO_3(s)$ −1130.7; $CO_2(g)$ −393.5; $H_2O(l)$ −285.8
Σ ΔfH(products) = (−1130.7) + (−393.5) + (−285.8) = −1810.0
Σ ΔfH(reactants) = 2 × (−950.8) = −1901.6
ΔrH = −1810.0 − (−1901.6) = **+91.6 kJ mol^{-1}** (positive, as a decomposition should be)

WORKED EXAMPLE: ΔFH FROM ENTHALPIES OF COMBUSTION

Find ΔfH of ethanol: $2C(s) + 3H_2(g) + ½O_2(g) \rightarrow C_2H_5OH(l)$. This can't be done directly.
ΔcH / kJ mol^{-1}: $C(s)$ −393.5; $H_2(g)$ −285.8; $C_2H_5OH(l)$ −1367.3
Cycle: elements at top left, ethanol at top right, $2CO_2 + 3H_2O$ at the bottom; all arrows point **down**.
ΔfH = Σ ΔcH(reactants) − Σ ΔcH(products)
= [2 × (−393.5) + 3 × (−285.8)] − (−1367.3)
= −1644.4 + 1367.3 = **−277.1 kJ mol^{-1}**
(Oxygen has no ΔcH: it is not a fuel.)

WORKED EXAMPLE: A CYCLE FROM TWO EXPERIMENTS

$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$ can't be measured directly. Instead react each solid with HCl:
ΔH_1: $MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + CO_2(g) + H_2O(l)$
ΔH_2: $MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$
Cycle: $MgCO_3(s) + 2HCl(aq)$ on the top left, $MgO(s) + CO_2(g) + 2HCl(aq)$ on the top right, $MgCl_2(aq) + H_2O(l) + CO_2(g)$ at the bottom.
Going the long way round: down by ΔH_1, then **up against** the ΔH_2 arrow.
$\Delta rH = \Delta H_1 - \Delta H_2$. With $\Delta H_1 = -50.3$ and $\Delta H_2 = -150.9$ kJ mol^{-1}:
ΔrH = (−50.3) − (−150.9) = **+100.6 kJ mol^{-1}** (positive, as a decomposition should be)

6.6 Combining equations (the algebra way)

You can also add and subtract equations like algebra. Reverse an equation → change the sign of its ΔH. Multiply an equation → multiply its ΔH. Cancel species that appear on both sides. Check that what's left is

the target equation.

EXAM TIP

Put **brackets round every value** so the minus signs survive: "−(−1367.3)", not "−−1367.3". Then check the sign makes sense (decomposition positive, combustion negative).

MARK-SCHEME LANGUAGE

- Cycle marks: "boxes / species correct with state symbols", "arrows correct", then "correct application and evaluation **including sign**".
- **No transferred error from an incorrect cycle**, but TE from an earlier numerical answer is allowed.
- Multiply by the coefficients: e.g. 2 × ΔfH(ethanoic acid) (WCH12 Jun 2021).

Often asked as: complete a formation or combustion cycle and calculate; a cycle built from two experiments ($CuSO_4$ hydration, $MgCO_3$ decomposition); combine three equations.

BEYOND THE SPEC, BUT EXAMINED

Cycles using the **enthalpy of solution** of an anhydrous and a hydrated salt to get the enthalpy of hydration (WCH12 Jan 2020, Oct 2024).
A cycle where the unknown is ΔfH of one species, here CO from phosgene data: −220.1 − (−107.6) = −112.5 kJ mol^{-1} (WCH12 Jun 2021).

COMMON MISTAKE

Drawing the cycle the wrong way round, or getting the subtraction backwards (products − reactants for ΔfH; reactants − products for ΔcH) (WCH12 Jun 2019, Jan 2020 examiner reports).
Forgetting to multiply by the coefficients, or leaving a ΔcH out of the sum (WCH12 Oct 2020 examiner report).
Copying values wrongly from the question into the working.

SPEC 6.7

Core Practical 2: an enthalpy change by Hess's law

WHAT THIS SPEC POINT ASKS

Find the enthalpy change of a reaction indirectly, using Hess's law and two measured enthalpy changes.

6.7 The idea

The target, e.g. the thermal decomposition $2KHCO_3(s) \rightarrow K_2CO_3(s) + CO_2(g) + H_2O(l)$, can't be measured: it needs strong heating, and you can't measure the temperature change of a solid being heated. Both solids react with hydrochloric acid at room temperature, though, and both give the **same products**, so a cycle links them.

- ΔH_1: $K_2CO_3(s) + 2HCl(aq) \rightarrow 2KCl(aq) + CO_2(g) + H_2O(l)$ (temperature **rises**)
- ΔH_2: $KHCO_3(s) + HCl(aq) \rightarrow KCl(aq) + CO_2(g) + H_2O(l)$ (temperature **falls**)
- Target: $\Delta rH = 2\Delta H_2 - \Delta H_1$

CORE PRACTICAL 2: METHOD IN BRIEF

1. Measure **30.0 cm^3 of 2 mol dm^{-3} HCl** (an excess) into a polystyrene cup standing in a beaker. Use a burette or measuring cylinder.
2. Weigh by difference about **3.5 g of anhydrous K_2CO_3** in a weighing bottle (record the exact mass).

3. Record the acid's temperature every 30 s for 2–3 minutes to get a steady starting value.
4. Add the solid at a known time, stir, and keep recording the temperature every 30 s for several minutes after the change stops. Add it **gradually** so the effervescence doesn't splash acid out.
5. Reweigh the empty bottle so you know exactly how much solid went in.
6. Repeat with **about 2.5 g of $KHCO_3$** and a fresh 30.0 cm^3 of acid.
7. Plot each temperature–time graph and extrapolate to the time of mixing to find each ΔT (see 6.8).

Safety: eye protection; 2 mol dm^{-3} HCl is an irritant.

WORKED EXAMPLE: PROCESSING THE RESULTS

K_2CO_3: 3.45 g (Mr 138.2) → 0.0250 mol. ΔT = +6.6 °C.
$q_1 = 30.0 \times 4.18 \times 6.6 = 827.6$ J → $\Delta H_1 = -0.8276 \div 0.0250 = -33.1$ kJ mol^{-1}
$KHCO_3$: 2.50 g (Mr 100.1) → 0.0250 mol. ΔT = −6.3 °C.
$q_2 = 30.0 \times 4.18 \times 6.3 = 790.0$ J → $\Delta H_2 = +0.7900 \div 0.0250 = +31.6$ kJ mol^{-1}
$\Delta rH = 2\Delta H_2 - \Delta H_1 = 2(+31.6) - (-33.1) =$ **+96.3 kJ mol^{-1}**

6.7 Other versions the exam uses

The same "two experiments, one cycle" idea appears with:

- **Copper(II) sulfate**: dissolve anhydrous $CuSO_4$ and hydrated $CuSO_4 \cdot 5H_2O$ separately in water; ΔH(hydration) = ΔH(anhydrous dissolving) − ΔH(hydrated dissolving). To keep the final solutions the same, use less water with the hydrated salt, because its crystals already contain 5 mol of water per mole (WCH12 Jan 2020).
- **Magnesium sulfate**, $MgSO_4$ and $MgSO_4 \cdot 7H_2O$ (WCH12 Oct 2024).
- **MgO and $MgCO_3$** (or CaO and $CaCO_3$) with HCl, to find the carbonate's decomposition enthalpy (WCH12 Jun 2025).

MARK-SCHEME LANGUAGE

Why the direct reaction can't be measured:

- anhydrous salt + water: "adding water … will produce a solution and does not form a solid", "some of the water may turn to steam", "difficult to know when exactly 5 mol of water has been added"
- decomposition: "the solid has to be heated", "it is difficult to measure the temperature of a solid / powder while it is being heated" (WCH12 Jun 2025)

How water dissolves an ionic solid: "breaks the lattice" and "hydrates the ions / ion–dipole attractions". DO NOT ACCEPT hydrogen bonds to Cu^{2+}, or "molecules of copper sulfate" (WCH12 Jan 2020).

Often asked as: draw the cycle for the two reactions, calculate ΔH_1 and ΔH_2 from data, combine them, and explain why the target can't be measured directly.

COMMON MISTAKE

Not recognising the practical in an unfamiliar setting: the cycle should be routine (WCH12 Jan 2020 examiner report).
Using the mass of the solid in mcΔT (WCH12 Oct 2022 examiner report).
Forgetting the factor of 2 for $2KHCO_3$, or the sign of ΔH_2 (it is positive because the temperature fell).

SPEC 6.8 Evaluating calorimetry results

WHAT THIS SPEC POINT ASKS

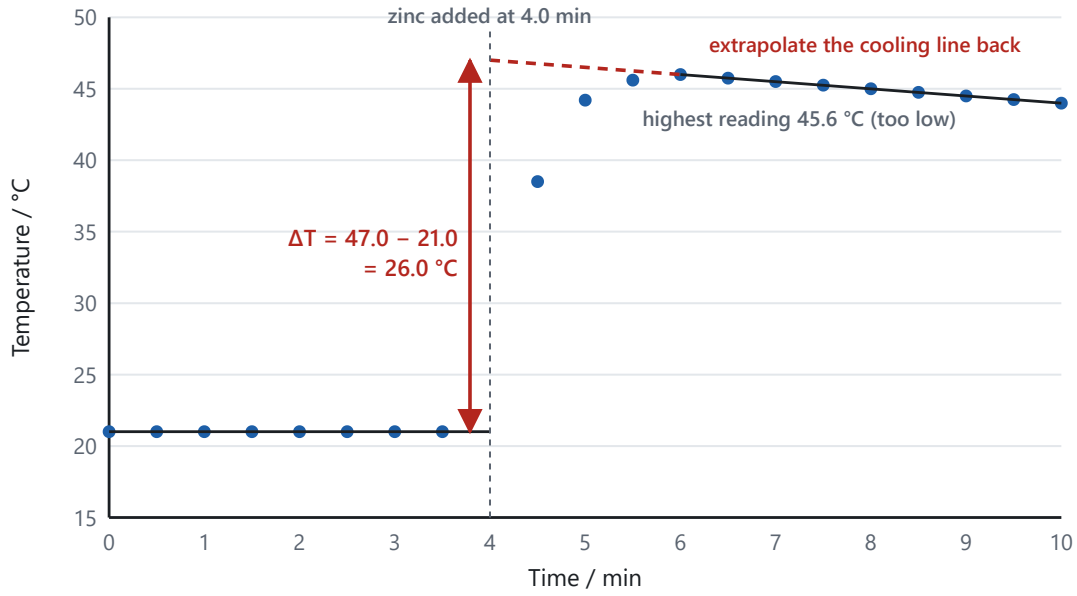
Evaluate results from mixing and combustion experiments: errors, uncertainties and assumptions. Draw suitable graphs and use cooling-curve corrections.

6.8 Assumptions behind the calculation

- The solution has the **density** and **specific heat capacity of water** (1.00 g cm^{-3}, 4.18 J g^{-1} $°C^{-1}$).
- **All the heat** goes into the water/solution: none is lost to the air or absorbed by the cup, thermometer or can.
- The reaction is **complete** (and, for combustion, the fuel burns **completely**).
- The other reagent is in excess, and the solid's own mass doesn't change the heat capacity.

6.8 The cooling-curve correction

The reaction doesn't happen instantly, so while the temperature is still rising, the mixture is **already losing heat** to the surroundings. The highest reading is therefore **lower** than the true value.



Draw two straight lines of best fit (before and after the reaction), extrapolate the cooling line back to the time of mixing and read ΔT there. Here ΔT = 26.0 °C, compared with only 24.6 °C from the highest reading.

1. Record the temperature for a few minutes **before** mixing to get a steady start temperature.
2. Mix at a recorded time (e.g. 4.0 min) without taking a reading at that moment.
3. Keep recording for several minutes after the maximum.
4. Draw a **straight line of best fit** through the cooling points and **extrapolate** it back to the time of mixing.
5. ΔT = extrapolated temperature − starting temperature at that time.

WORKED EXAMPLE: FROM THE GRAPH

50.0 cm^3 of 0.500 mol dm^{-3} $CuSO_4$ with excess zinc powder. ΔT from the graph = 26.0 °C.

q = 50.0 × 4.18 × 26.0 = 5434 J; n($CuSO_4$) = 0.500 × 0.0500 = 0.0250 mol

ΔrH = −5.434 ÷ 0.0250 = **−217 kJ mol^{-1}** for $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

6.8 Why combustion values come out too small

Measured ΔcH values are always **less exothermic** than data-book values:

Source of error	Effect	Improvement
Heat lost to the surroundings (air, clamp)	less heat reaches the water	draught shield, lid, flame close to the can, insulate the can
Heat absorbed by the **copper can** and thermometer	not counted in mcΔT	include the can's heat capacity in the calculation
Incomplete combustion (soot, CO)	less energy released per mole	better air supply
Fuel evaporates from the wick	mass loss is more than the mass burned	keep the cap on when weighing; weigh quickly
Water in the products is a **gas**, not liquid as in the standard state	less energy released	(built into the method)

6.8 Uncertainty

- Percentage uncertainty = (uncertainty ÷ value measured) × 100.
- A **temperature change** (or a mass by difference, or a titre) comes from **two readings**, so double the reading uncertainty.

WORKED EXAMPLE: PERCENTAGE UNCERTAINTY IN ΔT

Thermometer read to ±0.05 °C; ΔT = 13.2 °C.
% uncertainty = (2 × 0.05) ÷ 13.2 × 100 = **0.76%**
To reduce it: use a thermometer with finer graduations, or a bigger ΔT (more reagent, less water).

MARK-SCHEME LANGUAGE

- Combustion errors (choose those the question allows): "**heat loss** (to the surroundings)", "**incomplete combustion**" (ACCEPT insufficient oxygen / soot / carbon monoxide), "some **ethanol evaporates**", "heat capacity of the beaker / can not taken into account" (WCH12 Jun 2022).
- IGNORE "not all the ethanol was burned", "thermometer", "impure ethanol", "not stirred", "no lid", "human error", "non-standard conditions".
- Graph: two straight lines of best fit extrapolated to the **time of mixing**; points covering at least half the grid; axes labelled with units.
- Percentage uncertainty: (2 × 0.05 × 100) ÷ 13.2 = 0.76%. DO NOT AWARD 0.75 or 0.80.

Often asked as: give two reasons the value is less exothermic than the data-book value; draw the lines and find ΔT; calculate a percentage uncertainty; compare two methods.

BEYOND THE SPEC, BUT EXAMINED

When the stem rules out heat loss, the expected answers are incomplete combustion and evaporation of the fuel (WCH12 Oct 2021).
Comparing two methods: a calorimetric value is low because heat is lost or absorbed by the container; a gravimetric value is high if the product is wet or impure (WCH12 Oct 2020).
Reading the temperature at the stated time on a graph, and reading a maximum temperature then subtracting the start (WCH12 Jun 2024).

COMMON MISTAKE

Listing heat loss when the question says "other than heat loss" (WCH12 Oct 2021 examiner report).
Stock answers: "human error", "non-standard conditions", "the water changed state".
Comparing a data-book value with an experimental one when the question compares **two data sources** (WCH12 Jan 2025 examiner report).
Freehand or dotted lines of best fit, or a ΔT read from the highest point instead of the extrapolation.

SPEC 6.9

Bond enthalpy and mean bond enthalpy

WHAT THIS SPEC POINT ASKS

Understand bond enthalpy and mean bond enthalpy, use them to calculate enthalpy changes, and know the limitations of doing so.

6.9 Definitions

- **Bond enthalpy**: the enthalpy change to **break one mole of a particular bond** in a specific molecule, with everything in the **gas phase**. Always positive (breaking is endothermic).
- **Mean bond enthalpy**: the bond enthalpy for a type of bond **averaged over many different compounds**. The C–H bond in methane is not exactly the same strength as the C–H bonds in ethanol, so data books give an average.
- **Breaking bonds takes in energy (+). Making bonds gives out energy (−).**

6.9 The calculation

ΔrH = Σ(bond enthalpies of bonds broken) − Σ(bond enthalpies of bonds made)

1. Draw **displayed formulae** of everything, so you can count every bond.
2. List bonds broken (reactants) with numbers, and bonds made (products).
3. Multiply each by its bond enthalpy; add up each side.
4. Broken − made. Negative → exothermic.

WORKED EXAMPLE: COMBUSTION OF ETHANOL FROM MEAN BOND ENTHALPIES

$C_2H_5OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$
Mean bond enthalpies / kJ mol^{-1}: C–H 413, C–C 347, C–O 358, O–H 464, O=O 498, C=O 805
Broken: 5 C–H (2065) + 1 C–C (347) + 1 C–O (358) + 1 O–H (464) + 3 O=O (1494) = **4728**
Made: 4 C=O (3220) + 6 O–H (2784) = **6004**
ΔcH = 4728 − 6004 = **−1276 kJ mol^{-1}**
Note: 3 O_2, not 3.5. The alcohol already contains one O atom.

6.9 Limitations

The bond-enthalpy answer (−1276) differs from the data-book ΔcH (−1367) because:

1. **Mean values are averages** over many compounds, so they aren't exact for the bonds in this particular molecule.
2. Bond enthalpies apply to the **gas phase**, but in standard conditions ethanol and water are **liquids**. The enthalpy of vaporisation isn't included.

MARK-SCHEME LANGUAGE

- Definition: "energy **required to break one mole** of a (specific) bond", "in the **gaseous** phase / state", "**averaged over a number of (different) compounds**". DO NOT ACCEPT "averaged over different types of bond"; IGNORE "mean" alone (WCH12 Oct 2021).
- Layout: "energy to break bonds = +…", "energy released making bonds = −…", "ΔrH = broken − made".
- Limitation (two marks): "bond enthalpies are **averaged** over a (large) number of compounds" and "bond enthalpies refer to the **gas phase** but (octane / water) are **liquids**" (WCH12 Jun 2023). IGNORE standard conditions and experimental error answers.

Often asked as: combustion of an alcohol or alkane by bond enthalpies; define mean bond enthalpy; explain why the answer differs from the data-book value.

BEYOND THE SPEC, BUT EXAMINED

Each extra CH_2 in a homologous series means **one more C–C and two more C–H** broken, and **two more C=O and two more O–H** made, so ΔcH changes by a nearly constant amount (WCH12 Jan 2026).
Counting bonds for many molecules: 8 SF_6 contain 48 S–F bonds (WCH12 Jan 2024).

COMMON MISTAKE

Doing made − broken (gives the right number with the wrong sign), or adding the two totals (WCH12 Jan 2025 examiner report).
Using 3.5 or 6.5 O_2 for an alcohol because you forgot its own oxygen (WCH12 Oct 2019).
Miscounting bonds: draw the displayed formula every time.
Giving "experimental error" as a limitation. There's no experiment: the answer is averages and gas phase (WCH12 Jun 2023 examiner report).

SPEC 6.10

Calculating mean bond enthalpies from enthalpy changes

WHAT THIS SPEC POINT ASKS

Calculate a mean bond enthalpy from enthalpy changes of reaction.

6.10 The approach

Turn the bond-enthalpy equation round: if ΔH and all the other bond enthalpies are known, the unknown one is the only gap. Often you build a cycle with **gaseous atoms** at the top.

WORKED EXAMPLE: MEAN C–H BOND ENTHALPY IN METHANE

Data / kJ mol^{-1}: ΔfH(CH_4, g) = −74.8; ΔatH(C) = +716.7; ΔatH(H) = +218.0
Target: $CH_4(g) \rightarrow C(g) + 4H(g)$ (breaks all four C–H bonds)
Route: CH_4 → elements (reverse of formation, +74.8) → atoms (716.7 + 4 × 218.0)
ΔH = +74.8 + 716.7 + 872.0 = +1663.5 kJ mol^{-1}
Mean C–H = 1663.5 ÷ 4 = **+416 kJ mol^{-1}**

WORKED EXAMPLE: AN UNKNOWN BOND FROM ΔRH

$I_2(g) + Cl_2(g) \rightarrow 2ICl(g)$, $\Delta rH = -30$ kJ mol^{-1}. E(I–I) = 151, E(Cl–Cl) = 243 kJ mol^{-1}.
Broken = 151 + 243 = 394. Made = 2 × E(I–Cl).
−30 = 394 − 2E(I–Cl) → 2E(I–Cl) = 424 → **E(I–Cl) = 212 kJ mol^{-1}**

MARK-SCHEME LANGUAGE

- Divide by the number of bonds: "O–H = 928 ÷ 2 = 464", "C–H = 1740 ÷ 4 = 435" (WCH12 Jun 2022).
- For I–Cl: "151 + 243 = 394"; DO NOT AWARD 2 × 151 + 2 × 243 (WCH12 Jan 2026).

Often asked as: find C–H, then O–H, then C–O and C–C step by step from atomisation data; find one unknown bond from ΔrH.

COMMON MISTAKE

Forgetting to divide by the number of bonds of that type in the molecule.
Using the wrong number of moles: 2ICl contains **two** I–Cl bonds.

SPEC 6.11

Bond enthalpy, which bond breaks and how fast

WHAT THIS SPEC POINT ASKS

Understand that bond enthalpy data hint at which bond breaks first in a reaction, how hard it is to break, and so how quickly a reaction goes at room temperature.

6.11 The idea

- To react, at least one bond usually has to break. The **weakest bond** (lowest bond enthalpy) tends to **break first**.
- A weak bond means a **lower activation energy**, so more collisions have enough energy and the reaction is **faster at room temperature**.
- A very strong bond means a high activation energy: the substance is **unreactive** even if the overall reaction would be exothermic.

Example	Bond enthalpy / kJ mol^{-1}	Consequence
C–Cl, C–Br, C–I in halogenoalkanes	346, 290, 228	iodoalkanes hydrolyse **fastest** (10.13)
N≡N in nitrogen	945	N_2 is very unreactive: the Haber process needs high temperature and a catalyst
F–F	158 (unusually weak)	fluorine reacts violently even in the cold
O=O vs C–H in a fuel	both strong	methane needs a spark to start burning

MARK-SCHEME LANGUAGE

- "the **C–I bond is weaker** / has a lower bond enthalpy **than the C–Cl** bond", so it breaks more easily (WCH12 Jun 2022).
- Name the bond: never "iodine is weaker", never the H–I or H–Cl bond. DO NOT ACCEPT "C–I is easy to break" with no comparison. IGNORE bond length, electronegativity and size (WCH12 Oct 2024, Oct 2021).

Often asked as: why iodoalkanes react faster than chloroalkanes (the overlap with 10.13 is large); why N_2 is unreactive.

COMMON MISTAKE

Linking a **stronger** bond to a faster rate (WCH12 Jan 2022 examiner report).
Using polarity: C–Cl is the most polar but the **slowest** to hydrolyse. Bond strength wins.

Topic 7 – Intermolecular Forces

SPEC 7.1 The three types of intermolecular force

WHAT THIS SPEC POINT ASKS

Understand the nature of (i) London forces (instantaneous dipole–induced dipole), (ii) permanent dipole–permanent dipole interactions and (iii) hydrogen bonds.

7.1 Intermolecular is not intramolecular

- **Intramolecular** bonds are the covalent bonds **inside** a molecule. They are strong.
- **Intermolecular forces** act **between** separate molecules. They are much weaker (a hydrogen bond is roughly a tenth of a covalent bond, London forces weaker still).
- When a simple molecular substance melts or boils, **only the intermolecular forces are overcome**. The covalent bonds stay intact: boiling water gives H_2O molecules, not H and O atoms.

7.1 (i) London forces

1. Electrons in a molecule are always moving. At any instant they may be **unevenly spread**, giving an **instantaneous (temporary) dipole**.
2. That dipole repels or attracts the electrons of a neighbouring molecule, **inducing a dipole** in it.
3. The δ+ end of one attracts the δ− end of the other. The dipoles keep vanishing and re-forming, but on average there is an attraction.

- London forces act between **all** molecules (and atoms of noble gases), polar or not.
- They get **stronger with more electrons per molecule**: more electrons give bigger instantaneous dipoles.
- They are also stronger when molecules have **more points of contact** (long, straight molecules can lie close along their length).
- Other names: dispersion forces, van der Waals forces, instantaneous dipole–induced dipole forces.

7.1 (ii) Permanent dipole–permanent dipole interactions

- A molecule is **polar** if it contains polar bonds (atoms of different electronegativity) **and** its shape doesn't cancel them out. HCl, CH_3Cl and propanone are polar; CO_2 and CCl_4 are not (symmetric, so the dipoles cancel).
- The δ+ end of one polar molecule attracts the δ− end of the next. This acts **in addition to** London forces.
- For molecules with a similar number of electrons, pd–pd forces make the polar one boil higher: propanone (32 electrons, polar) boils at 56 °C; butane (34 electrons, non-polar) at about 0 °C.

7.1 (iii) Hydrogen bonds

- A hydrogen bond forms between a **δ+ hydrogen** that is covalently bonded to **N, O or F**, and a **lone pair on an N, O or F atom** of another molecule.
- N, O and F are very electronegative and small, so the N–H, O–H and F–H bonds are very polar, and the hydrogen (with no inner electrons) is a tiny, very exposed δ+ centre.
- The three atoms X–H···Y lie in a **straight line (180°)**, because the lone pair and the bonding pair repel each other as far apart as possible.
- Hydrogen bonds are the **strongest** intermolecular force, but still much weaker than covalent bonds.

Strength order (for molecules of similar size): hydrogen bonds > permanent dipole–dipole > London forces. But a big molecule's London forces can outweigh a small molecule's hydrogen bonds: I_2 (106

electrons, London only) is a solid, while water is a liquid.

MARK-SCHEME LANGUAGE

- London forces: "**instantaneous / temporary dipole**" caused by "an **uneven / asymmetrical distribution of electrons**", which "**induces a dipole**" in a neighbouring molecule (WCH12 Oct 2024). ACCEPT van der Waals, dispersion.
- Strength: "(hydrogen iodide has) **more electrons**". IGNORE "bigger molecule"; DO NOT AWARD with wrong extra reasons such as "more polarising".
- Comparisons must be comparative: "hydrogen bonds are **stronger than** London forces". "Hydrogen bonding is strong" alone scores nothing (WCH12 Jun 2022).
- Levels answers: "F_2 has **only** London forces; HCl has pd–pd as well; CH_3OH has hydrogen bonds" (WCH12 Jun 2021). Say "only" where it applies.
- Any mention of **breaking covalent bonds** on boiling loses the mark.

Often asked as: explain how London forces arise; a 6-mark comparison of boiling or liquefaction temperatures (F_2, HCl, CH_3OH; H_2 and NH_3); which forces are present in a named substance.

COMMON MISTAKE

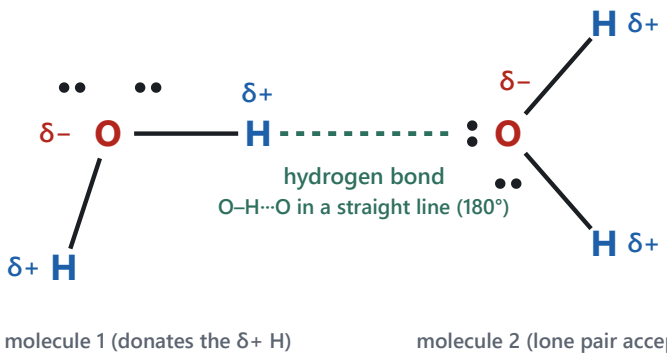
Describing the trend in London forces down a group when asked **how they arise** (WCH12 Oct 2024 examiner report).
Comparing Mr instead of **number of electrons** (WCH12 Jan 2025 examiner report).
Calling the covalent N–H bond a "hydrogen bond", or putting a hydrogen bond **inside** one molecule (WCH12 Jan 2022 examiner report).
Forgetting that polar and hydrogen-bonded molecules **also** have London forces.

SPEC 7.2

Hydrogen bonding in water, ammonia and hydrogen fluoride

WHAT THIS SPEC POINT ASKS

Understand the interactions in H_2O, liquid NH_3 and liquid HF that lead to hydrogen bonding.



A hydrogen bond (dotted) runs from the δ+ H of one water molecule to a lone pair on the O of the next. O–H···O is a straight line.

7.2 What each molecule brings

Molecule	δ+ H atoms (donors)	Lone pairs on N/O/F (acceptors)	Average hydrogen bonds per molecule
H_2O	2	2 (on O)	**2**
NH_3	3	1 (on N)	1 (limited by the single lone pair)
HF	1	3 (on F)	1 (limited by the single H)

- Each hydrogen bond needs one δ+ H **and** one lone pair. Water is the only one of the three with equal numbers (two of each), so it forms the **most hydrogen bonds per molecule**. That's why water boils highest of the three.
- **HF** forms the **strongest single** hydrogen bond, because F is the most electronegative element, so H–F is the most polar bond. In liquid HF the molecules form zig-zag chains, F–H···F–H···F.
- **NH_3**: N is less electronegative than O or F, so its hydrogen bonds are weaker. In liquid ammonia each N lone pair accepts an H from a neighbouring molecule.

7.2 Drawing a hydrogen bond (the marks)

1. Draw both molecules with all atoms (no "O_2H").
2. Put **δ+ on the H** and **δ− on the O (N, F)**.
3. Show the **lone pair** on the accepting O, N or F.
4. Draw the hydrogen bond as a **dotted or dashed line** from that lone pair to the H.
5. Make O–H···O **straight** and label it **180°**.

MARK-SCHEME LANGUAGE

- "Hydrogen bond from a **lone pair on the oxygen** of one molecule to the **(δ+) hydrogen** of another", "**linear** O–H···O, labelled **180°**", "δ+ on H, δ− on O". The bond must be dotted, not solid (WCH12 Oct 2019).
- Conditions: "H attached to N, O or F" plus "N, O or F with a lone pair".
- HF strongest: "F has the highest electronegativity, so H–F is the most polar bond" (WCH12 Jan 2020).
- Ammonia: "hydrogen bonds from the **N lone pair** to the δ+ H of a **different** NH_3 molecule" (WCH12 Jun 2023).

Often asked as: draw a hydrogen bond between two water molecules, two alcohol molecules, or ethanol and water; explain why HF boils higher than HCl.

BEYOND THE SPEC, BUT EXAMINED

Water forms more hydrogen bonds than NH_3 or HF because it has **two lone pairs and two H atoms** (a 2 : 1 ratio compared with NH_3's one lone pair) (WCH12 Jan 2025).
The N–H···N angle in ammonia need not be drawn as exactly 180° (WCH12 Jun 2023).

COMMON MISTAKE

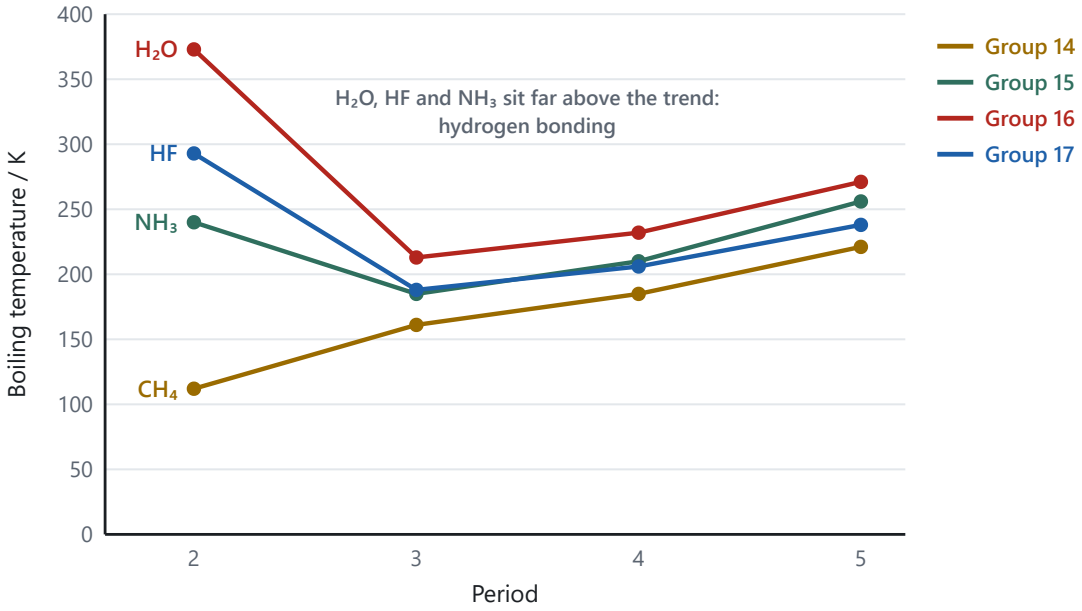
Missing lone pairs, missing the 180° label, or drawing H–O···H (wrong atoms lined up) (WCH12 Jan 2025 examiner report).
Drawing the hydrogen bond to the δ+ H of a C–H group: C–H is not polar enough.

SPEC 7.3 The anomalous properties of water

WHAT THIS SPEC POINT ASKS

Understand how hydrogen bonding explains (i) water's high melting and boiling temperatures compared with similar molecules and (ii) why ice is less dense than liquid water.

7.3 (i) High melting and boiling temperatures



From period 3 down, boiling temperature rises as the number of electrons (and so London forces) rises. H_2O, HF and NH_3 break the pattern because of hydrogen bonding. CH_4 (Group 14) has no hydrogen bonding and follows the trend.

- Going down Group 16 from H_2S to H_2Te, boiling temperature rises because the molecules have **more electrons**, so **stronger London forces**.
- Extending that trend back, H_2O "should" boil at around −80 °C. It actually boils at **100 °C**.
- Water molecules form **hydrogen bonds** (two per molecule), which are **much stronger** than the London and pd–pd forces in H_2S. **Much more energy** is needed to separate the molecules.
- The same reasoning explains why water is a liquid, not a gas, at room temperature, and why its melting temperature (0 °C) is so high for its size.

7.3 (ii) Ice is less dense than water

- In **ice**, every water molecule is hydrogen-bonded to **four** others in a fixed, tetrahedral arrangement. This makes an **open lattice** with a lot of empty space between the molecules.
- When ice melts, some hydrogen bonds **break**. The open structure collapses and the molecules can move **closer together**, so liquid water is **denser** (about 1.00 g cm^{-3}, compared with about 0.92 g cm^{-3} for ice).
- So ice **floats**. Ponds freeze from the top down, and the ice layer insulates the water below, so aquatic life survives.
- For almost every other substance, the solid is denser than the liquid.

MARK-SCHEME LANGUAGE

Water v ammonia (WCH12 Jan 2025, three marks):

- "similar number of electrons, so London forces are similar"
- "water forms **more hydrogen bonds** (per molecule) than ammonia"
- "density of water increases as it changes from solid to liquid" because the **open hydrogen-bonded structure of ice** collapses.

ACCEPT "hydrogen bonds in water are stronger because O is more electronegative than N". IGNORE comments on pd–pd forces.

Often asked as: explain water's boiling temperature compared with H_2S or NH_3; explain the density change on melting.

COMMON MISTAKE

Quoting Mr; saying ammonia can't hydrogen bond; comparing the density of water with the density of ammonia instead of ice with liquid water (WCH12 Jan 2025 examiner report: half the candidates scored zero).
Saying covalent O–H bonds break when water boils.

SPEC 7.4

Predicting hydrogen bonding in other molecules

WHAT THIS SPEC POINT ASKS

Predict whether hydrogen bonding is present in molecules similar to H_2O, NH_3 and HF.

7.4 The two-part test

A substance hydrogen-bonds **with itself** only if its molecules have **both**:

1. an **H atom bonded directly to N, O or F** (an O–H, N–H or H–F group), **and**
2. a **lone pair on an N, O or F** atom.

Molecule	O–H / N–H present?	Hydrogen bonds between its own molecules?
Alcohols, e.g. CH_3OH, C_2H_5OH	O–H	**yes**
Carboxylic acids, e.g. CH_3COOH	O–H	**yes** (strongly: they can pair up as dimers)
Amines, e.g. CH_3NH_2	N–H	**yes**
Urea, $(NH_2)_2CO$	N–H	**yes** (and pd–pd)
Propane-1,2,3-triol (glycerol)	three O–H	**yes**, three per molecule: very viscous
Propanone, CH_3COCH_3	no (only C–H)	**no**: pd–pd and London only
Ethers, aldehydes, esters	no	**no** with themselves
Halogenoalkanes, e.g. CH_3Cl	no	**no**: pd–pd and London
HCl, HBr, HI	H–X, but X is not N, O or F	**no**

- Molecules like **propanone** can't hydrogen-bond to each other, but the lone pairs on their O **can accept** a hydrogen bond from water. That's why propanone mixes with water (7.6).
- **Count the hydrogen-bonding groups.** More O–H groups → more hydrogen bonds per molecule → higher boiling temperature and viscosity (ethane-1,2-diol and glycerol boil much higher than ethanol).

MARK-SCHEME LANGUAGE

- Urea: "urea forms **hydrogen bonds**", "(and permanent) dipole–dipole forces", "these are **stronger than London forces**" (WCH12 Jan 2022).
- Propan-2-ol v propanone: "propan-2-ol forms hydrogen bonds (between its molecules); propanone has only London and pd–pd forces; hydrogen bonds are stronger". Both dissolve in water because both can form hydrogen bonds **with water** (WCH12 Oct 2025).
- Glycerol: "forms **three** hydrogen bonds per molecule" / "three OH groups" (WCH12 Jun 2023).

Often asked as: which of these molecules can form hydrogen bonds; compare the boiling temperatures of an alcohol and a ketone or alkane; explain the properties of a molecule in a Section C context.

COMMON MISTAKE

Drawing a hydrogen bond from an H bonded to carbon.
Forgetting London forces in a comparison: very few answers mention them (WCH12 Jan 2022 examiner report).
Saying HCl hydrogen-bonds. Cl is not N, O or F.

SPEC 7.5 Intermolecular forces and physical properties

WHAT THIS SPEC POINT ASKS

Explain, using intermolecular forces: (i) boiling temperatures of alkanes as chain length increases, (ii) the effect of branching, (iii) the low volatility of alcohols compared with alkanes with a similar number of electrons, and (iv) the boiling temperatures of HF to HI.

7.5 The answer pattern

Every boiling-point answer has the same three links:

1. **Name the force(s)** present in each substance.
2. **Compare** their strength, with a reason (number of electrons, contact area, hydrogen bonding).
3. **More energy is needed to overcome** the stronger intermolecular forces (to **separate the molecules**), so the boiling temperature is higher.

7.5 (i) Chain length in alkanes

Alkane	**CH_4**	**C_2H_6**	**C_3H_8**	**C_4H_{10}**	**C_5H_{12}**	**C_6H_{14}**
Boiling temperature / °C	−162	−89	−42	−1	36	69

- Each extra CH_2 adds **8 electrons** and makes the molecule longer, so there are **more electrons** and **more points of contact**. London forces get stronger, so more energy is needed to separate the molecules.
- Alkanes are non-polar: London forces are the **only** intermolecular force.

7.5 (ii) Branching

Isomer of C_5H_{12}	**pentane**	**2-methylbutane**	**2,2-dimethylpropane**
Shape	straight chain	one branch	most branched, nearly spherical
Boiling temperature / °C	36	28	10

- Isomers have the **same number of electrons**, so the difference is all about shape.

- Straight chains can lie **close together along their length**: **more points of contact / larger surface area of contact**, so stronger London forces.
- Branched molecules are more compact and can't get as close, so they have fewer points of contact, weaker London forces and a lower boiling temperature.

7.5 (iii) Alcohols v alkanes

Compound	Electrons	Forces	Boiling temperature / °C
propane, C_3H_8	26	London only	−42
ethanol, C_2H_5OH	26	London + **hydrogen bonds** (+ pd–pd)	78
butane, C_4H_{10}	34	London only	−1
propan-1-ol, C_3H_7OH	34	London + **hydrogen bonds** (+ pd–pd)	97

- With the **same number of electrons**, the London forces are **similar**. The alcohol also has **hydrogen bonds** between its O–H groups, which are **much stronger**, so it needs much more energy to boil. Alcohols are **less volatile** (evaporate less readily).

7.5 (iv) The hydrogen halides

	HF	HCl	HBr	HI
Electrons	10	18	36	54
Boiling temperature / °C	**20**	−85	−67	−35

- **HCl → HBr → HI**: boiling temperature **rises** because the number of electrons rises, so **London forces get stronger**. (The permanent dipole actually gets **smaller** from HCl to HI, as electronegativity falls, so the trend is driven by London forces, not pd–pd.)
- **HF** is far higher than expected because only HF forms **hydrogen bonds**, which are much stronger than the forces between HCl molecules.

MARK-SCHEME LANGUAGE

- Alkanes: "London forces **increase with the number of electrons**", "more energy needed to **overcome / break these intermolecular forces**". ALLOW "more points of contact / surface area". Mentioning hydrogen bonds or pd–pd loses the force marks; mentioning breaking covalent bonds loses the energy mark (WCH12 Oct 2020).
- Branching: "London forces are **stronger between straight chains**" because of "**greater surface area / more points of contact**". DO NOT AWARD longer carbon bonds or breaking covalent bonds (WCH12 Oct 2019).
- Hydrogen halides: "London forces increase from HCl to HI (more electrons)", "**only HF** has hydrogen bonding", "hydrogen bonds are (much) stronger" (WCH12 Jun 2022). DO NOT AWARD ions, or breaking H–Cl bonds.
- Group 15 hydrides: London forces rise with electrons; "(only) ammonia has hydrogen bonding" (WCH12 Jun 2025). DO NOT AWARD "London forces are similar down the group".

Often asked as: 6-mark comparisons of boiling temperatures; explain a data table or a graph of boiling temperature against period or chain length.

BEYOND THE SPEC, BUT EXAMINED

Iodine v bromine: "iodine has **more electrons per molecule**", so stronger London forces, so a higher melting temperature. DO NOT AWARD "between atoms" (WCH12 Jun 2021).
1-iodopropane has the strongest London forces of the halogenopropanes even though its C–I dipole is the smallest (WCH12 Oct 2021).
Pentane is a **liquid**, so part of the energy from burning it isn't measured in ΔcH calculated from gaseous bond enthalpies; boiling and combustion trends must not be confused (WCH12 Oct 2020).

COMMON MISTAKE

"Straight-chain isomers boil higher" with no reason; "London forces are present in both" with no comparison (WCH12 Oct 2019 examiner report).
Talking about size, not the **number of electrons** (WCH12 Oct 2021 examiner report).
Claiming hydrogen bonds in HCl, HBr, HI or in 1-iodopropane.

SPEC 7.6 Choosing a solvent

WHAT THIS SPEC POINT ASKS

Understand why: (i) water dissolves some ionic compounds (hydration of ions); (ii) water dissolves simple alcohols (hydrogen bonding); (iii) water is a poor solvent for compounds that can't hydrogen-bond with it, including polar ones like halogenoalkanes; (iv) non-aqueous solvents dissolve compounds with similar intermolecular forces.

7.6 The energy rule behind dissolving

To dissolve, you must **break** the forces between solute particles **and** some of the forces between solvent molecules. That costs energy. It is paid back by the **new attractions between solute and solvent**. A substance dissolves well when the new solute–solvent attractions are **similar in strength to (or stronger than)** the ones broken.

"Like dissolves like" is the short version, but on its own it earns nothing: always **name the forces**.

7.6 (i) Ionic compounds in water: hydration

- Water is polar. Its **δ− oxygen** is attracted to **positive ions**; its **δ+ hydrogens** are attracted to **negative ions**.
- Several water molecules surround each ion. These are **hydrated ions**, and forming them releases energy (**hydration enthalpy**).
- If the energy released by hydration is enough to compensate for breaking up the **ionic lattice**, the compound dissolves.
- If the lattice is very strong (small or highly charged ions, e.g. $MgCO_3$, $CaCO_3$, $BaSO_4$), hydration can't make up for it and the compound is **insoluble**.

When you draw hydrated ions: at least two water molecules round each ion, **O pointing at the cation**, **H pointing at the anion**. These are **ion–dipole** attractions, not hydrogen bonds.

7.6 (ii) Simple alcohols in water

- Methanol, ethanol and propan-1-ol mix with water in all proportions. The **O–H group forms hydrogen bonds with water molecules**, and these new hydrogen bonds are similar in strength to the ones broken in pure water and pure alcohol.

- As the hydrocarbon chain gets longer, solubility **falls**. The non-polar chain can only form London forces with water, and it gets in the way of water's own hydrogen bonding. Hexan-1-ol is only slightly soluble.

7.6 (iii) Water as a poor solvent

- **Halogenoalkanes** are polar (C–X dipole) but **can't form hydrogen bonds** with water. To dissolve, they would have to break water's strong hydrogen bonds and replace them with weaker pd–pd and London forces. So they are **immiscible** with water and form a separate layer.
- The same applies to **alkanes** and other non-polar compounds, which can only offer London forces.
- That's why halogenoalkane hydrolysis with silver nitrate is done in **ethanol**: ethanol mixes with both the halogenoalkane and the aqueous silver nitrate (10.8).

7.6 (iv) Non-aqueous solvents

- A non-polar solvent such as **hexane** or **cyclohexane** dissolves non-polar solutes (iodine, alkanes, fats, halogenoalkanes): solute and solvent both rely on **London forces**, so the new attractions match the old ones.
- **Iodine**: barely soluble in water, very soluble in cyclohexane, where it gives a **purple** solution.
- Ionic compounds don't dissolve in hexane: London forces with hexane release nowhere near enough energy to break the lattice.

Solute	Water	Ethanol	Hexane / cyclohexane
NaCl (ionic)	dissolves (hydration)	very slightly	no
Ethanol	mixes (hydrogen bonds)	—	dissolves (London forces with its chain)
1-bromobutane	immiscible	mixes	mixes
Iodine	very slightly (pale yellow/brown)	dissolves (brown)	dissolves (purple)

MARK-SCHEME LANGUAGE

- Hydration diagram: "the **oxygen** is closest to the (calcium) **cation**; the **hydrogen** is closest to the (chloride) **anion**", at least **two** water molecules per ion (WCH12 Jun 2021). DO NOT AWARD O_2H. Penalise a labelled "hydrogen bond" or full charges on water (WCH12 Oct 2024).
- Insoluble ionic solid: "**large** amount of energy needed to **break the ionic lattice**" but "**smaller** amount released on **hydration**" (WCH12 Jun 2019). DO NOT AWARD molecules, atoms or London forces.
- Iodine in cyclohexane: "cyclohexane and iodine form **London forces**"; "hydrogen bonds between water molecules are stronger than the London forces between iodine and water" (WCH12 Jun 2019).
- Ethanol and water mix because "they form hydrogen bonds with each other", "from the lone pair on an O of one to the δ+ H of the other", and the new forces are **similar in strength** to those broken (WCH12 Jun 2021).

Often asked as: draw ions hydrated by water; explain why a solid is insoluble; why iodine dissolves in cyclohexane but not water; why ethanol and water mix.

BEYOND THE SPEC, BUT EXAMINED

Layer colours when iodine is shaken with water and cyclohexane: **pale yellow** aqueous layer, **pink / purple / violet** organic layer (WCH12 Jun 2019).
Explaining insolubility by reversing the dissolving argument: compare solute–solute, solvent–solvent and solute–solvent forces (WCH12 Jun 2019 examiner report).

COMMON MISTAKE

"Like dissolves like" with no forces named (WCH12 Jun 2019, Oct 2019 examiner reports).
Saying ions form hydrogen bonds with water, or writing about intermolecular forces in an ionic solid.
Saying iodine forms hydrogen bonds with cyclohexane, or "reacts" with it.
Calling the aqueous layer colourless when "most", not all, of the iodine moves into the organic layer.

Topic 8 – Redox Chemistry and Groups 1, 2 and 7

SPEC 8.1 Oxidation numbers and the rules

WHAT THIS SPEC POINT ASKS

Know what an oxidation number is and the rules for assigning one.

8.1 What an oxidation number is

An **oxidation number** is the charge an atom **would have if the compound were made entirely of ions**. In a covalent bond, imagine the shared electrons all go to the more electronegative atom. It is a book-keeping tool: it lets you track electrons in any reaction, ionic or covalent.

Write the sign **before** the number: +2, −1. (A charge on an ion is written the other way round: Mg^{2+}.)

8.1 The rules, in the order you apply them

Rule	Example
An **uncombined element** is **0**	Na, Fe, O_2, Cl_2, S_8 are all 0
A **simple ion**: oxidation number = its charge	Na^+ +1, Mg^{2+} +2, Cl^- −1, O^{2-} −2
The numbers in a **neutral compound add up to 0**	H_2O: 2(+1) + (−2) = 0
The numbers in an **ion add up to the ion's charge**	SO_4^{2-}: (+6) + 4(−2) = −2
Group 1 metals always **+1**; **Group 2** always **+2**	Na in NaCl, Ca in $CaCO_3$
Fluorine is always **−1**	F in HF, CaF_2, F_2O
Hydrogen is **+1**, **except in metal hydrides (−1)**	H in H_2O is +1; in NaH, CaH_2 it is −1
Oxygen is **−2**, **except in peroxides (−1)** and with fluorine (+2 in F_2O)	O in H_2O_2 and Na_2O_2 is −1
Chlorine is **−1** in chlorides, but positive when bonded to O or F	Cl in ClO_3^- is +5
In a covalent bond, the **more electronegative** atom takes the **negative** value	In NH_3, N is −3 and H is +1

EXAM TIP

Fill in the "fixed" elements first (Group 1 and 2 metals, F, then H, then O), then the unknown is whatever makes the total equal the overall charge. Finally check the sum.

MARK-SCHEME LANGUAGE

- The meaning of an oxidation number: "the **charge** the atom **would have if the species were fully ionic**" (WCH12 Jun 2024).
- ALLOW "1" or "1+" for +1 (WCH12 Oct 2020), but writing the sign first is safest.

Often asked as: give oxidation numbers in a named reaction; explain what the Roman numeral in a name means.

COMMON MISTAKE

Multiplying instead of sharing out: in H_2O each H is **+1**, not +2 (WCH12 Oct 2021 examiner report). Giving the total for an ion (e.g. "NO_3^- changes from 5 to 4") without naming the **element** that changes (WCH12 Oct 2020 examiner report).

SPEC **8.2**

Calculating oxidation numbers, including peroxides and hydrides

WHAT THIS SPEC POINT ASKS

Calculate the oxidation number of an element in compounds and ions, including peroxides and metal hydrides.

WORKED EXAMPLES: COMPOUNDS AND IONS

SO_4^{2-}: S + 4(−2) = −2 → **S = +6**
$Cr_2O_7^{2-}$: 2Cr + 7(−2) = −2 → 2Cr = +12 → **Cr = +6**
MnO_4^-: Mn + 4(−2) = −1 → **Mn = +7**
NH_4^+: N + 4(+1) = +1 → **N = −3**
ClO_3^-: Cl + 3(−2) = −1 → **Cl = +5**
$KClO_4$: (+1) + Cl + 4(−2) = 0 → **Cl = +7**
$AgNO_3$: Ag^+ is +1 (it's the ion Ag^+ and NO_3^-); N + 3(−2) = −1 → **N = +5**
IO_3^-: I + 3(−2) = −1 → **I = +5**

WORKED EXAMPLES: THE EXCEPTIONS

H_2O_2 (hydrogen peroxide): 2(+1) + 2O = 0 → **O = −1**
Na_2O_2 (sodium peroxide): 2(+1) + 2O = 0 → **O = −1**
BaO_2 (barium peroxide): (+2) + 2O = 0 → **O = −1**
NaH (sodium hydride): Na is +1, so **H = −1**
CaH_2 (calcium hydride): (+2) + 2H = 0 → **H = −1**
$LiAlH_4$: Li +1, Al +3, so 4H = −4 → **H = −1**
How to spot them: a peroxide has an O–O bond (the formula has "too many" oxygens for O to be −2); a metal hydride is a metal joined to H only.

8.2 Fractional and average values

- In **$S_4O_6^{2-}$** (tetrathionate), 4S + 6(−2) = −2 → 4S = +10 → S = **+2.5**. That is an **average**: the sulfur atoms are actually in different environments.
- In **Fe_3O_4**, Fe averages **+8/3** (it contains both Fe^{2+} and Fe^{3+}).

MARK-SCHEME LANGUAGE

- Iodine species: "IO_3^- = +5, I^- = −1, I_2 = 0". DO NOT AWARD I^- = −5 (WCH12 Jun 2022).
- Manganese and carbon in MnO_4^- and $C_2O_4^{2-}$: Mn +7, C +3 (WCH12 Jan 2025).

Often asked as: oxidation number of an element in an unfamiliar ion; changes during a reaction.

BEYOND THE SPEC, BUT EXAMINED

Sulfur: **−2** in H_2S, **0** in S, **+4** in SO_2, **+6** in H_2SO_4 and SO_4^{2-}. You need these for the halide + concentrated sulfuric acid reactions (WCH12 Jun 2022).
Cobalt goes from +4 in CoO_2 to +3 in $LiCoO_2$ in a lithium-ion cell (WCH12 Oct 2024).

COMMON MISTAKE

Silver as +9 in $AgNO_3$: the +1 belongs to Ag, and the rest of the charge belongs to the nitrate ion (WCH12 Oct 2020 examiner report).
Chlorine as −2 in $CaCl_2$ (it's −1: two chlorines share the −2 total) (WCH12 Jan 2023 examiner report).
Forgetting that H is −1 in a metal hydride and O is −1 in a peroxide.

SPEC 8.3

Oxidation numbers as Roman numerals

WHAT THIS SPEC POINT ASKS

Show the oxidation number of an element in a compound or ion with a Roman numeral.

8.3 When and how

- Use a Roman numeral when an element can have **more than one** oxidation number: transition metals, and non-metals in oxyanions.
- The numeral goes in brackets **straight after the element it describes**, with no space: iron(III) chloride.
- For a negative ion ending in **-ate**, the numeral describes the central element: sulfate(VI), chlorate(V).
- Group 1 and 2 metals never need one: it's sodium chloride, not sodium(I) chloride.

Name	Formula	Element and its oxidation number
iron(II) sulfate	$FeSO_4$	Fe +2
iron(III) chloride	$FeCl_3$	Fe +3
copper(I) oxide	Cu_2O	Cu +1
manganese(IV) oxide	MnO_2	Mn +4
sodium chlorate(I)	$NaClO$	Cl +1 (the active ingredient in bleach)
sodium chlorate(V)	$NaClO_3$	Cl +5
potassium chlorate(VII)	$KClO_4$	Cl +7
potassium manganate(VII)	$KMnO_4$	Mn +7
potassium dichromate(VI)	$K_2Cr_2O_7$	Cr +6
sulfate(VI) / sulfate(IV)	SO_4^{2-} / SO_3^{2-}	S +6 / +4
nitrate(V) / nitrate(III)	NO_3^- / NO_2^-	N +5 / +3

Everyday names drop the numeral when there's no doubt: "sulfate" means sulfate(VI), "nitrate" means nitrate(V), "nitrite" is nitrate(III).

MARK-SCHEME LANGUAGE

- "(V) is the **oxidation number of the chlorine** (in the ion)" or "the charge chlorine would have if the ion were fully ionic". DO NOT AWARD "the charge on chlorate(V) is +5" (WCH12 Jun 2024).
- "Potassium chlorate(VII)" is $KClO_4$; "potassium chlorate" alone is not enough (WCH12 Jun 2021).

Often asked as: name a compound with its numeral; explain what the numeral means.

COMMON MISTAKE

Saying the numeral is the charge on the **ion**. Chlorate(V), ClO_3^-, has a charge of −1; the V refers to the chlorine.

SPEC 8.4 Writing formulae from oxidation numbers

WHAT THIS SPEC POINT ASKS

Write the formula of a compound or ion when given its name with oxidation numbers.

8.4 Method

1. Write the oxidation number of each element (from the numeral, or the fixed rules).
2. For an oxyanion, work out how many O atoms give the right charge; for a compound, balance the total to zero.
3. Use brackets round a polyatomic ion if you need more than one.

WORKED EXAMPLES

Tin(IV) oxide: Sn +4, O −2 → need 2 O → **SnO_2**
Iron(III) sulfate: Fe^{3+} and SO_4^{2-} → lowest common multiple 6 → **$Fe_2(SO_4)_3$**
Lead(II) nitrate: Pb^{2+} and NO_3^- → **$Pb(NO_3)_2$**
Sodium chlorate(V): Cl +5 with O −2. For an ion of charge −1: +5 + 3(−2) = −1 → ClO_3^- → **$NaClO_3$**
Potassium chromate(VI): Cr +6; +6 + 4(−2) = −2 → CrO_4^{2-} → **K_2CrO_4**
Sodium bromate(I): Br +1; +1 + (−2) = −1 → BrO^- → **NaBrO**

EXAM TIP

With an oxyanion, the usual charges are −1 (ClO_3^-, NO_3^-, MnO_4^-) or −2 (SO_4^{2-}, CO_3^{2-}, CrO_4^{2-}). Choose the number of oxygens that gives one of these, then check.

COMMON MISTAKE

Writing ClO_5 for chlorate(V). The V is the oxidation number, not the number of oxygens.

SPEC 8.5 Oxidation and reduction in s- and p-block reactions

WHAT THIS SPEC POINT ASKS

Understand oxidation and reduction as electron transfer and as changes in oxidation number, and apply this to reactions of s-block and p-block elements.

8.5 Two ways to spot redox

	Oxidation	Reduction
Electrons	**loss** of electrons	**gain** of electrons
Oxidation number	**increases**	**decreases**

Memory aid: **OIL RIG** (Oxidation Is Loss, Reduction Is Gain). A redox reaction always has both: electrons lost by one species are gained by another.

8.5 s-block and p-block examples

Reaction	Oxidised	Reduced
$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$	Na: 0 → +1	Cl: 0 → −1
$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	Mg: 0 → +2	O: 0 → −2
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	Mg: 0 → +2	H: +1 → 0
$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$	K: 0 → +1	H: +1 → 0
$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$	Br: −1 → 0	Cl: 0 → −1
$Ag(s) + 2HNO_3(aq) \rightarrow AgNO_3(aq) + NO_2(g) + H_2O(l)$	Ag: 0 → +1	N: +5 → +4

EXAM TIP

Write the oxidation number above **every** element first. Usually only two elements change. Name the **element**, give both numbers ("from 0 to +1") and say which is oxidised and which reduced.

MARK-SCHEME LANGUAGE

- "Silver is **oxidised from 0 to +1**; nitrogen is **reduced from +5 to +4**" (WCH12 Oct 2020).
- For K + water: "potassium oxidised 0 → +1; **hydrogen** reduced +1 → 0" (WCH12 Oct 2021).
- When the question says "in terms of oxidation numbers", answer with numbers, not electrons.

Often asked as: identify what is oxidised and reduced in a given equation, with oxidation numbers.

BEYOND THE SPEC, BUT EXAMINED

Redox in industrial and environmental contexts: in selective catalytic reduction, NH_3 is oxidised (N −3 → 0) and NO reduced (N +2 → 0) (WCH12 Jan 2020).

COMMON MISTAKE

Hydrogen "starting at +2" in water because there are two of them; naming water or oxygen as the species reduced in K + H_2O (WCH12 Oct 2021 examiner report).
Showing a change for a spectator such as calcium (WCH12 Jan 2023 examiner report).
Swapping oxidation and reduction.

SPEC 8.6 Oxidising agents and reducing agents

WHAT THIS SPEC POINT ASKS

Know that oxidising agents gain electrons and reducing agents lose electrons.

8.6 The definitions

- An **oxidising agent accepts (gains) electrons** from another species, so it oxidises that species. The oxidising agent is itself **reduced** (its oxidation number falls).
- A **reducing agent donates (loses) electrons**, so it reduces another species. The reducing agent is itself **oxidised** (its oxidation number rises).

In $2Na + Cl_2 \rightarrow 2NaCl$: Cl_2 is the oxidising agent (gains electrons); Na is the reducing agent (loses electrons).

Common oxidising agents	Common reducing agents
halogens ($F_2 > Cl_2 > Br_2 > I_2$)	s-block metals (Group 1, Group 2) and other reactive metals
O_2	H_2, C, CO
acidified MnO_4^- (manganate(VII))	halide ions ($I^- > Br^- > Cl^-$)
acidified $Cr_2O_7^{2-}$ (dichromate(VI))	Fe^{2+}, SO_2
concentrated H_2SO_4, HNO_3	
H_2O_2	H_2O_2 (with a stronger oxidising agent)

- **Hydrogen peroxide** can be both. As an oxidant, O goes $-1 \rightarrow -2$ (e.g. with I^- or Fe^{2+} in acid). As a reductant, O goes $-1 \rightarrow 0$ and O_2 is given off (e.g. with acidified MnO_4^-).

MARK-SCHEME LANGUAGE

- "Iodide ions are **better reducing agents** than chloride ions" because "**electrons are more easily lost** from iodide ions" (the outer electrons are further from the nucleus and more shielded) (WCH12 Jun 2025).
- DO NOT AWARD "iodine" or "chlorine" for the ions, or anything linking it to a better **oxidising** agent.

Often asked as: name the oxidising or reducing agent in an equation; compare the reducing power of two halide ions.

COMMON MISTAKE

Saying the oxidising agent is oxidised. It is the opposite: it **causes** oxidation and is itself reduced. Writing "iodine is a stronger reducing agent" when you mean **iodide** ions.

SPEC 8.7 Disproportionation

WHAT THIS SPEC POINT ASKS

Understand that in disproportionation an element in a single species is oxidised and reduced at the same time.

8.7 The definition

Disproportionation: a reaction in which **the same element in the same species** is **simultaneously oxidised and reduced**. One species goes in; the element comes out in two different oxidation states, one higher and one lower.

Reaction	Oxidation numbers of the element
$Cl_2 + H_2O \rightleftharpoons HCl + HClO$	Cl: 0 → −1 (reduced) and 0 → +1 (oxidised)
$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$ (cold, dilute)	Cl: 0 → −1 and 0 → +1
$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$ (hot)	Cl: 0 → −1 and 0 → +5
$2H_2O_2 \rightarrow 2H_2O + O_2$	O: −1 → −2 and −1 → 0
$2Cu^+(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$	Cu: +1 → 0 and +1 → +2
$3ClO^- \rightarrow 2Cl^- + ClO_3^-$ (chlorate(I) on heating)	Cl: +1 → −1 and +1 → +5

WORKED EXAMPLE: ANSWERING A DISPROPORTIONATION QUESTION

"Explain why $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$ is a disproportionation."

- It is a **disproportionation** reaction.
- Chlorine is **oxidised** from **0** in Cl_2 to **+1** in ClO^-.
- Chlorine is **reduced** from **0** in Cl_2 to **−1** in Cl^-.
- The same element in one species (Cl_2) is oxidised and reduced at the same time.

MARK-SCHEME LANGUAGE

- "disproportionation" (a stand-alone mark), then "chlorine **oxidised** from 0 to +1 (or +5)" and "**reduced** from 0 to −1". "Reference to 0 for Cl_2 required once only" (WCH12 Jun 2021, Oct 2025).
- Oxidation numbers may be written on the equation; three correct numbers without the oxidised/reduced links score one mark.
- Do **not** give changes for H or O: "Do not award … if changes of oxidation numbers of H and O given" (WCH12 Jun 2023).

Often asked as: Cl_2 with water, cold NaOH, hot NaOH or $Ca(OH)_2$; a 6-mark question on all three (WCH12 Jan 2022).

COMMON MISTAKE

Not saying it is **simultaneous** oxidation and reduction, or not linking "oxidised" to an increase and "reduced" to a decrease (WCH12 Jan 2023 examiner report).
Adding oxidation-number changes for H, O, Na or Ca, which don't change.

SPEC **8.8**

Using oxidation numbers to classify reactions

WHAT THIS SPEC POINT ASKS

Know that oxidation numbers let you classify a reaction as redox and as disproportionation.

8.8 The decision tree

1. Write oxidation numbers for every element on both sides.
2. **No element changes** → **not redox** (e.g. acid–base, precipitation, many thermal decompositions).
3. One element goes up and another goes down → **redox**.
4. The **same element in one species** goes both up and down → **disproportionation** (a special kind of redox).

Reaction	Changes?	Type
$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$	none	acid–base, **not redox**
$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	none	precipitation, **not redox**
$CaCO_3 \rightarrow CaO + CO_2$	none	thermal decomposition, **not redox**
$2NaBr + 2H_2SO_4 \rightarrow Br_2 + SO_2 + Na_2SO_4 + 2H_2O$	Br −1 → 0; S +6 → +4	**redox**
$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$	N +5 → +4; O −2 → 0	**redox** (but not disproportionation: two different elements)
$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$	I +5 → 0 and I −1 → 0	**redox, not disproportionation**: two different species of iodine end up in the same state

MARK-SCHEME LANGUAGE

- Not redox: "**no oxidation number has changed**" (ALLOW "no electrons transferred") (WCH12 Jun 2025).
- IO_3^- + I^-: not disproportionation because "**two different species** (of iodine) are oxidised and reduced (to form the same species)" / "there is not one species being oxidised and reduced" (WCH12 Jun 2022).

Often asked as: show that a reaction is (or is not) redox or disproportionation.

COMMON MISTAKE

Writing "compounds" when you mean species or ions (WCH12 Jun 2022 examiner report).
Saying both species are reduced in the iodate–iodide reaction.

SPEC 8.9

Metals form positive ions; non-metals form negative ions

WHAT THIS SPEC POINT ASKS

Understand that metals generally lose electrons to form positive ions (oxidation number rises), and non-metals generally gain electrons to form negative ions (oxidation number falls).

8.9 The pattern

- **Metals** have few outer electrons and relatively low ionisation energies. They **lose** electrons to form **positive ions**: $Na \rightarrow Na^+ + e^-$, $Ca \rightarrow Ca^{2+} + 2e^-$. Their oxidation number **increases**, so metals are **oxidised** and act as **reducing agents**.
- **Non-metals** have nearly full outer shells and attract electrons strongly. They **gain** electrons to form **negative ions**: $Cl_2 + 2e^- \rightarrow 2Cl^-$, $O_2 + 4e^- \rightarrow 2O^{2-}$. Their oxidation number **decreases**, so non-metals are **reduced** and act as **oxidising agents**.
- This is why Group 1 and 2 metals are reducing agents (8.12) and the halogens are oxidising agents (8.24).

EXAM TIP

A metal is never found with a negative oxidation number in a simple compound. If you get one, recheck.

COMMON MISTAKE

Saying "non-metals lose electrons" by analogy with metals. Non-metals (other than H when it bonds to a more electronegative atom) gain electrons when they react with metals.

SPEC 8.10

Ionic half-equations and full ionic equations

WHAT THIS SPEC POINT ASKS

Write ionic half-equations and combine them into a full ionic equation.

8.10 Writing a half-equation (acidic solution)

1. Write the species that changes and what it becomes. **Balance that element.**
2. Balance **O** by adding **H_2O**.
3. Balance **H** by adding **H^+**.
4. Balance **charge** by adding **electrons** (e^-) to the more positive side.
5. Check: atoms and total charge both balance.

WORKED EXAMPLE: MANGANATE(VII) TO MANGANESE(II)

MnO_4^- → Mn^{2+} (Mn already balanced)
MnO_4^- → Mn^{2+} + **$4H_2O$** (balance O)
MnO_4^- + **$8H^+$** → Mn^{2+} + $4H_2O$ (balance H)
Charge: left −1 + 8 = +7; right +2. Add $5e^-$ to the left:
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (reduction: electrons on the left)

8.10 Combining two half-equations

1. One half-equation is a reduction (electrons on the **left**); the other an oxidation (electrons on the **right**).
2. Multiply one or both so the **number of electrons is the same**.
3. Add them and **cancel** the electrons and anything else that appears on both sides (H^+, H_2O).

WORKED EXAMPLE: HYDROGEN PEROXIDE OXIDISES IODIDE

Oxidation: $2I^- \rightarrow I_2 + 2e^-$
Reduction: $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$
Electrons already equal (2), so add:
$H_2O_2 + 2H^+ + 2I^- \rightarrow I_2 + 2H_2O$

WORKED EXAMPLE: MANGANATE(VII) OXIDISES IRON(II)

Oxidation: $Fe^{2+} \rightarrow Fe^{3+} + e^-$ (× 5)
Reduction: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$
Check charge: left −1 + 8 + 10 = +17; right +2 + 15 = +17 ✓

WORKED EXAMPLE: CHLORINE IN COLD ALKALI

In alkali, balance with OH^- and H_2O instead of H^+.
Reduction: $Cl_2 + 2e^- \rightarrow 2Cl^-$
Oxidation: $Cl_2 + 4OH^- \rightarrow 2ClO^- + 2H_2O + 2e^-$
Add and divide by 2: **$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$**

8.10 Full ionic equations without half-equations

Write the full equation with state symbols, split soluble ionic compounds into ions, then **cancel the spectator ions** (ions unchanged on both sides).

- Precipitation: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$ becomes **$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$**
- Acid + carbonate: **$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$**
- Displacement: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ becomes **$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$** ($K^+$ is a spectator)

MARK-SCHEME LANGUAGE

- $H_2O_2 + I^-$: "$2I^- \rightarrow I_2 + 2e^-$" and "$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$"; overall "$2I^- + 2H^+ + H_2O_2 \rightarrow I_2 + 2H_2O$". "**Do not award uncancelled electrons**" (WCH12 Jun 2019).
- "$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$"; "$C_2H_2O_4 \rightarrow 2CO_2 + 2H^+ + 2e^-$" (WCH12 Jan 2025).
- Barium sulfate: "$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$", **state symbols needed**; DO NOT AWARD with spectator ions (WCH12 Oct 2024).
- Carbonate + acid: "$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$"; DO NOT AWARD H_2CO_3 as the product (WCH12 Oct 2019).
- Multiples of a correct equation are allowed.

Often asked as: combine two given half-equations; write the half-equation for an unfamiliar species; ionic equations for precipitation, neutralisation and displacement.

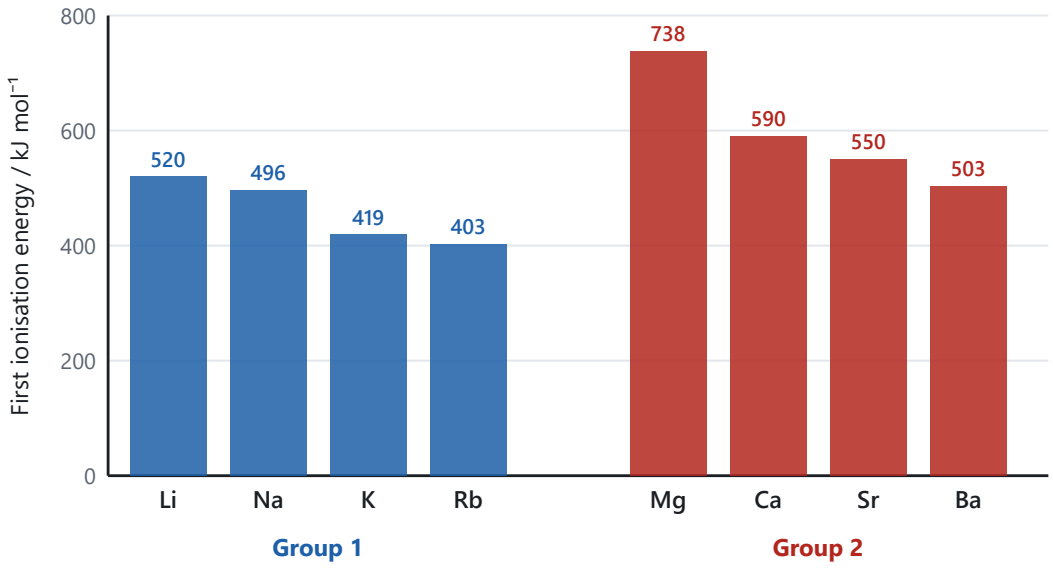
COMMON MISTAKE

Putting O^{2-} ions in an acidic half-equation: use H_2O and H^+ (WCH12 Jun 2019 examiner report).
Both half-equations with electrons on the same side. One must be a reduction and one an oxidation.
Leaving electrons or spectator ions in the final equation; wrong charges (CO_3^- instead of CO_3^{2-}).

SPEC 8.11 Ionisation energy down Groups 1 and 2

WHAT THIS SPEC POINT ASKS

Understand the reasons for the trend in ionisation energy down Groups 1 and 2.



First ionisation energy falls down both groups. Each Group 2 element has a higher value than the Group 1 element in the same period.

8.11 The trend and the reason

First ionisation energy decreases down both groups. Going down:

1. Each element has **one more occupied shell**, so the outer electron is **further from the nucleus** (larger atomic radius).
2. There are **more inner shells shielding** the outer electron from the nucleus.
3. The nuclear charge does increase, but the extra distance and shielding outweigh it. The **attraction between the nucleus and the outer electron gets weaker**, so **less energy is needed to remove it**.

8.11 Group 2 needs two electrons removed

- Group 2 metals form M^{2+}, so what matters for reactivity is the **sum of the first and second ionisation energies**. That sum also falls down the group: about 2190 kJ mol^{-1} for Mg down to about 1470 for Ba.
- The **second** ionisation energy is always bigger than the first: the electron is removed from a **positive ion**, so it is held more strongly (same shell, but the ion is smaller and the remaining electrons are pulled closer).
- Group 2 values are higher than Group 1 in the same period because the Group 2 atom has **one more proton** in the same shell, so a stronger attraction.

MARK-SCHEME LANGUAGE

- "Atomic radius increases / **more shells** / **more shielding**, so the outer electron is **less strongly attracted** (to the nucleus) and **easier to remove**".
- DO NOT AWARD "nuclear charge decreases". ALLOW "effective nuclear charge decreases" (WCH12 Jun 2022, by analogy with electronegativity).

Often asked as: rarely on its own; it is the first step of a reactivity explanation (8.12).

COMMON MISTAKE

Saying the nuclear charge falls down a group. It rises; shielding and distance win.
Using ionisation energy to explain thermal stability of carbonates and nitrates. That needs polarisation (8.16) (WCH12 Jan 2023 examiner report).

SPEC 8.12 Reactivity trends in Groups 1 and 2

WHAT THIS SPEC POINT ASKS

Understand why reactivity increases down Group 1 (Li to K) and Group 2 (Mg to Ba).

8.12 The explanation

- When a Group 1 or 2 metal reacts, it **loses its outer electron(s)** to form a positive ion. It is acting as a **reducing agent**.
- Down each group, the outer electrons are **further from the nucleus** and **more shielded**, so they are **lost more easily** (lower ionisation energy, 8.11).
- So reactivity **increases down the group**: K > Na > Li, and Ba > Sr > Ca > Mg.
- Group 1 metals are **more reactive** than the Group 2 metal in the same period: they need to lose only **one** electron, and their ionisation energy is lower.

EXAM TIP

A full answer: "Down the group the atoms have more shells, so the outer electron is further from the nucleus and more shielded. The attraction is weaker, so the electron is lost more easily, and the metal reacts more vigorously."

COMMON MISTAKE

"The atoms get bigger so they are more reactive", with no link to losing electrons.
Saying the metal **gains** electrons when it reacts.

SPEC 8.13 Reactions of Group 1 and 2 elements with oxygen, chlorine and water

WHAT THIS SPEC POINT ASKS

Know the reactions of Group 1 (Li to K) and Group 2 (Mg to Ba) with oxygen, chlorine and water.

8.13 With oxygen

Metal	Equation (normal oxide)	Observations
Li	$4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$	burns with a **red** flame, white solid
Na	$4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$	burns with a **yellow-orange** flame
K	$4K(s) + O_2(g) \rightarrow 2K_2O(s)$	burns with a **lilac** flame
Mg	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	**bright white** flame, white powder
Ca	$2Ca(s) + O_2(g) \rightarrow 2CaO(s)$	**brick-red** (orange-red) flame, white solid
Sr, Ba	$2M(s) + O_2(g) \rightarrow 2MO(s)$	red (Sr) and pale green (Ba) flames

- Group 1 metals **tarnish** quickly in air, faster down the group, so they are stored **under oil**.
- In excess oxygen, sodium mainly forms the **peroxide** Na_2O_2 and potassium the **superoxide** KO_2; barium can form BaO_2. The simple oxides above are what you'd normally be asked for.

8.13 With chlorine

All react on heating to give white ionic chlorides, more vigorously down each group:

- Group 1: **$2M(s) + Cl_2(g) \rightarrow 2MCl(s)$**, e.g. $2Na + Cl_2 \rightarrow 2NaCl$ (sodium burns with a yellow flame in chlorine).
- Group 2: **$M(s) + Cl_2(g) \rightarrow MCl_2(s)$**, e.g. $Mg + Cl_2 \rightarrow MgCl_2$.

8.13 With water

Group 1: $2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$. The solution is strongly alkaline.

Metal	What you see
Li	floats, **fizzes** steadily, moves slowly, gradually **disappears**
Na	floats, **melts into a silvery ball** (low melting point, exothermic reaction), whizzes around, fizzes, disappears
K	as sodium but more violent; the hydrogen **ignites with a lilac flame**; may spit

Group 2: $M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq \text{ or } s) + H_2(g)$

Metal	What you see
Mg	**very slow** with cold water (a few bubbles over days). With **steam** it burns with a white light: **Mg(s) + $H_2O(g)$ → MgO(s) + $H_2(g)$**
Ca	**fizzes steadily**, the water turns **cloudy white** (some $Ca(OH)_2$ doesn't dissolve), the mixture warms
Sr, Ba	faster and more vigorous; less cloudiness down the group because the hydroxides get more soluble

MARK-SCHEME LANGUAGE

- Ca + water: "$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$" (ALLOW $CaO + H_2$). The H_2 is often forgotten (WCH12 Oct 2019).
- K + water: "$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$", **state symbols marked** (WCH12 Oct 2021). Li: "$Li(s) + H_2O(l) \rightarrow LiOH(aq) + ½H_2(g)$" (WCH12 Oct 2024).
- Observations: "bubbles / fizzing / effervescence", "lithium disappears / gets smaller", "floats / moves on the surface". IGNORE "hydrogen produced", "exothermic"; DO NOT AWARD "white precipitate" or "smoke" for lithium (WCH12 Oct 2024).
- Ca + O_2: "$2Ca(s) + O_2(g) \rightarrow 2CaO(s)$" (WCH12 Oct 2025).

Often asked as: equation with state symbols and observations for a named metal and water; pH of the solution formed.

BEYOND THE SPEC, BUT EXAMINED

Lithium also reacts with **nitrogen** in air to form lithium nitride, **Li_3N** ($6Li + N_2 \rightarrow 2Li_3N$). Very few candidates could write it (WCH12 Oct 2024).
Identifying a Group 1 metal from a titration of its hydroxide: find moles of MOH, then Ar = mass ÷ moles (WCH12 Oct 2021).

COMMON MISTAKE

"Hydrogen was formed" as an observation: you see **bubbles**, you don't see hydrogen (WCH12 Oct 2024 examiner report).
LiOH(s) or $H_2O(aq)$ as state symbols.
Forgetting H_2 in the Group 2 equation.

SPEC 8.14

Reactions of oxides and hydroxides with water and acid

WHAT THIS SPEC POINT ASKS

Know the reactions of (i) Group 1 and 2 oxides with water and dilute acid and (ii) Group 1 and 2 hydroxides with dilute acid.

8.14 (i) Oxides with water

The oxide ion reacts with water to make hydroxide ions: **$O^{2-} + H_2O \rightarrow 2OH^-$**. So the solutions are **alkaline**.

Oxide	Equation	pH of the result
Na_2O	$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	about 14
Li_2O, K_2O	similar: $M_2O + H_2O \rightarrow 2MOH$	about 14
MgO	$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(s)$ (only slightly)	about 9–10: $Mg(OH)_2$ is barely soluble
CaO	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq/s)$ (very exothermic: "slaking" lime)	about 12
BaO	$BaO(s) + H_2O(l) \rightarrow Ba(OH)_2(aq)$	about 13

8.14 (i) Oxides with dilute acid

Oxides are **bases**: they neutralise acids to give a **salt and water**.

- $MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$; ionic: $MgO(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2O(l)$
- $Na_2O(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l)$
- $CaO(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + H_2O(l)$

8.14 (ii) Hydroxides with dilute acid

$OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$ in every case.

- $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
- $Ca(OH)_2(aq) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$
- $Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$ ($Mg(OH)_2$ in **antacids** neutralises excess stomach acid)

With **sulfuric acid**, calcium, strontium and especially barium compounds form a layer of **insoluble sulfate** on the solid, which slows or stops the reaction.

WORKED EXAMPLE: AN ANTACID

An antacid tablet neutralises 0.0240 mol of HCl. What mass of $Mg(OH)_2$ (Mr 58.3) does it contain?
$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$, so $n(Mg(OH)_2) = 0.0240 \div 2 = 0.0120$ mol
mass = 0.0120×58.3 = **0.700 g**

MARK-SCHEME LANGUAGE

- Mass from moles: "0.150 ÷ 2 = 0.075 mol", "Mr 58.3", "4.37 g" (WCH12 Oct 2019).
- $Ca(OH)_2$ solubility by titration: moles of HCl in the titre, ÷ 2, scale to 1 dm^3 (× 1000 ÷ 25.0), × 74.1 → g dm^{-3} (WCH12 Oct 2020).

Often asked as: equations with acid; titration calculations for $Ca(OH)_2$ or CaO; back titration for the purity of a solid.

BEYOND THE SPEC, BUT EXAMINED

Back titration for the purity of CaO: add a known excess of HCl, titrate the leftover acid with NaOH, subtract to find the acid used by CaO, ÷ 2, then mass and % purity (WCH12 Oct 2025, 6 marks).

COMMON MISTAKE

Forgetting the 1 : 2 ratio for $Ca(OH)_2$ or MgO with HCl (WCH12 Oct 2020 examiner report).
Writing O^{2-} ions in an equation for a reaction in acid (use the solid oxide or H_2O).

SPEC 8.15 Solubility of Group 2 hydroxides and sulfates

WHAT THIS SPEC POINT ASKS

Know the trends in solubility of the Group 2 hydroxides and sulfates.

8.15 Opposite trends

	Mg	Ca	Sr	Ba
Hydroxide, $M(OH)_2$	very sparingly soluble	sparingly soluble	fairly soluble	**soluble**
Sulfate, MSO_4	**soluble**	sparingly soluble	very sparingly soluble	**insoluble**

- **Hydroxides become more soluble down the group.** So the solutions contain more OH^- and are **more alkaline** further down: $Mg(OH)_2$ suspension about pH 10, saturated $Ba(OH)_2$ about pH 13+.
- **Sulfates become less soluble down the group.** $BaSO_4$ is effectively insoluble.

8.15 Uses that follow from the trends

- **$Mg(OH)_2$** ("milk of magnesia"): an antacid and laxative. Low solubility means it's only mildly alkaline, so safe to swallow.
- **$Ca(OH)_2$** (slaked lime): neutralises acidic soils and lakes; its saturated solution is **limewater**.
- **$BaSO_4$**: the "barium meal" for X-rays of the gut. Barium ions are **toxic**, but $BaSO_4$ is so insoluble that almost no Ba^{2+} is absorbed.
- **$BaCl_2(aq)$**: the test for sulfate ions (8.19), because $BaSO_4$ precipitates.

MARK-SCHEME LANGUAGE

- "Calcium hydroxide is **more soluble than** magnesium hydroxide" / "solubility of the hydroxides increases down the group", so the "**concentration of hydroxide ions is greater**" (WCH12 Oct 2019).
- With $Mg(OH)_2$ instead of $Ca(OH)_2$, "magnesium hydroxide is less soluble, so the OH^- concentration is lower and the titre would be smaller" (WCH12 Oct 2020). DO NOT AWARD "magnesium hydroxide is insoluble".
- Barium meal: "barium ions are precipitated as (insoluble) barium sulfate" (WCH12 Oct 2024).

Often asked as: compare the pH of two Group 2 hydroxide solutions; predict a titre; explain the barium meal.

COMMON MISTAKE

Mixing up the directions: it's hydroxides **up** in solubility, sulfates **down**. Memory aid: "**Ba**se gets **better** (more soluble hydroxide), **Ba**SO_4 is **bad** (insoluble)."
A vague pH statement with no link to OH^- concentration or solubility (WCH12 Oct 2019 examiner report).

SPEC 8.16 Thermal stability of nitrates and carbonates

WHAT THIS SPEC POINT ASKS

Understand the trends in thermal stability of Group 1 and 2 nitrates and carbonates in terms of the size and charge of the cations.

8.16 What happens on heating

Compound	Decomposition	Example equation
Group 2 carbonates	→ oxide + CO_2	$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$
Li_2CO_3	→ oxide + CO_2 (the only Group 1 carbonate that decomposes at Bunsen temperatures)	$Li_2CO_3(s) \rightarrow Li_2O(s) + CO_2(g)$
Na_2CO_3, K_2CO_3	**no decomposition** at Bunsen temperatures	—
Group 2 nitrates	→ oxide + **NO_2** + O_2	$2Mg(NO_3)_2(s) \rightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$
$LiNO_3$	→ oxide + NO_2 + O_2 (like Group 2)	$4LiNO_3(s) \rightarrow 2Li_2O(s) + 4NO_2(g) + O_2(g)$
$NaNO_3$, KNO_3	→ **nitrite** (nitrate(III)) + O_2 only	$2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$

- **Thermal stability increases down both groups**: $BaCO_3$ needs a much higher temperature than $MgCO_3$.
- **Group 1 compounds are more stable** than the Group 2 compounds in the same period.
- Nitrogen dioxide is a **brown, toxic, acidic** gas: decompose nitrates in a **fume cupboard**.

8.16 Why: polarisation of the anion

1. The **cation** attracts the electron cloud of the large **carbonate (or nitrate) anion** towards itself. This **polarises** (distorts) the anion.
2. The distortion **weakens the covalent C–O (or N–O) bonds inside the anion**, making it easier to break off CO_2 (or NO_2 and O_2).
3. A cation is more polarising when it has a **high charge density**: a **small ionic radius** and/or a **high charge**.
4. **Down a group**, the cation charge stays the same but the **ionic radius increases**, so charge density falls. The anion is **less polarised**, its bonds are **weakened less**, so **more energy (a higher temperature)** is needed to decompose it.
5. **Group 2 v Group 1**: M^{2+} ions have **twice the charge** and are smaller than M^+ ions in the same period, so they polarise more: Group 2 compounds decompose more easily.
6. **Lithium** is the exception in Group 1: Li^+ is very **small**, so its charge density is high enough (similar to Mg^{2+}) to polarise carbonate and nitrate ions as Group 2 ions do.

MARK-SCHEME LANGUAGE

Standard three-mark answer (WCH12 Jun 2019, Jun 2022, Jan 2026):

- "Group 2 carbonates **increase in (thermal) stability down the group**"
- "**size of the (metal) ion increases** / charge density decreases" (charge is the same)
- "so the metal ion is **less polarising** / the (electron cloud of the) **carbonate ion is less distorted**"
- "so the **(covalent) bonds in the carbonate ion are weakened less** / more energy is needed to break them"

Must refer to **ions** (DO NOT AWARD "magnesium carbonate is less polarising"). DO NOT AWARD "weaker ionic bond". IGNORE atomic radius, effective nuclear charge. Penalise missing "ion" once only.
Nitrates: "Li and Group 2 give oxide + NO_2 + O_2; Na and K give nitrite + O_2" (WCH12 Oct 2024).

Often asked as: explain the carbonate trend; compare Li, Na and Mg nitrates (6-mark); correct the errors in a given explanation.

BEYOND THE SPEC, BUT EXAMINED

Lithium and magnesium behave alike (Li^+ and Mg^{2+} have similar charge densities): both nitrates give NO_2, while sodium nitrate gives nitrite and O_2 (WCH12 Oct 2024, 6 marks).
Mass-loss experiments: heat a carbonate in a weighed tube to **constant mass**, find moles of CO_2 lost, then the Mr of MCO_3 and so Ar of M (WCH12 Jan 2026). The same method finds x in a hydrated salt (WCH12 Jun 2019).
Correcting errors: "the **nitrate ion is polarised by the calcium ion**" (not the other way round) and "it weakens the covalent bonds **within** the nitrate ion" (not the bond between the ions) (WCH12 Oct 2025).

COMMON MISTAKE

Thinking $MgCO_3$ is more stable than $CaCO_3$, or that Mg^{2+} and Ca^{2+} have different charges (WCH12 Jun 2022 examiner report).
Writing "atomic radius" instead of **ionic radius**; saying "the anion is polarised" without saying by what; saying the **ionic bond** is weakened.
Talking about the stability of the **element** instead of the compound.

SPEC 8.17

Flame colours

WHAT THIS SPEC POINT ASKS

Understand how Group 1 and 2 compounds produce characteristic flame colours through electron transitions, and know the colours.

8.17 Why there is a colour

1. Heat from the flame gives energy to electrons in the metal ions. Electrons are **excited (promoted) to higher energy levels**.
2. The excited electrons are unstable and **fall back to lower energy levels** (towards the ground state).
3. As they fall, they **release the energy as light**. The energy gap is fixed for each element, so the light has **specific frequencies (wavelengths)**.
4. If those frequencies are in the **visible** region, we see a colour. Each element has its own set of energy gaps, so its own colour.

Magnesium gives **no flame colour**: its electron transitions release energy outside the visible range (in the ultraviolet), so all you see is the colour of the flame itself.

Ion	Flame colour
Li^+	**red** (crimson)
Na^+	**yellow-orange** (intense, long-lasting)
K^+	**lilac**
Rb^+	red-violet
Cs^+	blue-violet
Mg^{2+}	**none**
Ca^{2+}	**brick-red** (orange-red)
Sr^{2+}	**red** (crimson)
Ba^{2+}	**pale (apple) green**

MARK-SCHEME LANGUAGE

- Explanation, three marks (WCH12 Oct 2025): "electrons are **excited / promoted** (by heat) to **higher energy levels**", "electrons **fall back** (to lower levels / the ground state)", "releasing (light) energy of a **specific wavelength / frequency**". IGNORE "emits a colour"; DO NOT AWARD "reflects a colour".
- Calcium: "**(brick) red / orange-red / yellow-red**"; DO NOT AWARD orange or yellow alone, scarlet or crimson (WCH12 Oct 2025).
- Potassium: "**lilac**" (ALLOW purple) (WCH12 Oct 2021).

Often asked as: the colour for a named ion; explain how a flame colour arises.

COMMON MISTAKE

Describing the atom being excited without mentioning **electrons**.
Saying "the electrons emit a colour" with no mention of energy or wavelength.
Calling calcium "orange" alone, or "crimson" (that's lithium and strontium).

SPEC 8.18

Practical procedures: thermal decomposition and flame tests

WHAT THIS SPEC POINT ASKS

Know experimental procedures to show (i) patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates (with tests for CO_2 and O_2, and recognising NO_2 by colour and acidity) and (ii) flame colours.

8.18 (i) Comparing thermal stability

SUGGESTED PRACTICAL: THERMAL DECOMPOSITION OF CARBONATES AND NITRATES

Carbonates

1. Put the **same number of moles** (e.g. 0.010 mol) of each carbonate into a boiling tube fitted with a bung and delivery tube.
2. Dip the end of the delivery tube into a fixed volume of **limewater**.
3. Heat with the **same Bunsen flame** (same height, same part of the flame).

4. Time how long it takes for the limewater to turn **milky (cloudy)**. Longer time → more stable carbonate.
5. Lift the delivery tube **out of the limewater before you stop heating**, or limewater is sucked back into the hot tube and cracks it.

Nitrates (in a fume cupboard)

1. Heat equal moles of each nitrate in a boiling tube with the same flame.
2. Time until **brown gas (NO_2)** appears (look against a white background), or test for O_2 with a **glowing splint**, which **relights**.

Safety: eye protection; NO_2 is toxic; don't heat a sealed tube.

Gas	Test	Result
CO_2	bubble through **limewater** (calcium hydroxide solution)	turns **milky / cloudy** (white precipitate of $CaCO_3$)
O_2	**glowing splint**	**relights**
NO_2	look at it; damp **blue litmus**	**brown** gas; turns blue litmus **red** (acidic)

Limewater equation: **$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$**

8.18 (ii) Flame tests

SUGGESTED PRACTICAL: FLAME TESTS

1. Clean a **nichrome (or platinum) wire** loop by dipping it in **concentrated hydrochloric acid** and holding it in a **hot (blue) Bunsen flame** until it gives no colour.
2. Dip the clean wire in the acid again, then into the solid sample.
3. Hold it at the **edge of the hottest (blue) part** of the flame and note the colour.
4. Clean the wire again between samples. Sodium contamination is the usual problem: its strong yellow masks other colours.

MARK-SCHEME LANGUAGE

- Limewater: "$CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$" (WCH12 Oct 2019).
- Safety: "safety glasses / goggles"; ALLOW gloves; IGNORE lab coat (WCH12 Jun 2023).
- Heating a hydrated nitrate: "magnesium nitrate **decomposes** when heated"; DO NOT AWARD "reacts with oxygen" (WCH12 Jun 2019).
- Heat a hydrated salt "**until no change in mass**" (to constant mass) (WCH12 Oct 2024).

Often asked as: tests for the gases from a decomposition; a plan for comparing stabilities or for a mass-loss experiment.

COMMON MISTAKE

Not recognising NO_2 as **acidic** (WCH12 Jan 2024 examiner report).
"Spitting" or "burning" as the explanation for mass loss from a heated nitrate (WCH12 Jun 2019 examiner report).
Using equal **masses** when the question wants a fair comparison per mole.

SPEC 8.19 Tests for carbonate, hydrogencarbonate, sulfate and ammonium ions

WHAT THIS SPEC POINT ASKS

Know the reactions and ionic equations for identifying (i) CO_3^{2-} and HCO_3^- with acid, (ii) SO_4^{2-} with acidified $BaCl_2$ and (iii) NH_4^+ with NaOH and warming.

Ion	Test	Positive result	Ionic equation
Carbonate, CO_3^{2-}	add **dilute acid** (e.g. HCl); bubble the gas through limewater	**effervescence**; limewater turns **milky**	$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$
Hydrogencarbonate, HCO_3^-	add **dilute acid**; gas through limewater	**effervescence**; limewater turns **milky**	$HCO_3^- + H^+ \rightarrow CO_2 + H_2O$
Sulfate, SO_4^{2-}	add **dilute HCl** (or HNO_3), then **barium chloride** solution	**white precipitate**	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
Ammonium, NH_4^+	add **NaOH(aq)** and **warm**	gas turns damp **red litmus blue**; gives **white smoke** with HCl fumes (from a glass rod dipped in conc. HCl)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$; then $NH_3 + HCl \rightarrow NH_4Cl$

8.19 Why the details matter

- The sulfate test is **acidified** first so that **carbonate** ions are removed (as CO_2). Otherwise $BaCO_3$, also a white precipitate, would give a false positive. Don't acidify with **sulfuric** acid: it adds sulfate ions.
- Litmus must be **damp**: ammonia acts as an alkali only when dissolved in water.
- Warming drives the ammonia out of solution.

EXAM TIP

Telling a carbonate from a hydrogencarbonate (both fizz with acid): add **magnesium sulfate** solution. A carbonate gives a white precipitate ($MgCO_3$) straight away; a hydrogencarbonate doesn't. Or heat the solid: a Group 1 hydrogencarbonate gives off CO_2 and water, but the Group 1 carbonate doesn't decompose.

MARK-SCHEME LANGUAGE

- Carbonate + acid: "$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$"; DO NOT AWARD H_2CO_3 as a product (WCH12 Oct 2019).
- $BaSO_4$: "$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$", with state symbols (WCH12 Oct 2024).
- Testing for an acid (–COOH) with a carbonate: "effervescence (and limewater turns cloudy)" (WCH12 Oct 2020).

Often asked as: name the reagent and the observation; write the ionic equation; part of an identification (Core Practical 8 style) question.

COMMON MISTAKE

Giving the reagent but not the **observation**, or the observation but not the reagent. You need both. "Gas turns limewater cloudy" without saying a gas was **seen** (effervescence / fizzing).

SPEC 8.20 Solution concentrations and acid–base titrations

WHAT THIS SPEC POINT ASKS

Calculate concentrations in mol dm^{-3} and g dm^{-3}, including from simple acid–base titrations using methyl orange and phenolphthalein.

8.20 The equations

- **n = c × V**, with V in **dm^3** (cm^3 ÷ 1000).
- **c = n ÷ V** (mol dm^{-3}).
- **Mass concentration (g dm^{-3}) = c (mol dm^{-3}) × M (g mol^{-1})**.

8.20 Indicators

Indicator	In acid	In alkali	End point	Use for
Methyl orange	**red**	**yellow**	**orange**	strong acid with strong or weak base (e.g. HCl with Na_2CO_3 or NH_3)
Phenolphthalein	**colourless**	**pink**	pale pink / just colourless	strong or weak acid with strong base (e.g. ethanoic acid with NaOH)

- For a strong acid + strong alkali, either works.
- Don't use **universal indicator**: it changes colour gradually, so there's no sharp end point.
- Describe the colour change **in the direction it happens**: alkali in the flask, acid added from the burette, phenolphthalein goes **pink → colourless**.

8.20 Titration calculation: the steps

1. Mean titre from the **concordant** results (within 0.10 cm^3 of each other).
2. Moles of the substance of known concentration: n = c × V.
3. Use the **mole ratio** from the balanced equation.
4. Concentration of the unknown: c = n ÷ V. Scale up if a sample was diluted.

WORKED EXAMPLE: CONCENTRATION OF NAOH, IN MOL DM^{-3} AND G DM^{-3}

25.0 cm^3 of NaOH(aq) needs a mean titre of 22.40 cm^3 of 0.100 mol dm^{-3} HCl.
n(HCl) = 0.100 × 22.40 ÷ 1000 = 2.240×10^{-3} mol
NaOH + HCl → NaCl + H_2O (1 : 1), so n(NaOH) = 2.240×10^{-3} mol
c(NaOH) = $2.240 \times 10^{-3} \div 0.0250$ = **0.0896 mol dm^{-3}**
Mass concentration = 0.0896 × 40.0 = **3.58 g dm^{-3}**

WORKED EXAMPLE: A 1 : 2 RATIO

25.0 cm^3 of Na_2CO_3(aq) needs 24.00 cm^3 of 0.150 mol dm^{-3} HCl (methyl orange).
$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$
n(HCl) = 0.150 × 0.02400 = 3.60×10^{-3} mol → n(Na_2CO_3) = 1.80×10^{-3} mol
c(Na_2CO_3) = $1.80 \times 10^{-3} \div 0.0250$ = **0.0720 mol dm^{-3}**

WORKED EXAMPLE: WATER OF CRYSTALLISATION

3.58 g of $Na_2CO_3 \cdot xH_2O$ is made up to 250 cm^3. 25.0 cm^3 portions need 25.00 cm^3 of 0.100 mol dm^{-3} HCl.
n(HCl) = 2.50 × 10^{-3} mol → n(Na_2CO_3) in 25.0 cm^3 = 1.25 × 10^{-3} mol
In 250 cm^3: × 10 = 1.25 × 10^{-2} mol
M of the hydrated salt = 3.58 ÷ 0.0125 = 286.4 g mol^{-1}
Mass of water per mole = 286.4 − 106.0 = 180.4 → x = 180.4 ÷ 18.0 = **10** ($Na_2CO_3 \cdot 10H_2O$)

MARK-SCHEME LANGUAGE

- Colour changes: "colourless to (pale) pink" (alkali added to acid) or "(pale) pink to colourless" (acid added to alkali). DO NOT AWARD red or purple; the reverse order scores only one (WCH12 Jun 2019, Oct 2021).
- Working: moles in titre → ratio → scale up (× 10 for a 250 cm^3 flask sampled with a 25.0 cm^3 pipette) → mass or Mr.
- x in a hydrated salt must be a **whole number** (WCH12 Jan 2020).
- Final answers to 2 or 3 s.f. with units.

Often asked as: concentration in both units; Mr or formula of a hydrated salt; identify a metal hydroxide; back titration.

BEYOND THE SPEC, BUT EXAMINED

Parts per million: e.g. mass of SO_2 in a volume of air using 24 dm^3 mol^{-1} (WCH12 Oct 2019).
Back titration to find the purity of a solid (WCH12 Oct 2025, Jun 2025).
Ideal gas equation, pV = nRT, with p in Pa and V in m^3, to find an Mr (WCH12 Oct 2024).

COMMON MISTAKE

Forgetting the mole ratio, or using the sample volume where the titre should go (WCH12 Oct 2020 examiner report).
Not scaling up by 10 when a 25.0 cm^3 portion was taken from 250 cm^3.
Rounding a calculated Mr too early, so the formula comes out wrong (WCH12 Oct 2024 examiner report).

SPEC 8.21

Core Practical 3: finding the concentration of hydrochloric acid

WHAT THIS SPEC POINT ASKS

Find the concentration of a hydrochloric acid solution by titration against a standard alkali.

CORE PRACTICAL 3: METHOD IN BRIEF

1. If the acid is too concentrated (around 1 mol dm^{-3}), dilute it first: **pipette 25.0 cm^3** into a **250 cm^3 volumetric flask**, make up to the mark with distilled water and invert to mix. This is a tenfold dilution.
2. **Rinse** the burette with a little of the (diluted) acid, then fill it. Make sure the **jet is full** (no air bubble) and remove the funnel. Read the bottom of the meniscus at eye level, to **0.05 cm^3**.
3. **Rinse** a pipette with the standard NaOH solution (e.g. 0.100 mol dm^{-3}), then use a **pipette filler** to transfer **25.0 cm^3** into a conical flask.
4. Add 2–3 drops of indicator (methyl orange or phenolphthalein). Stand the flask on a **white tile**.
5. Run in acid while **swirling**. Do a quick **rough (rangefinder)** titration first.

6. Repeat accurately: near the end point add acid **dropwise**, rinsing the flask walls with distilled water, until **one drop** gives a permanent colour change.
7. Repeat until you have **concordant** titres (within **0.10 cm^3**). Average only the concordant ones.

WORKED EXAMPLE: PROCESSING CP3 RESULTS

25.0 cm^3 of 0.100 mol dm^{-3} NaOH; titres of diluted HCl: rough 22.90, then 22.30, 22.50, 22.30 cm^3.
Concordant: 22.30 and 22.30 → mean **22.30 cm^3** (22.50 is not within 0.10 cm^3, so it's left out)
n(NaOH) = 0.100 × 0.0250 = 2.50 × 10^{-3} mol = n(HCl)
c(diluted HCl) = 2.50 × 10^{-3} ÷ 0.02230 = 0.112 mol dm^{-3}
c(original HCl) = 0.112 × 10 = **1.12 mol dm^{-3}**

8.21 Errors and why each step is there

Step	If you don't...
rinse the burette with the acid	water left inside dilutes the acid → titre too **large**
fill the jet (no air bubble)	the bubble escapes during the titration → reading too **large**
rinse the pipette with the alkali	the alkali is diluted → fewer moles → titre too **small**
add dropwise near the end point	you overshoot → titre too **large**
(water in the conical flask)	**no effect**: it doesn't change the moles of alkali

MARK-SCHEME LANGUAGE

- Odd titres: "first titre likely to be a **rangefinder**", "**air bubble** in the burette jet", "burette **rinsed with water** (rather than the solution)"; ALLOW "overshot / not dropwise near the end point". IGNORE parallax, water in the conical flask (WCH12 Jun 2019).
- Why no third titre: "**two concordant titres** had already been obtained" (or "only 10 cm^3 of solution was left"). DO NOT AWARD "titres within ±0.1 cm^3" alone (WCH12 Oct 2019).
- Describe a titration (two marks): "known volume of one solution", "known concentration of the other", "titrate until a colour change" (WCH12 Jan 2025).

Often asked as: process titration results; explain anomalous titres; describe the method; why only two titres were needed.

COMMON MISTAKE

Describing a **rates** experiment when asked how to do a titration (WCH12 Jan 2025 examiner report: over half scored zero).
Including the rough titre in the mean.
Recording burette readings to 0.1 cm^3: always 2 decimal places, ending in 0 or 5.

SPEC 8.22 Minimising uncertainty in volumetric analysis

WHAT THIS SPEC POINT ASKS

Understand how to minimise measurement uncertainty in volumetric analysis, and estimate the overall uncertainty in a calculated result.

8.22 Uncertainty of each piece of apparatus

Percentage uncertainty = (uncertainty ÷ quantity measured) × 100

Apparatus	Typical uncertainty	Notes
2 d.p. balance	±0.005 g per reading	weighing by difference = **two** readings → ±0.01 g
25.0 cm^3 pipette	±0.06 cm^3	one reading
250 cm^3 volumetric flask	±0.3 cm^3 (class B) or less	one reading
burette	±0.05 cm^3 per reading	a titre = **two** readings → ±0.10 cm^3

Use the values a question gives you; these are typical.

Overall uncertainty: add the **percentage** uncertainties of every measurement used in the calculation. Then apply that percentage to the answer.

8.22 How to reduce uncertainty

- Use a **larger mass** of solid, or a **larger titre** (e.g. by changing the concentrations), so the fixed uncertainty is a smaller percentage.
- Use more precise apparatus: a 3 d.p. balance; a pipette and volumetric flask rather than measuring cylinders.
- Repeat to get **concordant** titres and take a mean (this reduces random error, not the apparatus uncertainty).
- Avoid systematic errors: rinse apparatus with the solution it will hold, read the **bottom of the meniscus at eye level**, remove the funnel from the burette, fill the jet.

WORKED EXAMPLE: OVERALL UNCERTAINTY (CP4 DATA)

Mass by difference 2.45 g (±0.01 g): 0.01 ÷ 2.45 × 100 = 0.41%
250 cm^3 flask (±0.3 cm^3): 0.3 ÷ 250 × 100 = 0.12%
25.0 cm^3 pipette (±0.06 cm^3): 0.06 ÷ 25.0 × 100 = 0.24%
Titre 24.80 cm^3 (±0.10 cm^3): 0.10 ÷ 24.80 × 100 = 0.40%
Total ≈ **1.2%**. For c(NaOH) = 0.102 mol dm^{-3}: 1.2% of 0.102 = ±0.001, so 0.102 ± 0.001 mol dm^{-3}.

MARK-SCHEME LANGUAGE

- Percentage uncertainty for a quantity from two readings: "(2 × 0.05 × 100) ÷ 13.2 = 0.76%". DO NOT AWARD 0.75 or 0.80 (WCH12 Jun 2022).

Often asked as: calculate a percentage uncertainty; suggest how to reduce it; choose suitable apparatus.

COMMON MISTAKE

Using 0.05 cm^3 for a titre: a titre is the **difference** of two readings, so ±0.10 cm^3.
Thinking repeating the titration reduces the uncertainty of the burette itself. It improves reliability, not the apparatus's precision.

SPEC 8.23 Core Practical 4: a standard solution from a solid acid

WHAT THIS SPEC POINT ASKS

Make a standard solution of a solid acid and use it to find the concentration of a sodium hydroxide solution.

8.23 What a standard solution is

A **standard solution** has an **accurately known concentration**. You make it by dissolving an accurately weighed mass of a pure, stable solid in water and making it up to an exact volume. A suitable solid acid is **sulfamic acid**, H_2NSO_3H (Mr 97.1), which is monoprotic: $H_2NSO_3H + NaOH \rightarrow H_2NSO_3Na + H_2O$ (1 : 1).

CORE PRACTICAL 4: METHOD IN BRIEF

Making the standard solution

1. Weigh a weighing bottle containing about 2.5 g of the solid acid on a 2 d.p. (or 3 d.p.) balance.
2. Tip the solid into a **beaker**, then **reweigh** the bottle: mass used = difference (**weighing by difference**, so solid left in the bottle doesn't matter).
3. Dissolve the solid in about 100 cm^3 of distilled water, stirring with a glass rod.
4. Pour the solution through a funnel into a **250 cm^3 volumetric flask**. **Rinse** the beaker, rod and funnel with distilled water and add all the **rinsings** to the flask.
5. Add distilled water until the **bottom of the meniscus** is on the line (use a dropping pipette for the last few drops, read at eye level).
6. Stopper and **invert several times** to mix.

Titration

7. Pipette 25.0 cm^3 of the standard acid into a conical flask; add phenolphthalein.
8. Titrate with the NaOH from the burette until the solution turns **pale pink** (colourless → pink). Repeat to concordant titres.

WORKED EXAMPLE: PROCESSING CP4 RESULTS

Mass of sulfamic acid = 2.45 g → n = 2.45 ÷ 97.1 = 0.02523 mol in 250 cm^3
c(acid) = 0.02523 ÷ 0.250 = 0.1009 mol dm^{-3}
In 25.0 cm^3: n = 0.1009 × 0.0250 = 2.523 × 10^{-3} mol = n(NaOH) (1 : 1)
Mean titre of NaOH = 24.80 cm^3
c(NaOH) = 2.523 × 10^{-3} ÷ 0.02480 = **0.102 mol dm^{-3}**

8.23 Why each step matters

- **Rinsings** go into the flask so that **all** the weighed acid ends up in the solution.
- **Inverting** makes the concentration the same throughout: otherwise each 25.0 cm^3 portion would have a different amount of acid.
- Overfilling past the line can't be undone: the solution must be remade.
- NaOH can't be used as a standard itself: the solid absorbs **water** and **CO_2** from the air, so its mass isn't a reliable measure of the moles of NaOH. That's why its concentration has to be found by titration.

MARK-SCHEME LANGUAGE

- The calculation is what gets marked: moles of solid in the 250 cm^3 flask, ÷ 10 for each 25.0 cm^3 portion (or × 10 going the other way), then the mole ratio (WCH12 Jan 2020, a 250 cm^3 standard solution of a carbonate).
- No paper from 2019 to 2026 asked for the full preparation method, so learn it from the practical itself: weigh by difference, dissolve, transfer **with rinsings**, make up to the mark, invert.

Often asked as: calculate the concentration of the standard solution and then of the NaOH; explain a step in making the solution.

COMMON MISTAKE

Forgetting to add the rinsings, or making the solution up to the mark **before** the solid has fully dissolved. Using the mass in the whole 250 cm^3 for the moles in one 25.0 cm^3 portion.

SPEC 8.24 Group 7 trends: physical properties, electronegativity and reactivity

WHAT THIS SPEC POINT ASKS

Understand the reasons for the trends in Group 7 (chlorine, bromine, iodine) in (i) melting and boiling temperatures and state, (ii) electronegativity and (iii) reactivity.

8.24 (i) Melting and boiling temperatures, and state

	F_2	Cl_2	Br_2	I_2
State and colour at room temperature	pale yellow gas	**pale green** gas	**red-brown liquid** (gives orange-brown vapour)	**grey-black (shiny) solid**; gives a **purple vapour** when warmed
Boiling temperature / °C	−188	−34	59	184
Electrons per molecule	18	34	70	106

- Melting and boiling temperatures **increase down the group**.
- The halogens are simple molecules, X_2, held together by **London forces only**.
- Down the group there are **more electrons per molecule**, so the **London forces get stronger**, and **more energy is needed to separate the molecules**. So the state goes gas → liquid → solid.

8.24 (ii) Electronegativity

Electronegativity is the **ability of an atom to attract the bonding pair of electrons in a covalent bond**.

	F	Cl	Br	I
Electronegativity (Pauling)	4.0	3.0	2.8	2.5

Electronegativity **decreases down the group**: the atoms have **more shells**, so the bonding pair is **further from the nucleus** and **more shielded** by inner electrons. It is less strongly attracted, even though the nuclear charge is higher.

8.24 (iii) Reactivity

- Halogens react by **gaining an electron** to form X^- ions (or by attracting a shared pair). They are **oxidising agents**.
- Down the group, the incoming electron goes into a shell **further from the nucleus** with **more shielding**, so it is **attracted less strongly**.
- So **reactivity (oxidising power) decreases down the group**: $F_2 > Cl_2 > Br_2 > I_2$.

MARK-SCHEME LANGUAGE

- Iodine: "**(dark) grey / black** and **solid**". Both state and colour are needed; IGNORE shiny (WCH12 Jan 2022).
- Electronegativity definition: "the **ability of an atom to attract a bonding / shared pair of electrons** (in a covalent bond)" (WCH12 Jan 2022).
- Trend (two marks): "elements become **less electronegative** down the group"; "the outer electrons are **further from the nucleus / more shielding**, so the bonding pair is **less attracted**". DO NOT AWARD "nuclear charge decreases". IGNORE ionic radius. One paper ignored "atomic radius increases" on its own, another accepted it, so give the full chain (WCH12 Jan 2022, Jun 2022).

Often asked as: state and colour of a halogen; define electronegativity; explain the electronegativity or boiling-temperature trend.

BEYOND THE SPEC, BUT EXAMINED

Electronegativity across a period: bromine has one more proton than selenium in the same shell, so it attracts a bonding pair more strongly; compared with chlorine, bromine has an extra shell, so it attracts less strongly (WCH12 Jan 2026).

COMMON MISTAKE

Confusing electronegativity with electron affinity or ionisation energy (WCH12 Jun 2022 examiner report).
Writing about the size of the **ion** in an electronegativity answer.
Missing "pair of electrons" or "covalent bond" from the definition.
Saying the covalent X–X bonds break when a halogen boils.

SPEC 8.25

Halogen displacement reactions and colours

WHAT THIS SPEC POINT ASKS

Understand the trend in reactivity of Cl_2, Br_2 and I_2 through their redox reactions with halide ions in solution, and know the colours of the halogens as elements, in water and in a non-polar solvent.

8.25 Displacement

A **more reactive halogen displaces** (oxidises) the ion of a **less reactive** halogen from solution.

Added halogen	$Cl^-(aq)$	$Br^-(aq)$	$I^-(aq)$
$Cl_2(aq)$	—	**yes**: yellow/orange	**yes**: brown
$Br_2(aq)$	no reaction	—	**yes**: brown
$I_2(aq)$	no reaction	no reaction	—

- $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$
- $Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$
- $Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$

In each: the halogen is **reduced** (0 → −1) and acts as the oxidising agent; the halide ion is **oxidised** (−1 → 0). The metal ion (K^+, Na^+) is a spectator.

This shows **oxidising power decreases down the group** ($Cl_2 > Br_2 > I_2$), and so the **reducing power of the halide ions increases** ($I^- > Br^- > Cl^-$).

8.25 Colours you must know

Halogen	As the element	In water	In a non-polar solvent (cyclohexane, hexane)
Cl_2	pale green gas	very pale green (almost colourless)	very pale green (almost colourless)
Br_2	red-brown liquid	**yellow / orange**	**orange (red-orange)**
I_2	grey-black solid	**brown** (pale yellow when very dilute)	**purple / violet / pink**

Brown in water could be bromine or iodine. To tell them apart, **add cyclohexane and shake**: the halogen moves into the upper organic layer, which is **orange** for bromine and **purple** for iodine.

MARK-SCHEME LANGUAGE

- "$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$" (ALLOW halves); DO NOT AWARD uncancelled K^+ (WCH12 Jun 2025).
- Chlorine added to bromide: the solution turns "**orange / yellow**"; DO NOT AWARD brown or green (WCH12 Jan 2026).
- Iodine shaken with water and cyclohexane: "(pale) yellow aqueous layer", "pink / purple / violet cyclohexane layer". DO NOT AWARD brown alone, colourless, red or grey (WCH12 Jun 2019).

Often asked as: equation and colour change for a displacement; colours of the layers; which halide is oxidised first.

BEYOND THE SPEC, BUT EXAMINED

Competitive displacement: when chlorine is added to a mixture of iodide and bromide, **iodide is oxidised first** (it is the stronger reducing agent). E.g. 0.200 mol Cl_2 with 0.300 mol I^- and 0.300 mol Br^-: 0.150 mol Cl_2 makes 0.150 mol I_2, and the remaining 0.050 mol Cl_2 makes 0.050 mol Br_2 (WCH12 Jan 2026 style).

COMMON MISTAKE

Saying "chlorine displaces bromine" when the reactant is bromide: chlorine displaces bromine **from bromide ions**.
Calling the aqueous layer colourless when only "most" of the iodine moved to the organic layer (WCH12 Jun 2019 examiner report).
Writing a reaction for $Br_2 + Cl^-$ or $I_2 + Br^-$: there isn't one.

SPEC 8.26 Redox reactions of the halogens

WHAT THIS SPEC POINT ASKS

Understand, in terms of oxidation number changes: (i) oxidation of Group 1 and 2 metals; (ii) disproportionation of chlorine with water and its use in water treatment; (iii) with cold, dilute NaOH to make bleach; (iv) with hot alkali; (v) analogous reactions.

8.26 (i) With Group 1 and 2 metals

The halogen is reduced (0 → −1); the metal is oxidised (0 → +1 or +2):

- $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

- $Ca(s) + Cl_2(g) \rightarrow CaCl_2(s)$
- $Mg(s) + Br_2(l) \rightarrow MgBr_2(s)$
- $2K(s) + I_2(s) \rightarrow 2KI(s)$

Reactions get less vigorous with the halogen going down Group 7, and more vigorous with the metal going down Groups 1 and 2.

8.26 (ii) Chlorine with water; water treatment

$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$

- Chlorine: 0 → **−1** in HCl (reduced) and 0 → **+1** in HClO, chloric(I) acid (oxidised). A **disproportionation**.
- **HClO** (and the ClO^- ion it forms) **kills bacteria** and other microorganisms, which is why chlorine is added to drinking water and swimming pools.
- **Benefits**: prevents waterborne diseases such as cholera and typhoid, and the chlorine stays in the water to protect it in the pipes.
- **Risks**: chlorine is **toxic**; it can react with organic matter in water to form **chlorinated organic compounds** (e.g. chloroalkanes) that may cause cancer. The benefit to public health is judged to far outweigh the risk, at low concentrations.
- Adding **alkali** removes H^+ (from HCl and HClO), so the equilibrium shifts **to the right**.

8.26 (iii) Chlorine with cold, dilute sodium hydroxide: bleach

$Cl_2(g) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$ (about 15 °C)
Ionic: $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$

- Chlorine: 0 → **−1** (Cl^-, reduced) and 0 → **+1** (ClO^-, chlorate(I), oxidised).
- The solution of **sodium chlorate(I)** (sodium hypochlorite) is household **bleach**; ClO^- is the active ingredient.

8.26 (iv) Chlorine with hot alkali

$3Cl_2(g) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$ (about 70 °C)
Ionic: $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$

- Chlorine: 0 → **−1** (five Cl atoms, reduced) and 0 → **+5** (one Cl atom in ClO_3^-, chlorate(V), oxidised). Electrons balance: 5 × 1 gained = 1 × 5 lost.
- In hot alkali, chlorate(I) ions themselves disproportionate: $3ClO^- \rightarrow 2Cl^- + ClO_3^-$.

8.26 (v) Analogous reactions

The same patterns work with other alkalis and halogens. Work out the products by analogy and check the oxidation numbers.

- Bromine, cold: $Br_2 + 2NaOH \rightarrow NaBr + NaBrO + H_2O$. Hot: $3Br_2 + 6NaOH \rightarrow 5NaBr + NaBrO_3 + 3H_2O$.
- Iodine behaves similarly; iodate(I) disproportionates very readily, so IO_3^- forms even when cold.
- Calcium hydroxide, cold: $2Cl_2 + 2Ca(OH)_2 \rightarrow CaCl_2 + Ca(ClO)_2 + 2H_2O$ (bleaching powder).
- Calcium hydroxide, hot: $6Cl_2 + 6Ca(OH)_2 \rightarrow 5CaCl_2 + Ca(ClO_3)_2 + 6H_2O$.

WORKED EXAMPLE: AN UNFAMILIAR DISPROPORTIONATION

$3Ca(OH)_2 + 3Cl_2 + KCl \rightarrow KClO_3 + 3CaCl_2 + 3H_2O$ (a route to potassium chlorate(V))
Chlorine in Cl_2: 0. In $KClO_3$: K +1, O 3 × (−2) = −6, so Cl = **+5** (oxidised). In $CaCl_2$: **−1** (reduced).
Same element, one species (Cl_2), oxidised and reduced: **disproportionation**. Calcium stays +2, so don't mention it.

MARK-SCHEME LANGUAGE

- Water: "$Cl_2 + H_2O \rightleftharpoons HCl + HOCl$" (ALLOW HClO, $H^+ + Cl^- + HOCl$) (WCH12 Jan 2022).
- Cold: "$Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$"; hot: "$3Cl_2 + 6NaOH \rightarrow NaClO_3 + 5NaCl + 3H_2O$". Ionic forms allowed. Cl oxidised to **+1, +1 and +5** respectively and reduced to **−1** in all three; name **disproportionation** (WCH12 Jan 2022, 6 marks).
- Alkali on the equilibrium: NaOH "reacts with / removes the HCl / H^+", so the position shifts "to the right" (WCH12 Oct 2024).
- Acid on bleach releases chlorine: "$NaClO + 2HCl \rightarrow NaCl + H_2O + Cl_2$"; DO NOT AWARD $NaClO + HCl \rightarrow NaOH + Cl_2$ (WCH12 Jun 2023).
- "$3Ca(OH)_2 + 3Cl_2 + KCl \rightarrow KClO_3 + 3CaCl_2 + 3H_2O$" (WCH12 Jun 2021).

Often asked as: equations and oxidation-number changes for the three chlorine reactions; benefits and risks of chlorinating water; why mixing acid with bleach is dangerous.

BEYOND THE SPEC, BUT EXAMINED

Atom economy of making a chlorate: e.g. about 24% for the $Ca(OH)_2$/KCl route (WCH12 Jun 2021, Oct 2025).
Chlorates(V) and (VII) and chlorine dioxide, ClO_2 (an unpaired electron), in an unfamiliar context (WCH12 Jun 2024).

COMMON MISTAKE

Showing an oxidation-number change for Ca or Na (WCH12 Jan 2023 examiner report).
Writing a sulfuric acid reaction when the question says "alkali" (WCH12 Jan 2022 examiner report).
Mixing up which conditions give ClO^- (cold, dilute) and which give ClO_3^- (hot).

SPEC 8.27

Reactions of halides and hydrogen halides

WHAT THIS SPEC POINT ASKS

Understand: (i) solid Group 1 halides with concentrated sulfuric acid, showing the trend in reducing ability of the hydrogen halides; (ii) precipitation of Cl^-, Br^- and I^- with acidified silver nitrate, and the solubility of the precipitates in ammonia; (iii) hydrogen halides with ammonia gas and with water.

8.27 (i) Solid halides with concentrated sulfuric acid

All three first make the hydrogen halide (an **acid–base** reaction, not redox):
$NaX(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HX(g) \rightarrow$ **misty (steamy) fumes**

Then HBr and HI, which are stronger **reducing agents**, go on to **reduce the sulfuric acid**:

Halide	Further reaction	Sulfur product (oxidation number)	What you see
NaCl	none: HCl can't reduce H_2SO_4	— (stays +6)	**misty fumes** of HCl only
NaBr	$2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$	**SO_2 (+4)**	misty fumes; **red-brown / orange fumes** of Br_2; colourless choking gas (SO_2)
NaI	$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$	SO_2 (+4)	misty fumes; **purple vapour / black solid** (I_2)
	$6HI + H_2SO_4 \rightarrow 3I_2 + S + 4H_2O$	**S (0)**	**yellow solid**
	$8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$	**H_2S (−2)**	**rotten-egg smell**

The trend: reducing power **increases** $Cl^- < Br^- < I^-$ (HCl < HBr < HI). The bigger the halide ion, the further its outer electrons are from the nucleus and the more shielded they are, so they are **lost more easily**. Iodide reduces sulfur all the way from +6 to −2; bromide only to +4; chloride not at all.

Iodide ionic equation (no spectators): **$8I^- + 10H^+ + SO_4^{2-} \rightarrow 4I_2 + H_2S + 4H_2O$** (or $8I^- + 8H^+ + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$).

8.27 (ii) Silver nitrate test for halide ions

1. Add **dilute nitric acid** to the solution. This removes ions such as CO_3^{2-} and OH^- that would also give a precipitate with Ag^+. (Not HCl, which adds Cl^-; not H_2SO_4.)
2. Add **silver nitrate solution**.
3. Then test the precipitate's solubility in **ammonia solution**.

Ion	Precipitate	Ionic equation	Dilute $NH_3(aq)$	Concentrated $NH_3(aq)$
Cl^-	**white** AgCl	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	**dissolves**	dissolves
Br^-	**cream** AgBr	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	doesn't dissolve	**dissolves**
I^-	**(pale) yellow** AgI	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	doesn't dissolve	**doesn't dissolve**

The colours are hard to tell apart (white v cream v pale yellow), which is why the ammonia step is needed. The precipitate dissolves because ammonia forms the soluble complex ion $[Ag(NH_3)_2]^+$. Fluoride gives **no precipitate**: AgF is soluble.

8.27 (iii) Hydrogen halides with ammonia and with water

- With **ammonia gas**: a **white smoke** of the ammonium halide forms. **$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$**. This is the test for HCl fumes (and for ammonia).
- With **water**: hydrogen halides are very soluble and react to form **acidic solutions**: **$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$** (hydrochloric acid). HBr and HI give hydrobromic and hydroiodic acid. The "misty fumes" are droplets of acid formed with moisture in the air.

MARK-SCHEME LANGUAGE

Iodide + conc. H_2SO_4 (WCH12 Oct 2025, 6 marks): "misty fumes (white smoke with NH_3) = HI", "purple vapour = I_2", "rotten-egg smell = H_2S", "I: −1 → 0; S: +6 → −2", "chloride doesn't change the oxidation number of S, so iodide is the stronger reducing agent", "$8I^- + 8H^+ + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$". DO NOT AWARD SO_2 when the stem mentions a rotten-egg smell.
KCl v KBr (WCH12 Jan 2020): similarities "both produce hydrogen halides / misty fumes" (acid–base, protonation); differences "**only the KBr reaction is redox**", "bromide is a stronger reducing agent", "(red-)brown fumes / SO_2". DO NOT AWARD white smoke, rotten-egg smell or yellow solid for bromide.
Lowest-oxidation-state sulfur **compound**: SO_2 from HBr, **H_2S** from HI (sulfur itself is not a compound) (WCH12 Jun 2022).
Silver nitrate: "add (dilute) **nitric acid** and silver nitrate", "white precipitate", "dissolves in **dilute** ammonia" (WCH12 Jun 2021). Bromide: "cream / off-white precipitate", "$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$", dissolves in **concentrated** ammonia (WCH12 Jan 2026). DO NOT AWARD HCl or H_2SO_4 as the acid.

Often asked as: observations and equations for a halide with conc. H_2SO_4; the silver nitrate test; explain the trend in reducing power.

BEYOND THE SPEC, BUT EXAMINED

White smoke forms only where the fumes meet ammonia: use it to identify HX fumes (WCH12 Jun 2019).
The silver nitrate test after hydrolysing halogenoalkanes: **yellow** precipitate for an iodoalkane, **white** for a chloroalkane (WCH12 Oct 2024; see 10.10).

COMMON MISTAKE

Mixing up the silver nitrate test with Tollens' reagent and adding NaOH (WCH12 Jan 2021 examiner report).
Confusing hydrogen halides (HX) with halogens (X_2), or listing sulfur as a "compound".
Saying chloride ions reduce sulfuric acid.

SPEC 8.28

Predicting the properties of fluorine and astatine

WHAT THIS SPEC POINT ASKS

Use the trends in Group 7 to predict the properties of fluorine and astatine and their compounds.

8.28 Extrapolate every trend

Property	Trend down the group	Fluorine (top)	Astatine (bottom)
State and colour	darker, gas → solid	pale yellow **gas**	**black solid**
Boiling temperature	increases (more electrons, stronger London forces)	lowest (−188 °C)	highest
Electronegativity	decreases	**highest of all elements (4.0)**	lowest in the group
Oxidising power of X_2	decreases	**strongest** oxidising agent	weakest
Displacement	each halogen displaces those below it	displaces all other halogens from their halides	displaces none; is displaced by all the others
Reducing power of X^-	increases	F^- the **weakest** reducing agent	At^- the **strongest** reducing agent
X^- + conc. H_2SO_4	more reduction of S	HF only (misty fumes), no redox	expected to reduce S to H_2S, like or beyond iodide
Silver halide	colour darkens, solubility in NH_3 falls	**AgF is soluble**: no precipitate	AgAt: a precipitate darker than AgI, insoluble in concentrated NH_3
HX thermal stability	decreases (H–X bond weaker)	HF very stable	HAt very unstable

8.28 Two predictions worth writing out

- Fluorine is so strong an oxidising agent that it **oxidises water**: $2F_2(g) + 2H_2O(l) \rightarrow 4HF(aq) + O_2(g)$. (Chlorine only reacts reversibly with water; see 8.26.)
- Astatine is predicted to react with hydrogen only slowly and reversibly, like iodine, and to form HAt, which, by analogy with HI, would dissolve in water to give an acid: $HAt(g) + H_2O(l) \rightarrow H_3O^+(aq) + At^-(aq)$.

EXAM TIP

For any "predict" question, state the **prediction**, then the **trend** you used ("because oxidising power decreases down the group"), then the **reason** for the trend (more shells, more shielding, weaker attraction for an incoming electron).

MARK-SCHEME LANGUAGE

- No WCH12 paper from 2019 to 2026 asked about fluorine or astatine directly. Expect them in multiple-choice questions or as a "predict" part using the trends from 8.24 to 8.27.

Often asked as: predict the state, colour, or a reaction of astatine or fluorine; predict the result of the silver nitrate test with fluoride or astatide.

COMMON MISTAKE

Predicting a precipitate with $AgNO_3$ for fluoride. AgF is soluble.
Predicting that astatine is a stronger oxidising agent than iodine: the trend runs the other way.

Topic 9 – Introduction to Kinetics and Equilibria

SPEC 9.1 Collision theory: concentration, temperature, pressure and surface area

WHAT THIS SPEC POINT ASKS

Use collision theory to explain how concentration, temperature, pressure and surface area change the rate of a reaction.

9.1 What has to happen for particles to react

- Particles must **collide**.
- The collision must have **energy at least equal to the activation energy** (Ea), so bonds can start to break (9.2).
- The particles must meet with the **right orientation**, so the reacting parts of the particles actually touch.
- A collision that meets these conditions is a **successful** (or effective) collision. Most collisions aren't successful: the particles just bounce apart.
- **Rate** depends on the **number of successful collisions per unit time** (the *frequency* of successful collisions), not just on how many collisions there are in total.

9.1 The four factors

Change	What happens to the particles	Effect on rate
Higher concentration (solutions)	more particles **in the same volume**, so they collide more often	more successful collisions **per unit time**
Higher pressure (gases only)	the same gas particles are squeezed into a smaller volume: more particles per unit volume	more successful collisions per unit time
Larger surface area (solids, e.g. powder instead of lumps)	more particles are **exposed** at the surface and available to collide	more successful collisions per unit time
Higher temperature	particles have more **kinetic energy**: (1) a much **greater proportion have energy ≥ Ea**; (2) they also move faster, so collide a little more often	many more successful collisions per unit time; effect (1) is by far the bigger one (see 9.4)

- Concentration, pressure and surface area change **how often** particles collide. They don't change the energy of the particles, so the **fraction** of collisions that succeed stays the same.
- Temperature is different: it changes the **energy** of the particles, so it changes the fraction of collisions that succeed. That's why a 10 °C rise can roughly double the rate of many reactions.
- Pressure only matters when at least one reactant is a **gas**. Changing the pressure on a reaction in solution has no real effect.

EXAM TIP

Build every "explain the effect on rate" answer from three links: **what changes about the particles** → **more successful collisions** → **per unit time** (or "more frequent"). Leaving out "per unit time" or "successful" is the most common way to drop a mark.

MARK-SCHEME LANGUAGE

- Concentration: "more particles **in the same volume**" and "more (successful) collisions **per unit time**". ALLOW "particles are closer", "frequency of collisions". IGNORE "chance / probability of collisions". "More particles so more successful collisions" scores only 1 (WCH12 Jan 2025).
- Surface area: "larger surface area", ALLOW "more (active) sites". IGNORE "greater area of contact".
- Temperature: "particles have more kinetic energy", "**more particles have energy ≥ Ea**", "more successful collisions per unit time".
- Pressure: "more collisions per unit time". Any reference to **atoms** (instead of molecules or particles) can lose the mark.

Often asked as: explain why a change in concentration, pressure or catalyst particle size speeds up a reaction, usually inside an industrial-process question.

COMMON MISTAKE

"Particles collide more" with no time reference. Say **more frequent** collisions or **more collisions per second**.
"Higher concentration gives particles more energy." It doesn't: only temperature changes particle energy.
Mixing up the explanation for **rate** with the explanation for **yield**. Pressure can raise both, but for different reasons: collisions for rate, Le Chatelier for yield (9.10). In a two-mark answer the examiners want one consistent route.

SPEC 9.2

Activation energy

WHAT THIS SPEC POINT ASKS

Understand that a collision only leads to reaction if it has enough energy: at least the activation energy.

9.2 Definition

- **Activation energy (Ea)** is the **minimum energy** that colliding particles need for a reaction to happen.
- It is the energy needed to start breaking bonds in the reactants. On a reaction profile it's the height of the "hill" from the **reactants** up to the **peak** (the transition state).
- Every reaction has an Ea, whether it's exothermic or endothermic. Its size isn't set by ΔH: a very exothermic reaction can still have a large Ea (that's why fuels don't catch fire until you light them).
- For an **endothermic** reaction Ea must be at least as big as ΔH, because the products sit higher than the reactants.

9.2 Reading a reaction profile

- x-axis: **progress of reaction**; y-axis: **enthalpy** (or energy).
- Ea: arrow from the **reactant line up to the peak**.
- ΔH: arrow from the **reactant line to the product line** (points down for exothermic, up for endothermic).
- See the figure in 9.6 for a labelled example.

MARK-SCHEME LANGUAGE

- "Particles with energy ≥ Ea" (ALLOW "collisions with energy greater than Ea").
- On a profile, Ea is drawn as a rise **from the reactants to the peak**. An arrow starting at the products, or ending below the peak, loses the mark (WCH12 Jan 2022).

Often asked as: label Ea on a profile (often for an endothermic reaction); use Ea inside a Maxwell–Boltzmann or catalyst explanation.

COMMON MISTAKE

Drawing Ea from the products to the peak (that's the Ea of the **reverse** reaction).
Calling Ea "the energy needed for the reaction": say **minimum** energy, and that it's needed for a **successful collision**.

SPEC 9.3

Calculating rate: 1/time and gradients

WHAT THIS SPEC POINT ASKS

Calculate a rate (i) as 1/time from the time taken for a reaction, and (ii) from the gradient of a graph, by drawing a tangent, for the initial rate or the rate at time t.

9.3 What rate means

- **Rate of reaction** = change in amount (or concentration) of a reactant or product **per unit time**.
- Units follow what you measured: mol dm^{-3} s^{-1} for concentration, cm^3 s^{-1} for a gas volume, g s^{-1} for a mass.

9.3 Ways to follow a reaction

Method	Use it when	Example
Gas volume (gas syringe or upturned measuring cylinder/burette over water)	a gas is made	marble chips + HCl; decomposition of H_2O_2
Mass loss (open flask on a balance)	a **dense** gas escapes (CO_2 works; H_2 is too light to measure well)	$CaCO_3$ + HCl
Time for a set change ("clock" method)	a colour appears or a precipitate hides a mark	sodium thiosulfate + HCl (cloudy sulfur hides a cross)
Sampling and titration	concentration changes but nothing is visible	take samples, **quench** them (stop the reaction, e.g. by cooling or diluting), titrate

9.3 (i) Rate = 1/time

- In a clock reaction you measure the **time** for a fixed amount of reaction (for example, the cross disappearing).
- A shorter time means a faster rate, so **rate ∝ 1/time**. The value 1/t (units s^{-1}) is used as a measure of the rate.
- Plot 1/t against concentration (or temperature) to see how rate depends on it. A straight line through the origin means rate is proportional to concentration.

WORKED EXAMPLE: RATE FROM 1/TIME

Sodium thiosulfate reacts with dilute HCl: $S_2O_3^{2-}(aq) + 2H^+(aq) \rightarrow S(s) + SO_2(g) + H_2O(l)$. The sulfur makes the mixture cloudy.

The cross under the flask disappears after **40 s** with 0.10 mol dm^{-3} thiosulfate, and after **80 s** with 0.05 mol dm^{-3}.

Rate (0.10 mol dm^{-3}) = 1/40 = **0.025 s^{-1}** (2.5 × 10^{-2} s^{-1})

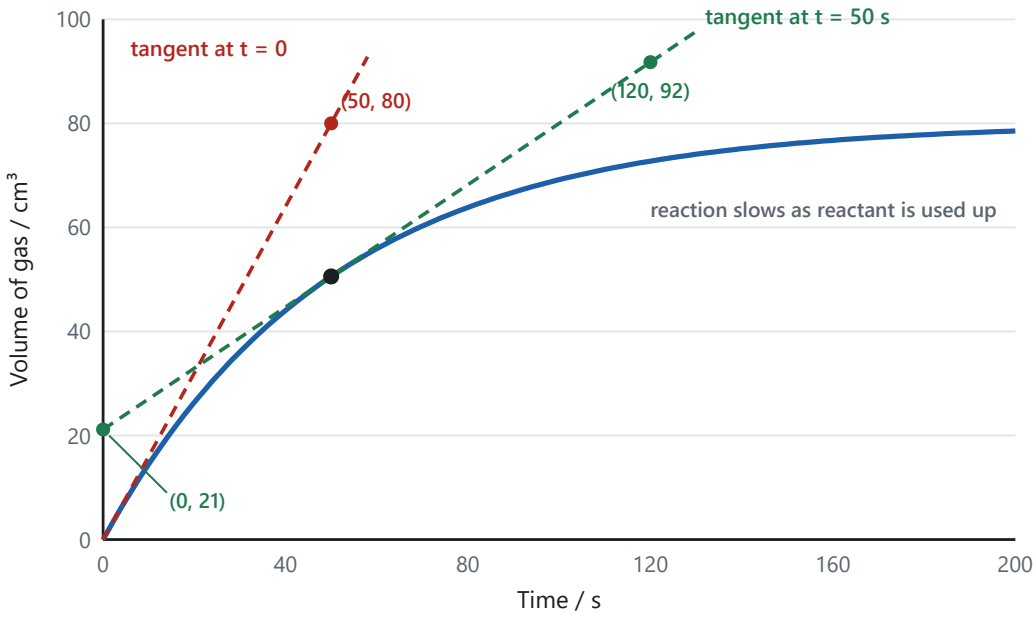
Rate (0.05 mol dm^{-3}) = 1/80 = **0.0125 s^{-1}** (1.25 × 10^{-2} s^{-1})

Halving the concentration halved 1/t, so rate is proportional to the thiosulfate concentration.

Exam data often uses (1/t) × 10^3 to give tidy numbers: 1/40 = 0.025, so (1/t) × 10^3 = 25.

9.3 (ii) Rate from a gradient

- On a graph of amount (or volume, or concentration) against time, the **rate at any moment = gradient of the curve** at that moment.
- The curve gets less steep with time because reactants are used up, so collisions become less frequent.
- To find the gradient of a curve, draw a **tangent**: a straight line that just touches the curve at that point, with the same slope.
- **Initial rate** = gradient of the tangent at **t = 0**. It's the fairest way to compare experiments, because nothing has been used up yet.



Volume of gas against time. The tangent at t = 0 gives the initial rate; the tangent at t = 50 s gives the rate at that moment.

WORKED EXAMPLE: INITIAL RATE AND RATE AT T = 50 S

Initial rate (red tangent): it passes through (0, 0) and (50, 80).

Gradient = (80 − 0) / (50 − 0) = **1.6 cm^3 s^{-1}**

Rate at 50 s (green tangent): read two points far apart on the tangent, (0, 21) and (120, 92).

Gradient = (92 − 21) / (120 − 0) = 71 / 120 = **0.59 cm^3 s^{-1}**

The rate has fallen to about a third of its starting value because the acid is being used up.

EXAM TIP

Draw the tangent **long**: at least half the width of the grid. Read two points **far apart** that sit on grid lines. A short tangent magnifies any reading error. Always write the units: they come from the axes ($cm^3 \div s \rightarrow cm^3\ s^{-1}$).

MARK-SCHEME LANGUAGE

- Tangent at t = 0 must **touch** the curve and be long enough (in Jun 2022 it had to reach at least 20 cm^3). Gradient accepted within a range (0.25–0.33 there). The unit **$cm^3\ s^{-1}$** was a separate mark.
- Graph marks: sensible scale using at least half the grid; **axes labelled with units**; points within ½ small square; **smooth curve** of best fit (not dot-to-dot).
- Halving the acid concentration (acid limiting): "initial rate halves **and** final volume of gas halves".

Often asked as: calculate 1/t for a table (to 3 s.f.) and plot it; draw a tangent and work out the initial rate; predict how initial rate and final volume change if the concentration or amount is changed.

BEYOND THE SPEC, BUT EXAMINED

- Choosing the amount of reactant so the gas fits a **100 cm^3 gas syringe**, using **24 $dm^3\ mol^{-1}$** for the molar volume at room temperature (WCH12 Jun 2022).
- If the acid is the limiting reactant, halving its concentration halves the **final volume** of gas as well as the rate. If the acid is in excess, the final volume doesn't change (WCH12 Jun 2022).
- Values of 1/t in a table rounded to **3 significant figures** (WCH12 Jan 2025).

COMMON MISTAKE

No tangent drawn: the gradient was worked out from the curve's end points (that gives an *average* rate, not the rate at an instant).
Saying the final volume of gas stays the same after halving a **limiting** reactant: fewer moles of acid means less gas.
Missing units on axes, or joining points dot-to-dot.

SPEC 9.4 Maxwell–Boltzmann distribution and temperature

WHAT THIS SPEC POINT ASKS

Use the Maxwell–Boltzmann distribution of molecular energies to explain, qualitatively, how temperature changes the rate of reaction.

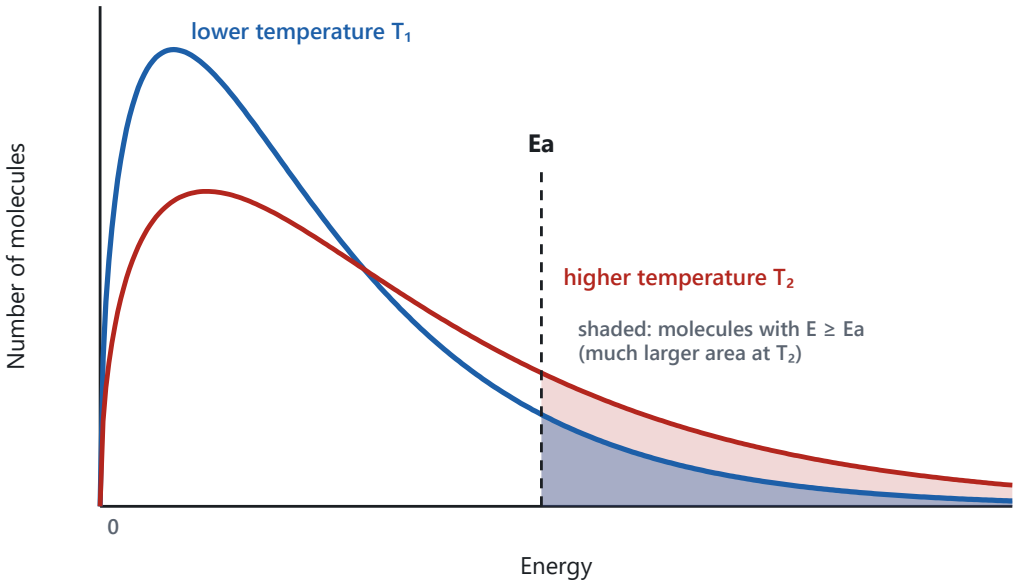
9.4 What the curve shows

- In a gas (or solution) at one temperature, the molecules don't all have the same energy. They collide and swap energy all the time, so there's a **spread** of energies.
- The **Maxwell–Boltzmann distribution** plots **number (or fraction) of molecules** (y-axis) against **energy** (x-axis).
- Features you must draw correctly:
  - starts at the **origin**: no molecules have zero energy;
  - rises to a **peak**: the most likely energy;
  - has a long tail to the right that gets closer and closer to the x-axis but **never touches it** (an **asymptote**): a few molecules have very high energy;

- the **area under the curve** = the total number of molecules.

- Mark **Ea** to the **right of the peak**. The area under the curve to the right of Ea is the number of molecules with **enough energy to react**.

9.4 Raising the temperature



At the higher temperature T_2 the peak is lower and further right, and the area beyond Ea is much larger. The number of molecules (total area) is the same.

- The curve **flattens and shifts right**: the peak moves to a **higher energy** and becomes **lower**.
- The peak drops because the **total area stays the same**: the number of molecules hasn't changed, so if the curve spreads out it must get lower.
- The tail at high energy sits **above** the original curve, so the area beyond Ea is much bigger.
- So: at a higher temperature **many more molecules have energy ≥ Ea → more successful collisions per unit time** → faster rate. Molecules also move faster and collide more often, but that is a much smaller effect.

MARK-SCHEME LANGUAGE

Drawing the higher-temperature curve:

- "starts at the **origin**", "**lower maximum**", "peak **moved to the right**", "**asymptotic** and **above** the original line" at high energy.
- DO NOT AWARD if any part of the new curve to the right of the peak falls below the original line, if the curves cross more than once, or if the tail ends too high (more than about a quarter of the new peak height) (WCH12 Jun 2023, Jan 2025).
- Axes: y "number / fraction / proportion of molecules", x "energy" (kinetic energy accepted).

Explaining: "more molecules / collisions have **energy greater than (or equal to) the activation energy**" or "the **area under the curve beyond Ea** increases", and "more **successful collisions per unit time**".

Often asked as: draw a second curve on a given one at a higher (or lower) temperature, then explain the rate change; part of a 6-mark question with a catalyst (9.8).

COMMON MISTAKE

Curve not starting at the origin, or the tail touching (or crossing) the x-axis.
Drawing the hotter curve with a **higher** peak: the area must stay the same.
The two curves crossing twice, or the hot curve dropping below the cold curve in the tail.
A good diagram but no **reference to it** in the written explanation: say "the area to the right of Ea is larger".
Forgetting "per unit time" in the final step.

SPEC 9.5

Catalysts: an alternative route with lower activation energy

WHAT THIS SPEC POINT ASKS

Understand that a catalyst works by providing an alternative reaction pathway with a lower activation energy.

9.5 How a catalyst works

- A **catalyst** increases the rate of a reaction **without being used up** overall. It's chemically unchanged at the end, so it can be reused.
- It provides an **alternative route (mechanism)** for the reaction, with a **lower activation energy**.
- It does **not** give particles more energy, and doesn't change ΔH. It lets **more of the collisions that already happen** succeed, because more molecules now have energy ≥ the new, lower Ea (9.8).
- The catalyst **does** take part in the reaction: it forms an **intermediate** with a reactant and is then regenerated (9.6). "Takes no part in the reaction" is wrong.
- A catalyst speeds up the forward and backward reactions equally, so it does **not** change the position of equilibrium or the yield; it just gets you to equilibrium faster (9.10).

9.5 Two kinds

	Homogeneous	**Heterogeneous**
Phase	same phase as the reactants	different phase (usually a solid with gas or liquid reactants)
How	forms an intermediate in solution	reactants **adsorb** onto the surface, bonds weaken and react, products **desorb**
Examples	H^+ in esterification; Fe^{2+}/Fe^{3+} in the iodide–peroxodisulfate reaction	**iron** in the Haber process; **V_2O_5** in the Contact process; **Pt/Rh/Pd** in catalytic converters

- Heterogeneous catalysts are used as **powders, pellets or fine meshes** on a support to give a **large surface area**, so there are more sites where reaction can happen.
- They can be **poisoned**: an impurity (e.g. lead or sulfur) bonds strongly to the surface and blocks it.

MARK-SCHEME LANGUAGE

- "A catalyst **decreases the activation energy** by providing an **alternative pathway / route / mechanism**" (WCH12 Oct 2025, Jun 2025).
- Then link it to rate: "more molecules have energy ≥ Ea" and "frequency of successful collisions increases".
- IGNORE "not used up" on its own when the question asks how it works; it's not the mechanism.

Often asked as: explain how a catalyst increases the rate; say why a substance in a process is (or isn't) a catalyst.

COMMON MISTAKE

"Catalysts do not take part in the reaction" (WCH12 Jan 2022 report). They do, and are regenerated.
"A catalyst gives the particles more energy" or "increases the yield". Neither is true.
Just writing "lowers Ea" with no mention of an **alternative route**, or no link to successful collisions.
In Jan 2022 the reactant (urea) in a catalytic converter was called a catalyst: it is used up, so it isn't.

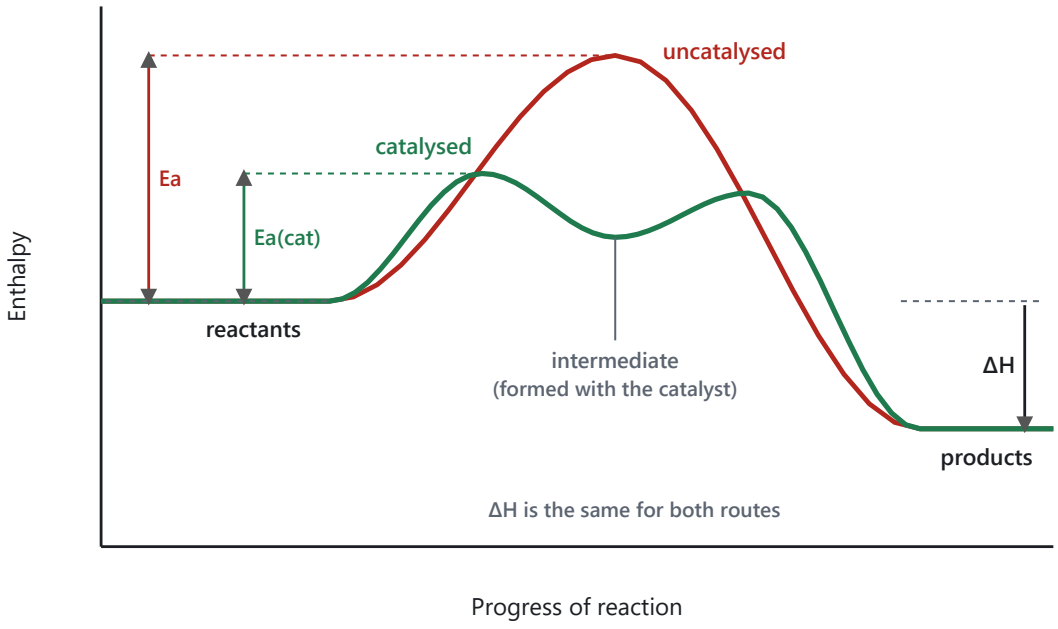
SPEC 9.6

Reaction profiles: uncatalysed and catalysed

WHAT THIS SPEC POINT ASKS

Draw reaction profiles for a reaction with and without a catalyst, including the energy level of the intermediate formed with the catalyst.

9.6 Drawing the two profiles



Exothermic reaction. Uncatalysed: one high hump. Catalysed: two lower humps with a dip between them, where the intermediate forms.

- **Same start, same finish**: the reactants and products are at the same levels on both profiles, so **ΔH is unchanged**.
- **Uncatalysed**: a single high peak. Ea = reactants → top of the peak.

- **Catalysed**: the route goes through an **intermediate** (a species made from the reactant and the catalyst). This shows as a **dip (trough)** between **two smaller peaks**:
  - step 1: reactant + catalyst → intermediate (first peak);
  - step 2: intermediate → product + catalyst (second peak), which regenerates the catalyst.
- Ea(cat) is measured from the **reactants to the highest point** on the catalysed route. Both catalysed peaks must be **lower** than the uncatalysed peak.
- The intermediate's dip sits **above the reactants** in most sketches, because the intermediate is a short-lived, fairly high-energy species. Draw it clearly as a minimum between the two humps.

WORKED EXAMPLE: A CATALYSED STEP-BY-STEP ROUTE

Fe^{2+} catalyses $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$ (a reaction between two negative ions, which repel, so the uncatalysed Ea is high).
Step 1: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$
Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$
Fe^{3+} is the **intermediate** (the dip on the profile); Fe^{2+} is regenerated, so it's the catalyst. Each step involves oppositely charged ions, so each has a lower Ea.

MARK-SCHEME LANGUAGE

- Reactant and product levels **unchanged**; catalysed peak(s) **lower** than the uncatalysed one; Ea labelled from the **reactant line** to the peak.
- The research found no WCH12 question (2019–2026) that asked for the intermediate dip, but the spec names it, so learn it.

Often asked as: sketch a profile for a given (often endothermic) reaction and label Ea and ΔH; add a catalysed route.

COMMON MISTAKE

Moving the reactant or product level for the catalysed route (ΔH would change: it can't).
Drawing the catalysed route as one hump when the question asks for the intermediate.
Labelling Ea(cat) only to the first peak when the second peak is higher.

SPEC 9.7 Catalysts in industry: sustainability

WHAT THIS SPEC POINT ASKS

Understand how catalysts make industrial processes more sustainable by cutting energy use and/or raising atom economy.

9.7 Why industry relies on catalysts

- **Lower temperature (and sometimes pressure)** for the same rate → **less energy** needed for heating and compressing → **less fossil fuel burned**, lower CO_2 emissions and lower cost.
- **More selective (specific)** catalysts speed up only the reaction you want → **fewer side products** → **higher atom economy** and less waste to separate or dispose of.
- Allow a **different route** with fewer steps or better atom economy (e.g. an addition reaction, atom economy 100%, instead of a substitution that makes a by-product).
- They are **not used up**, so a small amount can be reused many times (though they need replacing when poisoned or worn).

- For an **exothermic** equilibrium, a catalyst lets the plant run at a **lower temperature**, which also gives a **better yield** (9.10) while keeping the rate acceptable.

Process	Catalyst	What it makes
Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	iron	ammonia (fertilisers)
Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	vanadium(V) oxide, V_2O_5	sulfuric acid
Hydration of ethene: $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$	phosphoric acid on silica	ethanol
Methanol: $CO + 2H_2 \rightleftharpoons CH_3OH$	copper (with ZnO/Al_2O_3)	methanol
Catalytic converter: $2CO + 2NO \rightarrow 2CO_2 + N_2$	platinum / rhodium / palladium	less toxic exhaust gases
Cracking of long-chain alkanes	zeolites	smaller alkanes and alkenes

MARK-SCHEME LANGUAGE

- "allows a **lower temperature** to be used / **less energy** is needed (for the same rate)".
- "**specific** for the product so there is less waste / fewer other products / **improves atom economy**".
- "not used up / can be used again".
- IGNORE "just increases the rate" or "lowers Ea" when the question asks about sustainability (WCH12 Oct 2020).

Often asked as: state two ways a catalyst makes a process more sustainable (copper in methanol manufacture, phosphoric acid in ethanol manufacture).

BEYOND THE SPEC, BUT EXAMINED

- Catalysts are used as **pellets or powder** for a larger surface area (WCH12 Oct 2021).
- Too much steam in ethene hydration can **wash the phosphoric acid catalyst off its support** (WCH12 Jun 2024).

COMMON MISTAKE

Answering a sustainability question with "it speeds up the reaction": link it to **energy** or **atom economy**. Saying a catalyst "increases the yield". For an exothermic equilibrium, it can *allow* a lower temperature that gives a better yield: say it that way.

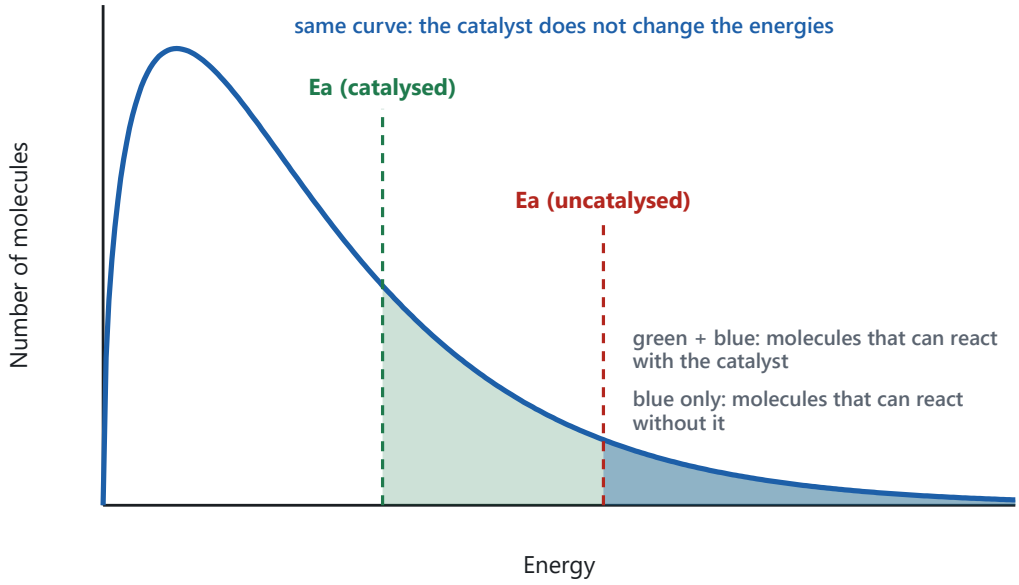
SPEC 9.8

Catalysts and the Maxwell–Boltzmann distribution

WHAT THIS SPEC POINT ASKS

Use the Maxwell–Boltzmann distribution to explain qualitatively how a catalyst increases the rate.

9.8 One curve, two activation energies



The catalyst doesn't change the curve. It moves Ea to the left, so a much larger area lies beyond it.

- The temperature hasn't changed, so the **distribution curve stays exactly the same**. Don't draw a second curve.
- Mark the **uncatalysed Ea** and, to its **left**, the **catalysed Ea**. Both must be to the **right of the peak**.
- The area under the curve beyond Ea(cat) is much bigger than the area beyond Ea, so **many more molecules have enough energy to react → more successful collisions per unit time** → faster rate.

9.8 Temperature v catalyst: a comparison

	Raise the temperature	Add a catalyst
Curve	changes: lower peak, shifted right	unchanged
Ea	unchanged	lowered (new line to the left)
Molecules with E ≥ Ea	more	more
Collision frequency	increases a little	unchanged

MARK-SCHEME LANGUAGE

Four-mark answer (WCH12 Oct 2021):

- both **axes labelled**;
- **Ea labelled**;
- catalysed Ea drawn **to the left** of the uncatalysed Ea (both to the right of the peak, on the same curve);
- "**more particles have energy greater than Ea**" / "greater area to the right of Ea(cat)".

Then for rate: "increases the number of successful collisions **per unit time**". Shading isn't required, but wrong shading loses the mark.
DO NOT AWARD "catalyst increases yield", "particles have more energy", "move faster" or "greater collision frequency".

Often asked as: a 6-mark question with both temperature and a catalyst on one diagram (WCH12 Oct 2019); draw catalysed Ea on a given curve and explain.

COMMON MISTAKE

Drawing a new curve for the catalyst.
Putting Ea(cat) at or to the left of the peak.
Losing the last mark by not saying **per unit time** or **frequency** (WCH12 Jan 2022 report).

SPEC 9.9 Reversible reactions and dynamic equilibrium

WHAT THIS SPEC POINT ASKS

Know that many reactions are reversible and can reach a dynamic equilibrium in which the forward and backward rates are equal and the concentrations of reactants and products stay constant.

9.9 Reversible reactions

- A reversible reaction can go both ways: reactants → products **and** products → reactants. The sign is ⇌.
- Example: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ (colourless ⇌ brown).
- Example: $CuSO_4{\cdot}5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$ (blue crystals ⇌ white powder; heat drives it right, adding water drives it left).

9.9 Dynamic equilibrium

Start with only reactants:

- At first the forward rate is high (lots of reactant) and the backward rate is zero (no product).
- As reactants are used up the forward rate falls; as products build up the backward rate rises.
- Eventually the two rates become **equal**. From then on, each reaction makes particles as fast as the other uses them up.

At **dynamic equilibrium**:

1. the **rate of the forward reaction equals the rate of the backward reaction**;
2. the **concentrations of reactants and products stay constant** (they are **not** usually equal).

- "Dynamic" because both reactions **are still happening**; at the particle level nothing has stopped.
- On the large scale nothing appears to change: the **colour, pressure, concentration** stay constant.
- It needs a **closed system** if any substance is a gas, so nothing can escape. (An open flask of solution can reach equilibrium; heating $CaCO_3$ in an open tube can't, because CO_2 escapes and the reaction goes to completion.)
- The same equilibrium can be reached **from either direction**: starting from products gives the same equilibrium mixture at the same conditions.

9.9 Suggested practicals

- **NO_2 / N_2O_4**: $2NO_2(g) \rightleftharpoons N_2O_4(g)$, $\Delta H = -57$ kJ mol^{-1}. NO_2 is **brown**, N_2O_4 is **colourless**. Sealed tubes of the mixture go **darker in hot water** and **paler in ice**; squeezing a syringe of the gas (higher pressure) makes the mixture paler once it settles (see 9.10).
- **ICl / ICl_3**: $ICl(l) + Cl_2(g) \rightleftharpoons ICl_3(s)$. ICl is a **brown liquid**, ICl_3 is a **yellow solid**. Passing more Cl_2 in gives more yellow solid; letting Cl_2 out (turning the U-tube upside down) gives back the brown liquid.

MARK-SCHEME LANGUAGE

- Equation questions: "products **and reversible arrow**" (ALLOW two full-headed arrows) (WCH12 Jan 2022).
- Definitions: "rate of forward reaction = rate of backward reaction" and "concentrations (of reactants and products) **remain constant**".

Often asked as: rarely as a bare definition. It hides inside industrial questions, where you must recognise that a reaction is an equilibrium and can't go to completion.

COMMON MISTAKE

"At equilibrium the concentrations of reactants and products are **equal**." They're **constant**, not equal.
"The reaction has stopped." Both reactions continue at the same rate.
Treating an equilibrium as if it goes to completion, e.g. "excess steam will react with all the methane" (WCH12 Jan 2023 report).

SPEC 9.10

Position of equilibrium: temperature, pressure and concentration

WHAT THIS SPEC POINT ASKS

Predict and justify how changing temperature, pressure and concentration shifts the position of a homogeneous equilibrium.

9.10 Le Chatelier's principle

- If a system at equilibrium is disturbed, the **position of equilibrium shifts to oppose the change**.
- "Position shifts **right**" = more products (higher yield). "Shifts **left**" = more reactants.
- **Homogeneous** = everything in the same phase (e.g. all gases, or all in solution).

9.10 The three changes

Change	Equilibrium shifts...	Why (opposes the change)
Increase concentration of a reactant	right	uses up some of the added reactant
Remove a product as it forms	right	replaces the product
Increase pressure (gases)	to the side with **fewer moles of gas**	fewer gas molecules → lowers the pressure
Decrease pressure	to the side with more moles of gas	more molecules → raises the pressure
Increase temperature	in the **endothermic** direction	absorbs the extra heat
Decrease temperature	in the **exothermic** direction	releases heat
Add a catalyst	**no shift**	speeds up both reactions equally; equilibrium is reached sooner

- Pressure: count **gas** moles only (ignore solids and liquids). If both sides have the same number of gas moles, pressure has **no effect** on the position.

- Temperature: use the **sign of ΔH** for the **forward** reaction. ΔH negative → forward is exothermic → a **lower** temperature shifts it right.
- Only temperature changes the value of the equilibrium constant, K (Unit 4). Concentration and pressure move the position but leave K unchanged.

WORKED EXAMPLE: THREE CHANGES ON ONE EQUILIBRIUM

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197$ kJ mol^{-1}

Higher pressure: left side has 3 moles of gas, right side has 2. The equilibrium shifts **right** (fewer gas moles), so the yield of SO_3 **increases**.

Higher temperature: forward reaction is **exothermic**, so the endothermic (backward) direction is favoured. Equilibrium shifts **left**; yield **decreases**.

Adding more O_2: equilibrium shifts **right** to use up the extra O_2; more SO_3.

WORKED EXAMPLE: A COLOUR CHANGE IN SOLUTION

$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$

yellow ⇌ orange

Add acid (more H^+): shifts **right** → the solution turns **orange**.

Add alkali (OH^- removes H^+): shifts **left** → turns **yellow**.

EXAM TIP

For a 2–3 mark "predict and justify" answer write: **direction of shift** + **the reason** (number of gas moles on each side, or sign of ΔH) + **effect on yield**. If you quote mole numbers they must be right.

MARK-SCHEME LANGUAGE

- Pressure: "higher pressure moves the position of equilibrium **to the right / forward** / increases the yield" and "**4 gaseous moles on the left and 2 on the right**" (Haber, WCH12 Oct 2020). A wrong direction scores 0. Quoted mole numbers must be correct.
- Temperature: "the (forward) reaction is **exothermic**", "a **lower temperature** would increase the yield / move the equilibrium to the right". If you call the forward reaction endothermic, neither mark is scored.
- Removing product: "remove the product (as it is formed)", e.g. "cool so ethanol (and water) condense".
- Adding NaOH to chlorine water moves its equilibrium right because the NaOH **removes H^+ / HCl** (WCH12 Oct 2024).

Often asked as: effect of pressure or temperature on the yield of an industrial process; why a product is removed; inside a 6-mark compromise question (9.11).

BEYOND THE SPEC, BUT EXAMINED

- More steam pushes ethene hydration to the right but **dilutes** the product and can wash the catalyst off its support (WCH12 Jun 2024).
- Removing CO_2 from a hydrogen-making process shifts the equilibrium towards more H_2 (WCH12 Jan 2023).

COMMON MISTAKE

"Higher temperature increases the yield because it increases the rate" (WCH12 Jan 2021 report). Rate and yield are different questions.
Counting all moles instead of **gas** moles, or saying "atoms".
Writing "shifts left" and then giving a reason that supports a shift right (WCH12 Oct 2022 report).
Assuming a reaction is endothermic just because it's heated (WCH12 Jan 2022 report): check the sign of ΔH.
Saying a catalyst moves the equilibrium.

SPEC 9.11 Industrial compromise: yield v rate

WHAT THIS SPEC POINT ASKS

Use data to explain why many industrial processes must compromise between a high yield and a fast rate.

9.11 Why a compromise is needed

For an **exothermic** reaction with **fewer gas moles on the product side** (very common):

Condition	Effect on yield (equilibrium)	Effect on rate (kinetics)	Other costs
High temperature	lower: favours the endothermic backward reaction	faster: more molecules with E ≥ Ea	more energy (fuel) to heat
Low temperature	higher	**too slow**	
High pressure	higher: shifts to fewer gas moles	faster: more collisions per unit time	expensive thick-walled plant and pumps, more energy to compress, safety risk
Catalyst	no effect	faster at a lower temperature	cost of catalyst, poisoning

- So a **compromise temperature** is chosen: low enough for an acceptable yield, high enough for an acceptable rate.
- A **compromise pressure** is chosen: high enough for a good yield and rate, but not so high that the plant is too expensive or dangerous.
- A **lower yield** each time through is acceptable because **unreacted gases are recycled**, so almost all the starting material is converted eventually. Removing the product as it forms also pulls the equilibrium right.
- Making product **fast** at a modest yield often gives more product **per day** than a high yield made very slowly.

Process	Temperature	Pressure	Catalyst	Yield per pass
Haber: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92$ kJ mol^{-1}	about 450 °C	about 200 atm	iron	about 15% (unreacted gas recycled)
Hydration of ethene: $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$, $\Delta H = -45$ kJ mol^{-1}	about 300 °C	about 60–70 atm	phosphoric acid on silica	about 5% (ethene recycled)
Contact: $2SO_2 + O_2 \rightleftharpoons 2SO_3$, $\Delta H = -197$ kJ mol^{-1}	about 450 °C	1–2 atm	V_2O_5	over 99% (so no high pressure needed)

WORKED EXAMPLE: MODEL ANSWER FOR A 6-MARK COMPROMISE QUESTION

Methanol: $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, $\Delta H = -91$ kJ mol^{-1}. Conditions used: 250 °C, 50–100 atm, copper catalyst. Explain the choice.

- **Temperature (rate)**: a high temperature gives a faster rate because more molecules have energy ≥ Ea, so there are more successful collisions per unit time.
- **Temperature (yield)**: the forward reaction is exothermic, so a high temperature moves the equilibrium left and lowers the yield. 250 °C is a **compromise** between rate and yield.
- **Pressure (yield)**: 3 moles of gas on the left, 1 on the right; high pressure moves the equilibrium to the right (fewer gas moles), increasing the yield. It also increases the rate.
- **Pressure (cost)**: very high pressure needs expensive, thick-walled equipment and a lot of energy for compressors, so a moderate pressure is used.
- **Catalyst**: copper provides an alternative route with lower Ea, so the rate is acceptable at the lower temperature; it doesn't affect the yield.
- **Recycling**: unreacted CO and H_2 are separated and recycled, so a moderate yield per pass is acceptable.

MARK-SCHEME LANGUAGE

Indicative points (WCH12 Jan 2021, Oct 2025, Oct 2021):

- "high temperature increases the rate (more particles with E ≥ Ea)";
- "a catalyst increases the rate by providing an alternative mechanism with lower Ea";
- "high temperature reduces yield / moves equilibrium left as the reaction is **exothermic**";
- "so the temperature is a **compromise**";
- "high pressure increases yield (equilibrium moves to the side of fewer gas molecules) **or** rate";
- "high pressure: **equipment cost**", "high temperature: **energy cost**";
- "low yield is acceptable because **unreacted reactants are recycled**" (ALLOW "product removed as it forms").

IGNORE environmental effects and atom economy in these questions.

Often asked as: a 6-mark question on a named process (ethanol, methanol, hydrogen) giving the conditions and ΔH; sometimes with a yield-against-pressure graph at several temperatures to evaluate.

BEYOND THE SPEC, BUT EXAMINED

- **Recycling** unreacted gases and **removing the product** (e.g. cooling so ethanol condenses) (WCH12 Jan 2021, Jun 2024).
- Engineering costs: **thicker pipe walls**, compressor energy, safety of high-pressure plant (WCH12 Oct 2025, Jun 2024).
- A higher-temperature route may be chosen because it makes a **useful by-product** (e.g. H_2) (WCH12 Jun 2023).

COMMON MISTAKE

Covering only temperature: unstructured answers miss a variable. Take **temperature, pressure, catalyst, recycling** in turn (WCH12 Jan 2021 report).
Explaining the cost of high pressure without saying **why** it costs more (thicker equipment, energy for compressors) (WCH12 Jan 2025 report).
Justifying a 5% yield because "a lot of product is made": the reason is **recycling**.

Topic 10 – Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

SPEC 10.1 Classifying reactions

WHAT THIS SPEC POINT ASKS

Classify organic reactions (including the Unit 1 ones) as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation.

10.1 The seven types

Type	What happens	Examples
Addition	two molecules join to make **one** product; a double bond opens	ethene + Br_2 → 1,2-dibromoethane; ethene + HBr; ethene + H_2O (steam)
Elimination	a small molecule (H_2O, HX) is **removed** from one molecule, leaving a **C=C**	halogenoalkane + ethanolic KOH → alkene; alcohol + conc. H_3PO_4 → alkene
Substitution	one atom or group is **swapped** for another	alkane + Cl_2 (UV) → chloroalkane + HCl; halogenoalkane + OH^- → alcohol + X^-
Oxidation	gain of O / loss of H (or electrons)	alcohol + acidified $K_2Cr_2O_7$; combustion; alkene + acidified $KMnO_4$ → diol
Reduction	gain of H / loss of O	alkene + H_2 (Ni catalyst) → alkane
Hydrolysis	a bond is **split by water** (or OH^-)	halogenoalkane + H_2O or OH^- → alcohol
Polymerisation	many small monomers join into a long chain	ethene → poly(ethene) (addition polymerisation)

- One reaction can fit more than one label:
  - halogenoalkane + OH^- → alcohol is **substitution** and **hydrolysis**;
  - alkene + H_2 is **addition** and **reduction**;
  - alcohol + conc. H_3PO_4 is **elimination**, also called **dehydration**.
- Answer with the label the question is after. If it asks for the *type of reaction*, one word (e.g. "elimination") is enough; if it asks for the *mechanism*, add the attacking species: "**nucleophilic** substitution", "**electrophilic** addition", "**free-radical** substitution".

MARK-SCHEME LANGUAGE

- A table of reagent, solvent and type (WCH12 Oct 2020): KOH(aq) → **substitution**; NH_3 in ethanol → substitution; KCN in ethanol → substitution; **KOH in ethanol → elimination**. "Nucleophilic" is ignored for the substitutions; "electrophilic" or "free radical" is penalised.
- "**elimination**": DO NOT AWARD "addition–elimination" or "nucleophilic elimination" (WCH12 Jan 2025, Jun 2024).
- IGNORE "dehydration" for elimination and "displacement" for substitution; DO NOT AWARD "dehydrogenation" (WCH12 Jun 2022).

Often asked as: name the type of reaction for a given equation or reagent, often in a reaction-scheme question.

COMMON MISTAKE

Calling the halogenoalkane + **ethanolic** KOH reaction "nucleophilic substitution". It's **elimination** (the most common wrong answer, WCH12 Jan 2025 report).
Giving a mechanism when the question asked only for the type.

SPEC 10.2

Reaction mechanisms

WHAT THIS SPEC POINT ASKS

Understand what a reaction mechanism is.

10.2 Equation v mechanism

- An **equation** shows only the starting materials and the products.
- A **mechanism** shows **how** the reaction happens, step by step: which bonds break, which form, and in what order, using **curly arrows** to show where electrons go.
- Mechanisms explain why reactions follow patterns: reagents that react the same way follow the same mechanism (all primary halogenoalkanes with OH^-, for example).

10.2 Curly arrows

- A **full-headed curly arrow** shows the movement of a **pair of electrons**. These are the arrows in every Topic 10 mechanism.
- A **half-headed arrow (fish-hook)** shows the movement of **one electron**, as in free-radical reactions (Unit 1).
- Rules:
  - an arrow **starts** at a **lone pair** or at the **middle of a covalent bond** (where the electrons are), never at a charge or an atom's symbol on its own;
  - it **ends** at the atom the electrons go to (to form a new bond) or just past an atom (when it leaves as an ion).
- Show **partial charges** (δ+ / δ−) on polar bonds and **full charges** on ions.

MARK-SCHEME LANGUAGE

- "The curly arrow should start from a **covalent bond or a lone pair**" (WCH12 Jun 2023 report). Half-arrows are penalised once.
- An arrow starting from the **negative charge** of OH^- instead of a lone pair loses the mark (WCH12 Oct 2021).

Often asked as: draw a mechanism; spot what's missing from a given mechanism ("identify the omissions", WCH12 Jan 2022).

COMMON MISTAKE

Arrows drawn to show where an **atom** moves, not where the electron pair moves (e.g. an arrow from H to O, when it should go from O's lone pair to H).
Fish-hook arrows in a nucleophilic substitution.

SPEC 10.3

Heterolytic bond breaking: electrophiles and nucleophiles

WHAT THIS SPEC POINT ASKS

Understand that heterolytic bond breaking produces species that act as electrophiles or nucleophiles.

10.3 Two ways a covalent bond can break

	Homolytic fission	Heterolytic fission
Electrons	**one** electron to each atom	**both** electrons to **one** atom
Products	two **radicals** (unpaired electron, e.g. Cl•)	a **positive ion** and a **negative ion**
Arrows	half-headed	full-headed
Typical when	non-polar bond (Cl–Cl, C–H) with UV light	polar bond (C–Br, H–Br)
Leads to	free-radical reactions	electrophilic and nucleophilic reactions

- In heterolytic fission the electrons go to the **more electronegative** atom. For C–Br: CH_3CH_2–Br → $CH_3CH_2^+$ + Br^-.
- The ions formed are reactive:
  - the **positive** species is electron-poor: it can **accept** a pair of electrons, so it's an **electrophile** (e.g. a carbocation, H^+, NO_2^+);
  - the **negative** species has a spare lone pair: it can **donate** a pair of electrons, so it's a **nucleophile** (e.g. Br^-, OH^-).
- A **carbocation** is an ion with a positive charge on a carbon atom. It has three bonding pairs and no lone pairs around the C^+, so it is **trigonal planar** (bond angle 120°).

MARK-SCHEME LANGUAGE

- "**heterolytic** (bond) fission" and "the (pair of) electrons in the C–Br bond are transferred to the **bromine** atom". There must be a mention of **two electrons** (WCH12 Jun 2023).
- Carbocation shape: "**trigonal planar**" because "three bond pairs (and no lone pairs) arranged to minimise repulsion" (WCH12 Jun 2023).

Often asked as: name the type of bond breaking in the first step of a mechanism; give the shape of a carbocation intermediate.

COMMON MISTAKE

Saying "the bond breaks and bromine takes an electron": it takes **both** electrons of the bond.
Stopping the shape explanation at "three bonded pairs". Say they **repel equally** and get as far apart as possible, giving 120°.

SPEC 10.4 Nucleophiles

WHAT THIS SPEC POINT ASKS

Know the definition of a nucleophile.

10.4 Definition

- A **nucleophile** is a species that **donates a lone pair of electrons** to an electron-deficient (δ+ or +) atom, forming a new **covalent bond**.
- "Nucleophile" means "nucleus-loving": it is attracted to positive centres, such as the **δ+ carbon** in a C–X bond.
- Every nucleophile has a **lone pair**. Many are negative ions, but some are neutral.

Nucleophile	Where the lone pair is	Product with a halogenoalkane
OH^-	on O	alcohol
CN^-	on C	nitrile
NH_3	on N (neutral molecule)	amine
H_2O	on O (neutral, weak)	alcohol
RNH_2 (an amine)	on N	secondary amine (unwanted side reaction)

- The opposite is an **electrophile**: a species that **accepts a pair of electrons** (e.g. H^+, $Br^{\delta+}$ in a polarised Br_2, a carbocation). Electrophiles attack electron-rich areas such as C=C (Unit 1).

MARK-SCHEME LANGUAGE

- "**electron pair donor**", or "(species with a) **lone pair** that attacks the δ+ carbon" (WCH12 Jan 2022, Jan 2026). "Acts as a nucleophile" is accepted for the role of OH^- in substitution; just repeating "nucleophilic substitution" is ignored.
- An amine is a nucleophile because of the "**lone pair on the nitrogen**"; it can be substituted further because it still has "**H atoms attached to N**" (WCH12 Jun 2025).

Often asked as: define a nucleophile; state the role of OH^- in a given reaction (nucleophile in substitution, **base** in elimination).

COMMON MISTAKE

"A nucleophile is a negative ion." NH_3 and H_2O are neutral nucleophiles. The key is the **lone pair**. "Electron donor" without "**pair**".

SPEC 10.5

Bond polarity and the type of mechanism

WHAT THIS SPEC POINT ASKS

Understand how bond polarity decides the type of mechanism a compound undergoes.

10.5 Match the bond to the attacker

Compound	Key bond	Polarity	Attacked by	Mechanism
Alkane	C–H, C–C	non-polar	radicals (Cl• from UV light)	**free-radical substitution** (homolytic fission)
Alkene	C=C	non-polar but a region of **high electron density** (the π bond)	**electrophiles** (Br_2, HBr)	**electrophilic addition**
Halogenoalkane	$C^{\delta+}$–$X^{\delta-}$	**polar**	**nucleophiles** (OH^-, CN^-, NH_3)	**nucleophilic substitution** (heterolytic fission)

- Halogens are **more electronegative** than carbon, so the C–X bonding pair is pulled towards X. The carbon becomes **δ+**, the halogen **δ−**.
- The δ+ carbon attracts nucleophiles. The C–X bond then breaks **heterolytically**, and the halogen leaves as a halide ion, X^-.
- So **the polarity of a bond tells you where a reagent will attack and whether the bond breaks evenly or unevenly**.

EXAM TIP

Electronegativity falls down Group 7 (F > Cl > Br > I), so C–F is the **most** polar bond and C–I the **least**. If polarity decided the rate, fluoroalkanes would react fastest. They don't: **bond enthalpy** decides the rate of hydrolysis (10.13).

MARK-SCHEME LANGUAGE

- Dipoles: "**δ+ on C, δ− on the halogen**", shown on the C–X bond in every substitution mechanism.

Often asked as: part of a mechanism question; explain why a halogenoalkane reacts with nucleophiles but an alkane doesn't.

COMMON MISTAKE

Putting the δ+ on the halogen. Writing full + and − charges on a polar bond in the starting molecule.

SPEC 10.6

Naming and drawing halogenoalkanes

WHAT THIS SPEC POINT ASKS

Name halogenoalkanes and draw their structural, displayed and skeletal formulae.

10.6 Naming rules

1. Find the **longest carbon chain** containing the C–X carbon: it gives the stem (meth-, eth-, prop-, but-, pent-, hex-) + **-ane**.
2. Halogens are **prefixes**: **fluoro-, chloro-, bromo-, iodo-**. Alkyl branches are prefixes too (methyl-, ethyl-).
3. Number the chain to give the **lowest numbers** to the substituents.
4. Put all prefixes in **alphabetical order** (bromo, chloro, ethyl, fluoro, iodo, methyl). Use di-, tri- for repeats (they don't count for alphabetical order).
5. Commas between numbers, hyphens between numbers and words.

Name	Structural formula	Class (10.7)
bromoethane	CH_3CH_2Br	primary
1-bromopropane	$CH_3CH_2CH_2Br$	primary
2-bromopropane	$CH_3CHBrCH_3$	secondary
1-bromo-2-methylpropane	$(CH_3)_2CHCH_2Br$	primary
2-chloro-2-methylpropane	$(CH_3)_3CCl$	tertiary
2-iodo-2-methylpropane	$(CH_3)_3CI$	tertiary
2-chloro-3-methylbutane	$CH_3CHClCH(CH_3)CH_3$	secondary
1,2-dibromoethane	CH_2BrCH_2Br	primary (both)

10.6 The three formula types

- **Structural**: enough to show the arrangement without drawing bonds, e.g. $CH_3CHBrCH_3$.
- **Displayed**: every atom and **every bond** drawn (including all C–H bonds).
- **Skeletal**: a zig-zag line; every corner and line end is a carbon; H atoms on carbon are **not shown**; the halogen is written at the end of its bond. 2-bromopropane is a "V" shape (three carbons) with Br on a line from the middle carbon.

MARK-SCHEME LANGUAGE

- "2-chloro-3-methylbutane" (WCH12 Jan 2025). Missing or extra hyphens and commas are ignored; wrong alphabetical order isn't credited.
- "2-iodo-2-methylpropane": DO NOT AWARD "2-iodine-..." (WCH12 Oct 2024).
- Nitrile names count the C of the C≡N in the chain: $(CH_3)_3CCN$ is **2,2-dimethylpropanenitrile** (WCH12 Jun 2023).

Often asked as: name a compound from its skeletal formula; draw a displayed or skeletal formula from a name.

COMMON MISTAKE

Writing "chloride", "chlorine" or "iodine" in the name instead of chloro- / iodo-.
Prefixes out of alphabetical order (methyl before chloro).
Missing H atoms in displayed formulae; drawing 2-bromopropane when 1-bromo-2-methylpropane was asked (WCH12 Jan 2024 report).

SPEC 10.7 Primary, secondary and tertiary halogenoalkanes

WHAT THIS SPEC POINT ASKS

Distinguish primary, secondary and tertiary halogenoalkanes.

10.7 Count the carbons on the C–X carbon

- Look at the carbon that carries the halogen. Count how many **other carbon atoms (alkyl groups)** are bonded to it:

Class	Carbons bonded to the C–X carbon	General	Example
Primary (1°)	one (or none, for CH_3X)	RCH_2X	1-bromobutane
Secondary (2°)	two	R_2CHX	2-bromobutane
Tertiary (3°)	three	R_3CX	2-bromo-2-methylpropane

- The four structural isomers of C_4H_9Br:
  - 1-bromobutane $CH_3CH_2CH_2CH_2Br$: **primary**;
  - 2-bromobutane $CH_3CHBrCH_2CH_3$: **secondary**;
  - 1-bromo-2-methylpropane $(CH_3)_2CHCH_2Br$: **primary** (the Br carbon has only one C attached, even though the chain is branched);
  - 2-bromo-2-methylpropane $(CH_3)_3CBr$: **tertiary**.
- The class controls how fast the halogenoalkane is hydrolysed (10.12).

MARK-SCHEME LANGUAGE

- Define by the number of **carbon atoms / alkyl groups attached to the carbon bonded to the halogen**. "Branched" is accepted for tertiary in rate questions (WCH12 Jun 2024).

Often asked as: classify a given halogenoalkane; draw isomers of a formula and classify each.

COMMON MISTAKE

Classifying by "how many halogens" or "where the halogen is on the chain".
Calling bromocyclohexane tertiary (it's **secondary**: the Br carbon has two C neighbours) (WCH12 Jun 2022 report).

SPEC 10.8 Reactions of halogenoalkanes

WHAT THIS SPEC POINT ASKS

Know the reactions of halogenoalkanes with (i) aqueous KOH (alcohols; OH^- as nucleophile), (ii) ethanolic KOH (alkenes by elimination; OH^- as base), (iii) aqueous silver nitrate in ethanol (water as nucleophile), (iv) ammonia in ethanol under pressure (amines), (v) KCN in ethanol (nitriles: lengthens the carbon chain).

10.8 Summary table

Reagent and conditions	Role of reagent	Type	Product (from bromoethane, or as stated)
KOH(aq) or NaOH(aq), **heat under reflux**	OH^- = **nucleophile**	nucleophilic substitution (hydrolysis)	**ethanol**, CH_3CH_2OH
KOH in ethanol (ethanolic), heat under reflux	OH^- = **base**	**elimination**	**alkene** (from 2-bromopropane: propene)
$AgNO_3$(aq) in ethanol, warm	H_2O = **nucleophile**	nucleophilic substitution (hydrolysis)	alcohol + silver halide precipitate
excess NH_3 in ethanol, heat under pressure (sealed tube)	NH_3 = **nucleophile**	nucleophilic substitution	**amine**: ethylamine $CH_3CH_2NH_2$
KCN in ethanol, heat under reflux	CN^- = **nucleophile**	nucleophilic substitution	**nitrile**: propanenitrile CH_3CH_2CN

10.8 (i) Aqueous alkali → alcohol

$CH_3CH_2Br + KOH \rightarrow CH_3CH_2OH + KBr$

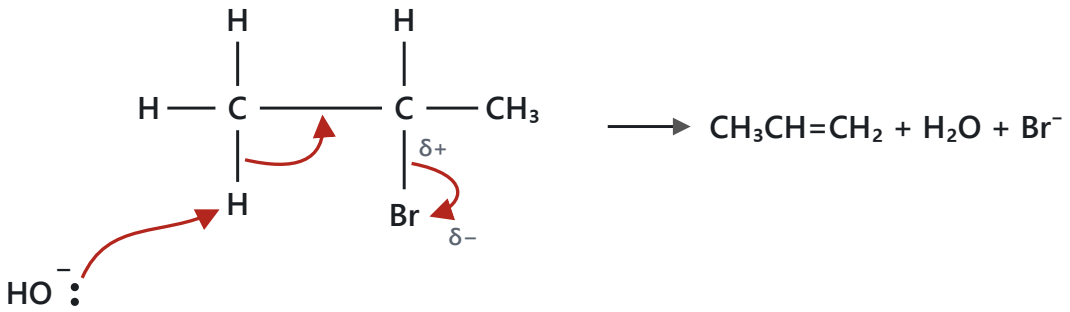
Ionic: $CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$

- **Water** is the solvent, so OH^- ions are surrounded by water and attack the δ+ carbon. A little ethanol is often added to help the halogenoalkane mix with the water.

10.8 (ii) Ethanolic KOH → alkene (elimination)

$CH_3CHBrCH_3 + KOH \rightarrow CH_2{=}CHCH_3 + KBr + H_2O$

- In **ethanol** (no water), OH^- acts as a **base**: it removes an **H^+** from a carbon **next to** the C–X carbon. H and X are lost from **neighbouring** carbons, and a **C=C** forms between them.
- The ethanol must be **dry**: any water would let OH^- act as a nucleophile and give the alcohol instead.
- If the C–X carbon has different neighbouring carbons with H atoms, you get **more than one alkene**. 2-bromobutane gives **but-1-ene** and **but-2-ene** (which exists as **E** and **Z** isomers): three alkenes in all.
- In practice substitution and elimination compete. Aqueous conditions and primary halogenoalkanes favour substitution; ethanolic conditions, heat and tertiary halogenoalkanes favour elimination.



Elimination from 2-bromopropane. Three arrows: lone pair on O → H; C–H bond → C–C (forming C=C); C–Br bond → Br.

10.8 (iii) Silver nitrate in ethanol → alcohol + precipitate

$R{-}X + H_2O \rightarrow R{-}OH + H^+ + X^-$ then $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$

- Water is a **weak nucleophile**, so this hydrolysis is slow enough to time. The halide ion released is caught by Ag^+ as a precipitate: **AgCl white, AgBr cream, AgI yellow**.
- **Ethanol** is a solvent for both the halogenoalkane (which doesn't mix with water) and the aqueous silver nitrate, so they mix and react.
- This is the basis of the rate comparisons in 10.10 and Core Practical 5.

10.8 (iv) Ammonia → amine

$CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

- Conditions: **excess** concentrated ammonia **in ethanol**, **heated under pressure** in a **sealed tube**. Reflux isn't used because ammonia gas would **escape through the condenser**.
- The second NH_3 removes H^+ from the intermediate (mechanism in 10.9), which is why the equation uses $2NH_3$.
- **Why excess ammonia?** The amine product also has a **lone pair on N**, so it is a nucleophile too and can attack more halogenoalkane, giving secondary and tertiary amines. A large excess of NH_3 makes it much more likely that each halogenoalkane molecule meets NH_3, not the amine.

10.8 (v) Potassium cyanide → nitrile

$CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN + KBr$

- Conditions: **KCN in ethanol**, heat under reflux.
- The product, **propanenitrile**, has **three** carbons; bromoethane had two. Making a new **C–C bond** is a way of **increasing the length of the carbon chain**.
- Name nitriles by counting the C of –C≡N in the chain: CH_3CH_2CN is propane**nitrile**.
- KCN is **highly toxic**, and acids release toxic HCN gas.

MARK-SCHEME LANGUAGE

- Substitution: "**aqueous** (solution)" (ALLOW aqueous ethanol); DO NOT AWARD "just water" (WCH12 Jun 2024).
- Elimination: "**ethanolic / alcoholic** (solution)" of KOH (ALLOW NaOH); DO NOT AWARD just "ethanol" or "alcohol" as the reagent. Heat/reflux ignored.
- Amine: "ammonia (ALLOW concentrated), **heat under pressure**, in **alcoholic solution**" or "in a **sealed tube**". DO NOT AWARD "heat under reflux" or "dilute / aqueous ammonia" (WCH12 Jun 2023, Jun 2025, Oct 2025).
- Nitrile: "$C_4H_9Cl + KCN \rightarrow C_4H_9CN + KCl$", "**pentanenitrile**", "a way of **extending the carbon chain** / forming a new C–C bond" (WCH12 Oct 2024, Oct 2025).
- Dry ethanol: "to remove water" so as to "prevent (nucleophilic) **substitution**" / stop OH^- acting as a nucleophile (WCH12 Oct 2025).
- 6-mark compare (WCH12 Jun 2022, Jun 2025): both involve OH^-; **aqueous** → **substitution**, OH^- is a **nucleophile**, alcohol formed; **ethanolic** → **elimination**, OH^- is a **base**, alkene(s) formed; two balanced equations. From 2-chlorobutane: butan-2-ol, and but-1-ene plus E- and Z-but-2-ene. Just "butene" scores only 1.

Often asked as: give reagents and conditions for a step in a reaction scheme; 6-mark comparison of aqueous and ethanolic KOH; state the role of OH^-; name the nitrile.

BEYOND THE SPEC, BUT EXAMINED

- **Elimination mechanism** with OH^- as a base: lone pair on O → H on the carbon next to C–X; the C–H bond pair → the C–C bond to make C=C; the C–X bond pair → X (WCH12 Jan 2022). See the figure above.
- Silver(I) oxide in water as a source of OH^-: $Ag_2O + H_2O \rightarrow 2Ag^+ + 2OH^-$; it's less corrosive than NaOH and can be recovered (WCH12 Jun 2023).
- Volumes, limiting reagent and % yield for a halogenoalkane preparation using density (WCH12 Jun 2019, Jun 2022).

COMMON MISTAKE

Writing "cyanide" or "hydroxide" without the metal: give **KCN** or **KOH** (WCH12 Oct 2020 report).
Missing "aqueous" for substitution, or "ethanol" / "under pressure" for the amine (WCH12 Jan 2024 report).
Suggesting NaOH to dehydrate an **alcohol**: that's halogenoalkane chemistry. Alcohols need conc. H_3PO_4 (10.17).
Drawing the nitrile as C–N or C=N, or naming it an amine or a "cyanide".

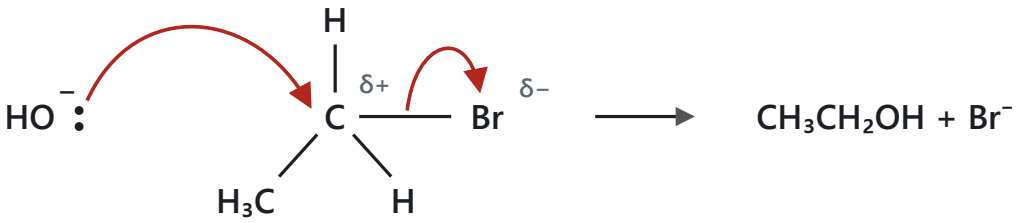
SPEC 10.9

Nucleophilic substitution mechanisms (OH^- and NH_3)

WHAT THIS SPEC POINT ASKS

Understand the mechanisms of nucleophilic substitution of primary halogenoalkanes with (i) aqueous KOH and (ii) ammonia. (SN1 and SN2 are named and tested in Unit 4.)

10.9 With hydroxide ions



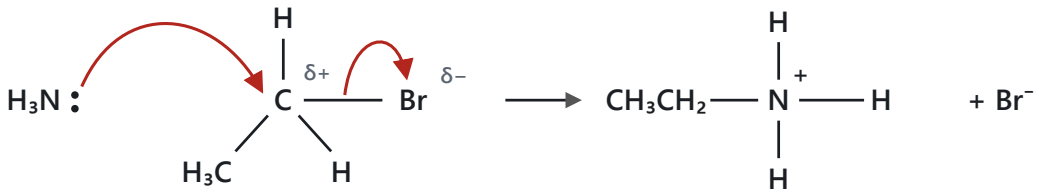
Bromoethane + OH^- → ethanol + Br^-. Two curly arrows.

1. Draw the **δ+** on the carbon and the **δ−** on the bromine of the C–Br bond.
2. The **lone pair on the oxygen** of OH^- is attracted to the δ+ carbon. Draw a curly arrow **from that lone pair to the carbon**.
3. At the same time the C–Br bond breaks **heterolytically**. Draw a curly arrow **from the middle of the C–Br bond to the Br** (or just past it).
4. Products: **CH_3CH_2OH** (the O is now bonded to C) and **Br^-**.

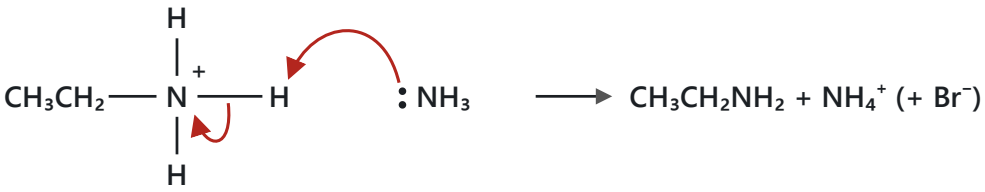
This happens in a single step: the new C–O bond forms as the C–Br bond breaks. (In Unit 4 you'll call this **SN2**.)

10.9 With ammonia

Step 1



Step 2



Step 1: ammonia substitutes for Br, giving an ammonium-type ion. Step 2: a second ammonia removes H^+.

Step 1: substitution

1. δ+ on C, δ− on Br.
2. Curly arrow from the **lone pair on the N of NH_3** to the δ+ carbon.
3. Curly arrow from the **C–Br bond to Br**.
4. Products: the intermediate ion **$CH_3CH_2NH_3^+$** (N now has four bonds and a **positive charge**) and **Br^-**.

Step 2: loss of H^+

1. A **second NH_3** molecule uses its lone pair to attack one of the **H atoms on the N^+**: curly arrow from the lone pair on N to that H.
2. Curly arrow from the **N–H bond** to the **N^+**: the bond pair stays on the nitrogen, cancelling its charge.
3. Products: **ethylamine, $CH_3CH_2NH_2$**, and **NH_4^+** (which pairs with Br^- as NH_4Br).

Overall: $CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

EXAM TIP

Mechanism marks are given for **separate items**, so draw every one: δ+/δ−; each lone pair the arrow starts from (two dots on O or N); each arrow starting at a lone pair or bond; the charge on the intermediate; **every** product including Br^- and NH_4^+.

MARK-SCHEME LANGUAGE

OH^- (WCH12 Jun 2019, Oct 2021):

- dipole on C–X and arrow from the **C–X bond to X** (or just beyond);
- arrow from the **lone pair on O of OH^-** to the δ+ C (not from the minus sign);
- both products. KCl is allowed if K^+ is shown. A covalent K–O bond in KOH loses the mark. A transition state is not needed.

NH_3 (WCH12 Jun 2025, six marking points for 3 marks):

- dipole on C–Br;
- arrow from the **lone pair on N** of NH_3 to the correct C;
- arrow from C–Br to Br;
- arrow from an **N–H bond of the intermediate to N**;
- arrow from the **lone pair on a second NH_3 to H** on the intermediate;
- **Br^- and NH_4^+** shown (or NH_4Br).

A missing lone pair, a charge on NH_3 and fish-hook arrows are each penalised once.
Often asked as: draw the mechanism (3–4 marks); find the omissions in a drawn mechanism ("lone pair on N", "δ+ on C").

BEYOND THE SPEC, BUT EXAMINED

SN1 for tertiary halogenoalkanes. WCH12 papers often use tertiary compounds and accept this mechanism (WCH12 Jun 2023, Oct 2021):

1. The C–Br bond in 2-bromo-2-methylpropane breaks **heterolytically** on its own: arrow from the C–Br bond to Br. This gives a **tertiary carbocation** $(CH_3)_3C^+$ (trigonal planar) and Br^-. This is the slow step.
2. The **lone pair on O of OH^-** (or H_2O) attacks the **C^+**: arrow from the lone pair to the positive carbon. Product: $(CH_3)_3COH$.

Tertiary carbocations are the most stable because the three alkyl groups **push electron density** towards the C^+ (WCH12 Jan 2020). Primary halogenoalkanes react by the one-step route above.
The bond formed when NH_3 accepts H^+ in step 2 is a **dative covalent** bond: N's lone pair goes into the empty orbital of H^+ (WCH12 Jun 2021).

COMMON MISTAKE

Drawing ammonia with a **negative charge**, or the lone pair on an H instead of N (WCH12 Oct 2020, Jun 2022, Jan 2024 reports).
Missing the second NH_3 step, or leaving out Br^- and NH_4^+.
Arrows that show an H atom moving (from H to N) instead of the electron pair.
Arrows aimed at the C–C bond instead of the carbon atom.

SPEC 10.10

Comparing rates of hydrolysis from experimental data

WHAT THIS SPEC POINT ASKS

Use observations and data to compare the relative rates of hydrolysis of (i) primary, secondary and tertiary isomers of a halogenoalkane, and (ii) primary chloro-, bromo- and iodoalkanes, using aqueous silver nitrate in ethanol.

10.10 How the comparison works

- The halogenoalkane is warmed with **aqueous silver nitrate dissolved in ethanol**. Water slowly hydrolyses it, releasing X^- ions:
  - $R–X + H_2O \rightarrow R–OH + H^+ + X^-$
  - $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$
- The **time for a precipitate to first appear** measures the rate: **shorter time → faster hydrolysis**. Rate ∝ 1/time (9.3).
- The precipitate colour tells you which halogen was present.

Halogenoalkane	Precipitate	Colour	Speed
1-chlorobutane	AgCl	**white**	slowest (may take a long time)
1-bromobutane	AgBr	**cream**	faster
1-iodobutane	AgI	**yellow**	fastest

Isomer of C_4H_9Br	Class	Speed
1-bromobutane	primary	slowest
2-bromobutane	secondary	in between
2-bromo-2-methylpropane	tertiary	fastest (precipitate almost at once)

- To make the comparison fair, keep the **same volume (or number of drops)** of each halogenoalkane, the **same volume and concentration** of $AgNO_3$, the **same volume of ethanol** and the **same temperature** (water bath).
- **Fluoroalkanes** give no precipitate in a reasonable time: the C–F bond is too strong.

MARK-SCHEME LANGUAGE

- What to measure: "**time taken for the (first appearance of the) precipitate** (of silver halide) to form" (WCH12 Oct 2021).
- 6-mark (WCH12 Oct 2020) on 1-bromobutane, 2-bromobutane and 2-bromo-2-methylbutane: nucleophilic substitution; silver halide precipitate; time depends on C–halogen **bond strength**; bond enthalpy decreases down Group 7 (larger atoms); rate **1° < 2° < 3°**; classify each compound. Wrong precipitate colours, or reasons based on **electronegativity** or **intermolecular forces**, lose a mark.
- Two opposing factors (WCH12 Jun 2022): 2-chloro-2-methylpropane "should react faster **because it is tertiary**"; 1-iodobutane "should react faster because the **C–I bond enthalpy is lower** than C–Cl"; so "it is **not possible to predict**" which wins.

Often asked as: describe how to compare rates; predict and explain the order; interpret times or a table of results; identify the halogen from the precipitate colour.

COMMON MISTAKE

"A stronger bond makes a faster reaction" (WCH12 Jan 2022 report): the **weaker** C–X bond breaks more easily.
Swapping the precipitate colours, or giving only formulae when colours were asked.
Describing the practical when the question asked you to compare and explain.
Calling the halogenoalkane itself a "carbocation".

SPEC 10.11 Core Practical 5: rates of hydrolysis of halogenoalkanes

WHAT THIS SPEC POINT ASKS

Investigate the rates of hydrolysis of some halogenoalkanes.

CORE PRACTICAL 5: METHOD IN BRIEF

Part 1: chloro v bromo v iodo

1. Label three test tubes. Put **1 cm^3 of ethanol** in each.
2. Add **the same number of drops** (e.g. 3) of 1-chlorobutane, 1-bromobutane and 1-iodobutane, one to each tube.
3. Stand the tubes, and a tube of **0.05 mol dm^{-3} silver nitrate solution**, in a **water bath at about 50–60 °C** for a few minutes so everything reaches the same temperature.
4. Add **1 cm^3 of the warm silver nitrate** to each tube, shake, and **start the clock**.
5. Record the **time for a precipitate to first appear** in each tube, and its colour.

Part 2: primary v secondary v tertiary

Repeat with 1-bromobutane, 2-bromobutane and 2-bromo-2-methylpropane (same halogen, so only the structure changes).

10.11 Variables

Independent	Dependent	Control
the halogen (part 1) or the class of halogenoalkane (part 2)	time for a precipitate to appear	volume of halogenoalkane, volume of ethanol, volume and concentration of $AgNO_3$, temperature of the water bath, size of test tubes

10.11 Processing results

WORKED EXAMPLE: FROM TIMES TO RELATIVE RATES

Times for a precipitate: 1-iodobutane 30 s, 1-bromobutane 240 s, 1-chlorobutane no precipitate after 1200 s.

Rate $\propto$ 1/t: iodo 1/30 = 0.033 s^{-1}; bromo 1/240 = 0.0042 s^{-1}; chloro < 1/1200 = 0.00083 s^{-1}.

Iodo is about 8 times faster than bromo. Order: **C–I > C–Br > C–Cl**, matching bond enthalpies 228 < 290 < 346 kJ mol^{-1}.

(These numbers are illustrative; your class results will differ, but the order shouldn't.)

10.11 Safety

- **Ethanol** and halogenoalkanes are **flammable**: heat with a water bath or electric heater, **no naked flames**.
- Halogenoalkanes are **harmful**; avoid skin contact and breathing the vapour.
- Silver nitrate is an **irritant** and **stains skin**; wear gloves and eye protection.

10.11 Errors and improvements

- Judging the **first appearance** of a precipitate is subjective, especially for the faint cream and yellow ones: view against a **black background**, or use a **colorimeter** / light sensor.
- Drops aren't all the same size: use a **graduated pipette** to add equal volumes (or equal **moles**, since the compounds have different densities and Mr).

- Temperature drops when the tubes are taken out: keep them **in the water bath** while timing.
- **Repeat** and take a mean.

EXAM TIP

The water (not OH^-) is the nucleophile here. That makes the reaction slow enough to compare by eye. If you used NaOH(aq) first, you'd have to **acidify with dilute nitric acid** before adding $AgNO_3$, otherwise brown silver oxide precipitates.

MARK-SCHEME LANGUAGE

- "Time for the precipitate to form" (ALLOW "how fast"); "same temperature (water bath)"; "same volume/amount of halogenoalkane"; colours AgCl white, AgBr cream, AgI yellow.

Often asked as: how to measure the rate; identify control variables; predict the times for an unfamiliar halogenoalkane; explain the order (10.12, 10.13).

SPEC 10.12 Reactivity trend: primary, secondary and tertiary

WHAT THIS SPEC POINT ASKS

Know the trend in reactivity of primary, secondary and tertiary halogenoalkanes.

10.12 The trend

- For the same halogen, the rate of hydrolysis is **tertiary > secondary > primary**.
- Example: 2-bromo-2-methylpropane gives a cream precipitate with $AgNO_3$ in ethanol almost straight away; 1-bromobutane takes much longer.

10.12 Why (beyond the spec, but useful)

- Tertiary halogenoalkanes react by forming a **carbocation** first (SN1, see 10.9). A **tertiary carbocation is the most stable**, because its three alkyl groups push electron density towards the C^+ and spread out the positive charge. So it forms most easily.
- Primary halogenoalkanes can't form a stable carbocation, so they react by the one-step route, where the nucleophile must reach the carbon directly.

MARK-SCHEME LANGUAGE

- "**tertiary** halogenoalkanes react more quickly than primary" (WCH12 Jun 2024). ALLOW "tertiary carbocation is more stable than primary" and "methyl stabilisation of the positive charge" (WCH12 Jan 2022).
- DO NOT AWARD "tertiary halogenoalkanes react faster because they are **more stable**" (it's the **carbocation** that is more stable). IGNORE "more branched".

Often asked as: put isomers in order of rate; explain which of two compounds reacts faster (often with a second factor, the halogen, as well).

COMMON MISTAKE

Saying the tertiary **halogenoalkane** is more stable. It's the tertiary **carbocation** that is stabilised.
For a "state" question, writing a long carbocation explanation: just give the order (WCH12 Oct 2024 report).

SPEC 10.13 Reactivity trend: chloro-, bromo- and iodoalkanes (bond enthalpy)

WHAT THIS SPEC POINT ASKS

Explain, using bond enthalpy, the trend in reactivity of chloro-, bromo- and iodoalkanes.

10.13 Bond enthalpy decides

Bond	Mean bond enthalpy / kJ mol^{-1}	Polarity	Rate of hydrolysis
C–F	467	most polar	no reaction
C–Cl	346		slow
C–Br	290		faster
C–I	228	least polar	fastest

- Down Group 7 the halogen atom gets **larger**, so the C–X bond is **longer** and the shared pair is further from the halogen nucleus. The bond gets **weaker** (lower bond enthalpy).
- A weaker bond needs **less energy to break**, so the activation energy is lower and more collisions succeed. **Iodoalkanes react fastest.**
- **Polarity** would predict the opposite order: C–F is the most polar, so its carbon is the most δ+ and should attract nucleophiles best. The observed order shows that **bond enthalpy matters more than polarity**.

MARK-SCHEME LANGUAGE

- "the **C–I bond is weaker** than C–Cl" / "C–I has a **lower bond enthalpy**", so iodoalkanes react faster (WCH12 Oct 2024, Jun 2024, Jan 2022).
- "C–Cl has a higher bond enthalpy (than C–Br)" so 1-chlorobutane is slower (WCH12 Jan 2022). Mark points may depend on the correct order.

Often asked as: explain why an iodoalkane reacts faster than a chloroalkane; combine with 10.12 for two compounds that differ in both class and halogen.

COMMON MISTAKE

Explaining with **electronegativity** or polarity, or with "reactivity of the halogen" or "size of the element" without linking to bond strength (WCH12 Jan 2025 report).
Vague "the iodine bond is weaker": say "the **C–I** bond".

SPEC 10.14 Core Practical 6: chlorination of 2-methylpropan-2-ol

WHAT THIS SPEC POINT ASKS

Prepare 2-chloro-2-methylpropane from 2-methylpropan-2-ol and concentrated hydrochloric acid.

10.14 The reaction

$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$

- 2-methylpropan-2-ol is a **tertiary** alcohol. Tertiary alcohols react with **concentrated HCl at room temperature**, because they form a stable tertiary carbocation. Primary and secondary alcohols barely react this way (use PCl_5 for those: 10.17).

- 2-chloro-2-methylpropane boils at about **51 °C** and has density about 0.84 g cm^{-3}, so it forms the **upper** layer above the aqueous acid.

CORE PRACTICAL 6: METHOD IN BRIEF

1. In a **fume cupboard**, put about 10 cm^3 of 2-methylpropan-2-ol and about 35 cm^3 of **concentrated hydrochloric acid** into a **separating funnel**.
2. Stopper, **shake** for several minutes (about 20 minutes in total), **inverting and opening the tap often** to release the pressure.
3. Leave to stand until two layers separate. Run off and discard the **lower (aqueous) layer**.
4. Add **sodium hydrogencarbonate solution** to the organic layer to **neutralise leftover acid**. Shake gently, venting often: CO_2 is produced. Discard the aqueous layer. Wash once with water.
5. Run the organic layer into a dry conical flask and add **anhydrous sodium sulfate** (or $MgSO_4$ / $CaCl_2$). Swirl until the liquid is **clear**.
6. Decant or filter into a clean flask and **distil**, collecting the fraction that boils at about **49–53 °C**.

10.14 Why each step

Step	Purpose
Separating funnel and shaking	the two liquids don't mix; shaking brings them into contact
Releasing pressure	HCl gas and (later) CO_2 build up pressure
$NaHCO_3$ wash	removes acid: $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$
Anhydrous salt	removes **water** left in the organic layer
Distillation	separates the product from unreacted alcohol (b.p. 82 °C) and other impurities

WORKED EXAMPLE: PERCENTAGE YIELD

10.0 cm^3 of 2-methylpropan-2-ol (density 0.786 g cm^{-3}, Mr 74.0) gave 6.50 g of 2-chloro-2-methylpropane (Mr 92.5).
Mass of alcohol = 10.0 × 0.786 = 7.86 g
Moles of alcohol = 7.86 ÷ 74.0 = 0.1062 mol (1 : 1 ratio, HCl in excess)
Theoretical mass of product = 0.1062 × 92.5 = 9.83 g
% yield = 6.50 ÷ 9.83 × 100 = **66.2%**
Yield is lost when the layers are separated, in the washing, in the drying agent and in the distillation flask.

MARK-SCHEME LANGUAGE

- Drying agent: "(anhydrous) **sodium sulfate** / magnesium sulfate / calcium chloride / calcium sulfate / calcium oxide" (ALLOW silica gel). DO NOT AWARD concentrated sulfuric acid, phosphoric acid, $CuSO_4$ or $CaCO_3$ (WCH12 Jun 2019).

Often asked as: name a drying agent; explain a step ($NaHCO_3$ wash, venting, distillation range); calculate the yield. No WCH12 paper (2019–2026) asked for the whole method.

COMMON MISTAKE

Offering **anhydrous copper(II) sulfate** as a drying agent: it's a test for water, not a drying agent (WCH12 Jun 2019 report).
Giving two drying agents where one is wrong.
Not venting the funnel after adding $NaHCO_3$.

SPEC 10.15

Naming and drawing alcohols

WHAT THIS SPEC POINT ASKS

Name alcohols and draw their structural, displayed and skeletal formulae.

10.15 Naming rules

- The functional group is **–OH** (hydroxyl). General formula of the alcohol series: **$C_nH_{2n+1}OH$**.
- Name = stem for the longest chain containing the C–OH + **-an-** + **position number** + **-ol**: propan-1-ol, propan-2-ol.
- Number the chain to give the **–OH the lowest number** (it takes priority over alkyl and halogen prefixes).
- Two –OH groups: keep the "e": ethane-1,2-diol.
- Branches go in front as prefixes: 2-methylpropan-2-ol.

Name	Structural formula
methanol	CH_3OH
ethanol	CH_3CH_2OH
propan-1-ol	$CH_3CH_2CH_2OH$
propan-2-ol	$CH_3CH(OH)CH_3$
butan-2-ol	$CH_3CH(OH)CH_2CH_3$
2-methylpropan-1-ol	$(CH_3)_2CHCH_2OH$
2-methylpropan-2-ol	$(CH_3)_3COH$
4-methylhexan-2-ol	$CH_3CH(OH)CH_2CH(CH_3)CH_2CH_3$
ethane-1,2-diol	$HOCH_2CH_2OH$

- **Displayed formula**: show the **O–H bond** as well as every C–H; the O must join to C (C–O–H), not H to C.
- **Skeletal formula**: zig-zag carbon chain with **OH** written at the end of a line from the right carbon (write "OH", not just "O").

MARK-SCHEME LANGUAGE

- "4-methylhexan-2-ol" (ALLOW 4-methyl-2-hexanol, 4-methylhexane-2-ol); DO NOT AWARD "4-methylhex-2-ol" (WCH12 Jun 2021).
- "2-methylpropan-1-ol", "butan-2-ol", "2-methylpropan-2-ol" (WCH12 Oct 2019). If a skeletal formula is asked for, displayed formulae aren't credited.
- A molecular formula like C_3H_7OH is not a structure and scores zero (WCH12 Jun 2023).

Often asked as: name or draw isomeric alcohols ($C_4H_{10}O$, $C_6H_{13}OH$); give the skeletal formula of a named alcohol.

COMMON MISTAKE

"3-methylpropan-1-ol": a methyl on the end carbon just lengthens the chain. It's **butan-1-ol** (WCH12 Oct 2019 report). The branched one is **2-methylpropan-1-ol**.
Leaving out the position number (butanol for butan-1-ol); drawing the same isomer twice (just rotated or flipped); drawing a cyclic alcohol when the formula doesn't allow it.
Joining the OH by its H in a displayed or skeletal formula.

SPEC 10.16 Primary, secondary and tertiary alcohols

WHAT THIS SPEC POINT ASKS

Distinguish primary, secondary and tertiary alcohols.

10.16 Classifying

Class	C–OH carbon bonded to	H atoms on the C–OH carbon	Example
Primary	one other C (or none, methanol)	two (three for methanol)	ethanol, propan-1-ol
Secondary	two other C	one	propan-2-ol, butan-2-ol
Tertiary	three other C	none	2-methylpropan-2-ol

The four alcohols with formula C_4H_9OH:

Name	Class
butan-1-ol	primary
2-methylpropan-1-ol	primary
butan-2-ol	secondary
2-methylpropan-2-ol	tertiary

- The class decides what oxidation gives (10.18): **primary → aldehyde → carboxylic acid; secondary → ketone; tertiary → no reaction**. The number of **H atoms on the C–OH carbon** is what matters, because oxidation removes one of them.

MARK-SCHEME LANGUAGE

- Secondary: "the carbon (atom) attached to the OH group is attached to **two other carbon atoms / alkyl groups**", or "has **one hydrogen** atom" (WCH12 Oct 2025). DO NOT AWARD "hydroxide group" or any mention of a carbocation.
- Jan 2026: define all three classes (by H atoms or by R groups) and give a correct example of each.

Often asked as: classify an alcohol; draw one example of each class for a given formula; explain why a tertiary alcohol isn't oxidised.

COMMON MISTAKE

"Primary means the OH is on the end carbon" (WCH12 Oct 2022 report). It's about how many **carbons** are attached to the C–OH carbon. 2-methylpropan-1-ol has a branched chain but is still primary. Writing "hydroxide" for the –OH group: hydroxide is the OH^- ion.

SPEC 10.17 Reactions of alcohols: combustion, halogenation, dehydration

WHAT THIS SPEC POINT ASKS

Know the reactions of alcohols with (i) oxygen (combustion), (ii) halogenating agents: PCl_5 (also a test for –OH), 50% concentrated sulfuric acid with KBr, and red phosphorus with iodine, and (iii) concentrated phosphoric acid to make alkenes by elimination. Mechanisms not needed.

10.17 (i) Combustion

- Alcohols burn in plenty of oxygen to give **CO_2 and H_2O**:
  - $CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
  - $C_3H_7OH(l) + 4½O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
- Balance O last: remember the alcohol already contains one O.
- Ethanol is used as a (bio)fuel; it burns with a clean blue flame.

10.17 (ii) Halogenation: replacing –OH with a halogen

Halogenoalkane made	Reagent and conditions	Equation (ethanol, or as stated)
Chloroalkane	**PCl_5**, room temperature, dry	$CH_3CH_2OH + PCl_5 \rightarrow CH_3CH_2Cl + POCl_3 + HCl$
Bromoalkane	**KBr + 50% concentrated H_2SO_4**, heat (HBr made in situ)	$KBr + H_2SO_4 \rightarrow KHSO_4 + HBr$; $CH_3CH_2OH + HBr \rightarrow CH_3CH_2Br + H_2O$
Iodoalkane	**red phosphorus + iodine**, warm under reflux (PI_3 made in situ)	$2P + 3I_2 \rightarrow 2PI_3$; $3CH_3CH_2OH + PI_3 \rightarrow 3CH_3CH_2I + H_3PO_3$
Chloroalkane (**tertiary** only)	**concentrated HCl**, room temperature	$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$ (Core Practical 6)

- **PCl_5 as a test for –OH**: add solid PCl_5 to the dry sample. **Steamy (misty) white fumes of HCl** are given off. They turn **damp blue litmus red** and form **white smoke** (NH_4Cl) with ammonia on a glass rod.
  - Water and carboxylic acids (which also contain –OH) give the same result, so the sample must be **dry**, and the test shows –OH, not specifically an alcohol.
- **Why 50% sulfuric acid?** Concentrated H_2SO_4 is an oxidising agent: it would oxidise some of the HBr to **bromine** (Topic 8). Diluting it to about 50% reduces this.

10.17 (iii) Dehydration (elimination) with concentrated phosphoric acid

$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$ (conc. H_3PO_4, heat; write the catalyst **above the arrow**)

- A molecule of **water** is eliminated from neighbouring carbons (the –OH and an H next to it), leaving a C=C.
- Concentrated H_3PO_4 is a **catalyst**: it's preferred to conc. H_2SO_4 because it causes fewer side reactions (H_2SO_4 is oxidising and chars the alcohol).
- The alkene is usually **distilled off** as it forms.
- If the C–OH carbon has two different neighbouring carbons with H atoms, you get a mixture: butan-2-ol → **but-1-ene + E-but-2-ene + Z-but-2-ene**.

MARK-SCHEME LANGUAGE

- PCl_5 test: "**misty fumes** evolved (that turn damp blue litmus red / form white smoke with ammonia)". ALLOW steamy / white fumes; PCl_3 allowed. DO NOT AWARD "white smoke" unless ammonia is involved. IGNORE "gas / hydrogen given off" (WCH12 Jun 2019).
- Equation: "$C_5H_{11}OH + PCl_5 \rightarrow C_5H_{11}Cl$ + **$POCl_3$** + **HCl**" (ACCEPT PCl_3O) (WCH12 Jan 2020).
- Bromination: "**50% / concentrated sulfuric acid and potassium bromide**". ALLOW P + Br_2, PBr_3, PBr_5. DO NOT AWARD dilute sulfuric acid (WCH12 Jun 2021).
- Chlorination: PCl_5, or **concentrated** HCl, $SOCl_2$, PCl_3. IGNORE just "HCl" (WCH12 Jun 2021).
- Dehydration: "(concentrated) **phosphoric(V) acid** / H_3PO_4"; ACCEPT concentrated H_2SO_4 or aluminium oxide. DO NOT AWARD dilute sulfuric acid (WCH12 Jan 2020).

Often asked as: reagents for converting an alcohol to a halogenoalkane or alkene; equations; the test for –OH; 6-mark choice of the best test for an alcohol group.

BEYOND THE SPEC, BUT EXAMINED

- Making a chloroalkane from an alcohol with PCl_5 is better than adding HCl to an alkene: it's **one step** and gives **no isomeric by-products** (WCH12 Jan 2020).
- Choosing a test for –OH (WCH12 Jun 2024): acidified dichromate fails for a **tertiary** alcohol; acidified $KMnO_4$ also reacts with **C=C**; PCl_5 works for all three classes.
- Other –OH tests: **sodium** gives effervescence (H_2); Lucas' reagent turns cloudy at once with tertiary alcohols (WCH12 Jun 2019).
- Bioethanol: renewable; fermentation must be **anaerobic** so the ethanol isn't oxidised; water in fuel ethanol causes corrosion and lowers energy per litre (WCH12 Oct 2021).

COMMON MISTAKE

"Add HCl" without "**concentrated**" (WCH12 Jan 2021 report).
Suggesting **NaOH** to dehydrate an alcohol (WCH12 Jan 2020 report).
Drawing the same alkene twice (rotated) as two isomers; claiming E/Z isomers for an alkene that can't have them.

SPEC 10.18 Oxidation of alcohols with acidified potassium dichromate(VI)

WHAT THIS SPEC POINT ASKS

Know that acidified potassium dichromate(VI) oxidises (i) primary alcohols to aldehydes (positive with Benedict's or Fehling's) if distilled as they form, (ii) primary alcohols to carboxylic acids (positive with Na_2CO_3 or $NaHCO_3$) if heated under reflux, and (iii) secondary alcohols to ketones. [O] may be used in equations.

10.18 The oxidising agent

- **Potassium dichromate(VI), $K_2Cr_2O_7$, in dilute sulfuric acid**. Name the acid: "acidified" alone isn't enough.
- Colour change: **orange** ($Cr_2O_7^{2-}$, Cr +6) → **green** (Cr^{3+}, Cr +3). The dichromate is **reduced** while the alcohol is oxidised.
- Half-equation: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
- In organic equations write the oxidising agent as **[O]**.

10.18 What each class gives

Alcohol	Product	Conditions	Equation (example)
Primary	**aldehyde**	**distil off as it forms**; alcohol in excess	$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$
Primary	**carboxylic acid**	**heat under reflux**; **excess** oxidising agent	$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$
Secondary	**ketone**	heat under reflux	$CH_3CH(OH)CH_3 + [O] \rightarrow CH_3COCH_3 + H_2O$
Tertiary	**no reaction**	stays orange	–

- **Aldehyde v acid: why the conditions matter.** The aldehyde is oxidised further to the acid if it stays in the flask. An aldehyde has a much **lower boiling point** than the alcohol (aldehyde molecules can't hydrogen-bond to each other), so it **distils out as soon as it forms** and escapes further oxidation. Under **reflux**, the aldehyde vapour condenses and runs back, so it is oxidised all the way to the acid.
- **Tertiary alcohols** have **no H atom on the C–OH carbon**, so they can't be oxidised this way: the mixture stays orange.
- **Ketones** aren't oxidised further, for the same reason.

10.18 Testing the products

Product	Test	Positive result
Aldehyde	warm with **Benedict's or Fehling's solution**	**blue** solution → **red(-brown) precipitate** (Cu_2O)
Ketone	Benedict's / Fehling's	stays blue (no reaction)
Carboxylic acid	add **sodium carbonate** or **sodium hydrogencarbonate**	**effervescence**; the gas (CO_2) turns **limewater milky**

- Acid with hydrogencarbonate: $CH_3CH_2COOH + NaHCO_3 \rightarrow CH_3CH_2COONa + H_2O + CO_2$

MARK-SCHEME LANGUAGE

- Reagents for the acid: "sodium / potassium **dichromate(VI)**" **and** "**(dilute) sulfuric acid**" **and** "(heat under) **reflux**". DO NOT AWARD potassium manganate(VII) or hydrochloric acid (WCH12 Oct 2020, Jun 2025).
- For the aldehyde: "(heat with) potassium dichromate(VI) and sulfuric acid", "**distil**", "so the aldehyde is **not oxidised further** / does not form a carboxylic acid". DO NOT AWARD reflux (WCH12 Jan 2026).
- Equations: "$CH_3OH + 2[O] \rightarrow HCOOH + H_2O$"; "$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$"; going via the aldehyde in two equations is allowed.
- Colour change: "**orange to green**" (ALLOW orange to blue); DO NOT AWARD the reverse.
- Tertiary alcohols "cannot be oxidised"; a primary alcohol in the body is oxidised to a toxic aldehyde (WCH12 Jan 2020).

Often asked as: reagents and conditions for an aldehyde or acid; equations with [O]; colour change; tests for the products; why a tertiary alcohol doesn't react; 6-mark on making propanoic acid.

BEYOND THE SPEC, BUT EXAMINED

- 6-mark (WCH12 Jun 2019): higher concentration or temperature → faster rate (more frequent collisions, more particles with E ≥ Ea); **excess** oxidising agent and **reflux** ensure **complete** oxidation to the acid.
- IR can follow the oxidation: the **O–H** peak (3750–3200 cm^{-1}) shrinks and a **C=O** peak (1740–1700 cm^{-1}) appears (WCH12 Oct 2025).
- The orange/green chromium colours mask an indicator, so the acid made can't be titrated directly in the reaction mixture (WCH12 Jun 2019).

COMMON MISTAKE

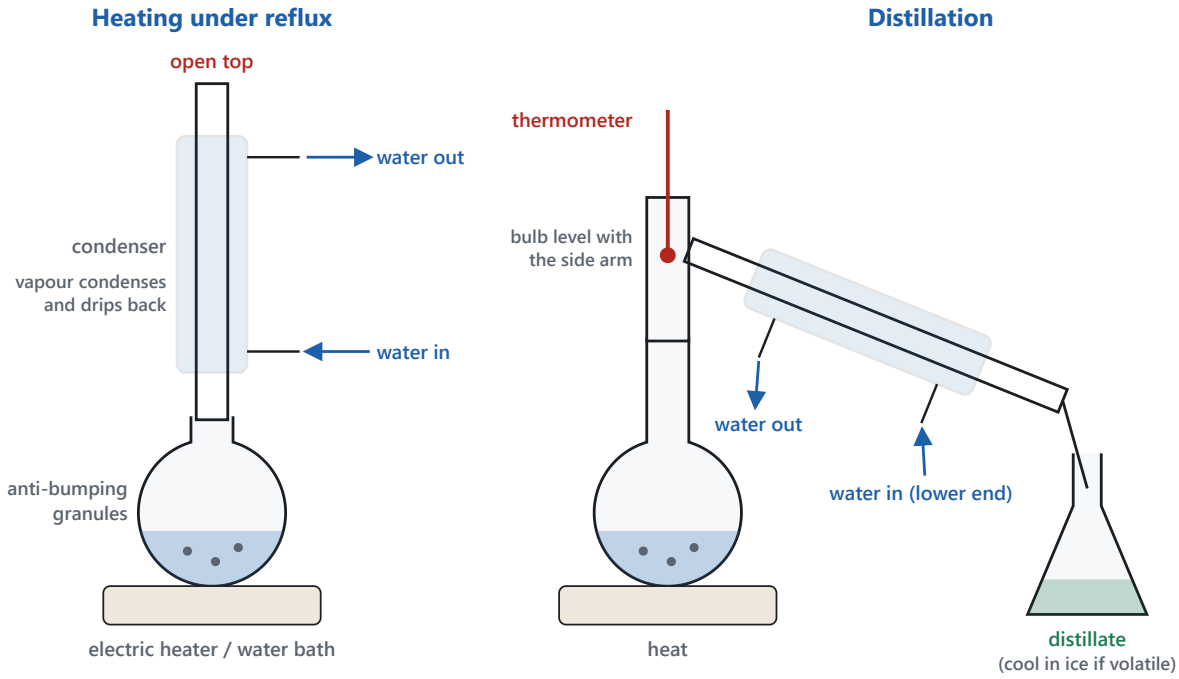
"Acidified potassium dichromate" with **no acid named** (WCH12 Jan 2025 report).
Using **reflux** to make an aldehyde (WCH12 Jan 2025 report).
Colour change backwards, or a wrong oxidation number for Cr.
Forgetting that tertiary alcohols **don't react** (WCH12 Oct 2022 report: over half scored nothing).

SPEC **10.19**

Techniques for preparing and purifying a liquid organic product

WHAT THIS SPEC POINT ASKS

Understand (i) heating under reflux, (ii) solvent extraction with a separating funnel, (iii) distillation, (iv) drying with an anhydrous salt, (v) measuring a boiling temperature.



Left: reflux (vertical condenser, open at the top). Right: distillation (thermometer bulb level with the side arm; water enters the condenser at the lower end).

10.19 (i) Heating under reflux

- **What**: heat the mixture in a flask with a **vertical condenser** on top.
- **Why**: organic reactions are slow and need heating for a long time. The volatile reactants and products **evaporate, condense and drip back** into the flask, so nothing is lost and the reaction can go to completion.

- Details:
  - add **anti-bumping granules** for smooth boiling;
  - water goes **in at the bottom** of the condenser and out at the top, so the jacket stays full;
  - the top is **open**: never heat a sealed system (pressure would build up and it could explode);
  - heat with a **water bath or electric heater**, not a Bunsen, because organic vapours are flammable.

10.19 (ii) Extraction with a separating funnel

- Used to separate two liquids that **don't mix** (immiscible), e.g. an organic layer from an aqueous layer, or to **wash** an organic product.
- Shake with the stopper on, **inverting and opening the tap** regularly to release pressure (especially with $NaHCO_3$, which makes CO_2).
- Let the layers settle. The **denser** liquid is the **lower** layer. Run it off through the tap.
- Not sure which layer is which? Add a few drops of **water**: the layer it joins is the aqueous one.
- **Washing**: shake with **sodium hydrogencarbonate** (or sodium carbonate) solution to remove acid, then with water.

10.19 (iii) Distillation

- Separates a liquid from **non-volatile** impurities or from liquids with **clearly different boiling points**.
- The **thermometer bulb** is level with the **side arm**, so it reads the temperature of the vapour that is actually distilling.
- **Sloping condenser**, water in at the **lower** end.
- **Collect the fraction** that boils within about **2 °C** of the product's boiling point; liquid coming over below or above that is a different substance.
- **Fractional distillation** (with a fractionating column) is used when boiling points are close.

10.19 (iv) Drying with an anhydrous salt

- Organic layers from a separating funnel contain a little water. Add an anhydrous salt: **anhydrous sodium sulfate, magnesium sulfate or calcium chloride**.
- Swirl. If the salt **clumps together**, add more; the liquid is dry when it turns from **cloudy to clear** and fresh salt moves freely.
- **Decant or filter** the dried liquid off the solid.

10.19 (v) Boiling temperature determination

- A **pure** liquid boils at a **fixed, sharp** temperature. Compare it with a data-book value to identify the product and check purity.
- **Impurities** change the boiling temperature and make it boil over a **range**.
- Method: distil the product (or heat a small sample in a tube in an oil bath) and record the steady thermometer reading while the liquid boils.

10.19 The usual purification order

reaction (reflux) → separate (separating funnel) → wash ($NaHCO_3$ / water) → dry (anhydrous salt) → distil (collect at the boiling point) → check purity (boiling temperature)

MARK-SCHEME LANGUAGE

- Name the apparatus: "(reflux / Liebig) **condenser**". An answer that suggests a **sealed** system is not awarded (WCH12 Oct 2025).
- Why distillation fails to separate: "the boiling temperatures are **(too) similar / close**" (WCH12 Jun 2025).
- Why not reflux for the ammonia reaction: "ammonia (gas) would **escape** (from the condenser)" (WCH12 Jun 2022).
- Purifying bioethanol: "(fractional) distillation" (WCH12 Oct 2021).

Often asked as: reflux v distillation; name a piece of apparatus; why the system isn't sealed; choose a drying agent; why a separation won't work.

COMMON MISTAKE

Sealing the top of a reflux condenser.
Water entering the condenser at the top (the jacket never fills).
Thermometer bulb dipping into the liquid during distillation.
Copper(II) sulfate as a drying agent.

SPEC 10.20 Core Practical 7: oxidation of propan-1-ol to propanal and propanoic acid

WHAT THIS SPEC POINT ASKS

Oxidise propan-1-ol to make propanal (by distillation) and propanoic acid (by heating under reflux).

10.20 Boiling points decide the method

Compound	b.p. / °C	Why
propanal, CH_3CH_2CHO	49	no hydrogen bonding between its molecules
propan-1-ol, $CH_3CH_2CH_2OH$	97	hydrogen bonds
propanoic acid, CH_3CH_2COOH	141	hydrogen bonds (and a larger molecule)

CORE PRACTICAL 7: METHOD IN BRIEF

Propanal (distil as it forms)

1. Make the oxidising mixture: potassium dichromate(VI) dissolved in **dilute sulfuric acid**, with anti-bumping granules, in a flask set up for **distillation**.
2. Warm the mixture gently. Add **propan-1-ol** (in excess, so there isn't enough oxidant for a second oxidation) **dropwise** from a tap funnel.
3. Keep the thermometer reading **below about 60 °C** and collect the **distillate** in a receiver **cooled in ice** (propanal is very volatile).
4. Test the distillate with **Benedict's or Fehling's** solution: blue → **red precipitate** on warming shows an aldehyde.

Propanoic acid (heat under reflux)

1. Put propan-1-ol with an **excess** of acidified potassium dichromate(VI) and anti-bumping granules in a flask with a **reflux** condenser.
2. Heat under reflux for about 15–20 minutes. The colour turns from **orange to green**.

3. Rearrange for **distillation** and distil off the propanoic acid (with some water).
4. Test the distillate: add **sodium hydrogencarbonate** (or sodium carbonate): **effervescence**, gas turns limewater milky. Universal indicator shows it's weakly acidic.

10.20 Equations

$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$
$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$

10.20 Safety

- Potassium dichromate(VI) is **toxic, carcinogenic** and an oxidiser: wear gloves and eye protection.
- Propan-1-ol and propanal are **flammable**: use a water bath or electric heater.
- Concentrated reagents and hot liquids: anti-bumping granules, never heat a sealed apparatus.

MARK-SCHEME LANGUAGE

- Propanal: "**distil**" so it is "not oxidised further"; reflux is not credited. Propanoic acid: "(heat under) **reflux**" with **excess** oxidant (WCH12 Jun 2021, Jan 2026).
- Colour change "orange to green"; condenser named; IR shows O–H (alcohol) disappearing and C=O appearing (WCH12 Oct 2025).
- Propanoic acid as a food preservative: below the limit "the food would go mouldy (it wouldn't preserve)", above it "the food tastes bad / becomes toxic" (WCH12 Jun 2019).

Often asked as: distil v reflux to get each product; equations; tests for each product; a Section C context on propanoic acid with a titration.

COMMON MISTAKE

Using reflux for propanal, or distillation straight away for propanoic acid.
"Acidified dichromate" without naming sulfuric acid.

SPEC 10.21 Mass spectra of organic compounds

WHAT THIS SPEC POINT ASKS

Use the m/z of the molecular ion and the fragmentation pattern in a mass spectrum to suggest possible structures of simple organic compounds.

10.21 Reading a spectrum

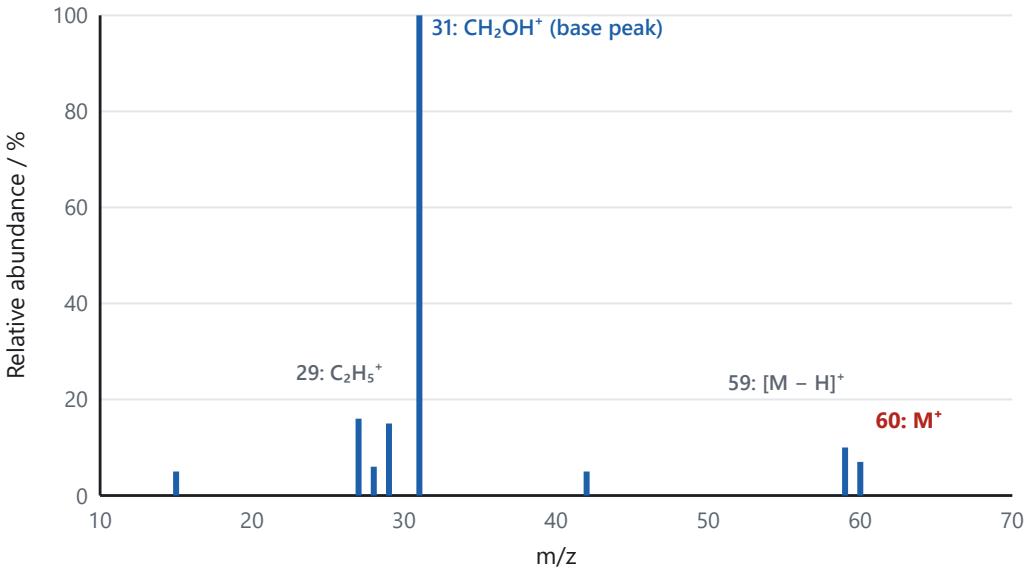
- Molecules are ionised by losing an electron: $M \rightarrow \mathbf{M^+} + e^-$. M^+ is the **molecular ion**.
- The **molecular ion peak** is the peak with the **highest m/z** (ignore the tiny M+1 peak from ^{13}C). Its m/z = **relative molecular mass, Mr**.
- Molecular ions are unstable and **fragment**: $M^+ \rightarrow X^+ + Y\bullet$. Only the **positive ion** X^+ is detected; the neutral radical Y• isn't.
- The **base peak** is the tallest peak (set to 100%): the most abundant (most stable) ion.
- The fragments show which groups are in the molecule. Mr alone only gives the molecular formula, and **isomers** have the same Mr: the fragments tell them apart.

10.21 Common fragment ions and losses

m/z	Fragment ion		Mass lost from M^+	Group lost
15	CH_3^+		15	CH_3
29	$C_2H_5^+$ or CHO^+		17	OH
31	CH_2OH^+		18	H_2O (alcohols)
43	$C_3H_7^+$ or CH_3CO^+		29	C_2H_5 or CHO
45	$COOH^+$ or CH_3CHOH^+		35 / 37	Cl
57	$C_4H_9^+$		79 / 81	Br

10.21 Halogen isotope patterns

- Chlorine is about **75% ^{35}Cl and 25% ^{37}Cl**, so a compound with **one Cl** shows two molecular-ion peaks **2 apart**, heights **3 : 1** (e.g. chloroethane: 64 and 66).
- Bromine is about **50% ^{79}Br and 50% ^{81}Br**, so **one Br** gives two peaks **2 apart** of **equal height**, **1 : 1** (e.g. bromomethane: 94 and 96).
- **Two Cl atoms** give M, M+2 and M+4 in the ratio **9 : 6 : 1**.



Simplified mass spectrum of propan-1-ol, $CH_3CH_2CH_2OH$ (Mr 60). The base peak at 31 is CH_2OH^+, which shows the –OH is on the end carbon.

WORKED EXAMPLE: PROPAN-1-OL OR PROPAN-2-OL?

Both have Mr = 60 (C_3H_8O), so the molecular ion at m/z 60 can't tell them apart.

Propan-1-ol, CH_3CH_2–CH_2OH: breaking the C–C bond next to the C–OH gives **CH_2OH^+, m/z 31** (base peak). Also $C_2H_5^+$ at 29.

Propan-2-ol, CH_3–CH(OH)–CH_3: losing a CH_3 gives **CH_3CHOH^+, m/z 45** (base peak). There is no big peak at 31.

So a large peak at 31 → propan-1-ol; a large peak at 45 → propan-2-ol.

WORKED EXAMPLE: PROPANAL OR PROPANONE?

Both C_3H_6O, Mr = 58.
Propanone, CH_3COCH_3: strong peak at **43** (CH_3CO^+, loss of CH_3) and at 15 (CH_3^+).
Propanal, CH_3CH_2CHO: strong peaks at **29** ($C_2H_5^+$ and CHO^+) and 28; 57 (loss of H).

EXAM TIP

When you identify a fragment, write its **formula with a + charge** ("CH_2OH^+", not "CH_2OH" and not just "m/z 31"). Missing charges are penalised.

MARK-SCHEME LANGUAGE

- Fragments must be written as **positive ions** with a structure or formula, e.g. "m/z = 53 is $(CH_2{=}C{-}CH{=}CH_2)^+$, loss of CH_3" (WCH12 Jun 2024). Missing charges penalised once.
- Isotopes: "chlorine isotopes (35 and 37) are in the ratio **3 : 1**", so "two chlorine atoms give the ratio **9 : 6 : 1**" (WCH12 Jun 2021). DO NOT AWARD ^{36}Cl.

Often asked as: identify the species causing given peaks; use M^+ to choose between candidate compounds; explain a pattern of peaks for a compound containing Cl or Br; 6-mark combined MS + IR identification (WCH12 Jan 2025).

BEYOND THE SPEC, BUT EXAMINED

- Find Mr first from gas data (pV = nRT, convert kPa to Pa and cm^3 to m^3) or from % composition, then match to M^+ (WCH12 Oct 2024, Jun 2025).
- Peaks for $COCl_2$ at 98, 100, 102 in the ratio 9 : 6 : 1, and the fragment $CO^{35}Cl^+$ (WCH12 Jun 2021). (That mark scheme writes the abundances the wrong way round: chlorine is 75% ^{35}**Cl**.)

COMMON MISTAKE

Taking the tallest peak (base peak) as the molecular ion.
Calculating the Mr of the candidate compounds wrongly (WCH12 Jan 2025 report): check your arithmetic.
Fragments written without a positive charge.

SPEC 10.22 Infrared spectra

WHAT THIS SPEC POINT ASKS

Use IR spectra (or IR data) to identify functional groups and predict absorptions for: C–H (alkanes, alkenes, aldehydes), C=C, O–H (alcohols, carboxylic acids), C=O (aldehydes, ketones, carboxylic acids), C–X and N–H (amines).

10.22 How IR works

- Covalent bonds **vibrate** (stretch and bend). Each type of bond absorbs **infrared radiation** at its own frequency, depending on the **bond strength** and the **masses of the atoms**.
- An IR spectrum plots **% transmittance** (y) against **wavenumber / cm^{-1}** (x, running from 4000 on the left down to about 500). Absorptions show as **dips** (downward "peaks").
- Describe a peak by **position** (wavenumber), **width** (broad or sharp) and **intensity** (strong or weak).
- Below about 1500 cm^{-1} is the **fingerprint region**: a complicated pattern unique to each compound. Matching it to a database spectrum confirms an identity.

10.22 Key absorptions (check against your Data Booklet)

Bond	Where	Wavenumber / cm^{-1}	Look
O–H	alcohols	3750–3200 (hydrogen-bonded: about 3550–3230)	strong, **broad**
O–H	carboxylic acids	3300–2500	**very broad**, overlaps C–H
N–H	amines	3500–3300	medium, sharper than O–H
C–H	alkenes (=C–H)	3095–3010	medium
C–H	alkanes	2962–2853	strong (in almost every organic compound)
C–H	aldehydes (–CHO)	2900–2820 and 2775–2700	two weak peaks
C=O	aldehydes	1740–1720	strong, **sharp**
C=O	ketones	1720–1700	strong, sharp
C=O	carboxylic acids	1725–1700	strong, sharp
C=C	alkenes	1669–1645	variable
C–O	alcohols, acids, esters	1300–1000	strong
C–Cl	chloroalkanes	about 800–600	strong
C–Br / **C–I**	bromo- / iodoalkanes	about 600–500 / about 500	

10.22 Using it

- Look first for the big diagnostic peaks: **O–H (broad, above 3200)**, **C=O (sharp, about 1700)**, **N–H**, **C=C**.
- Combine them:

Peaks seen	Compound type
broad O–H 3750–3200, no C=O	**alcohol**
C=O about 1720, no O–H	**aldehyde or ketone** (aldehyde if C–H at 2900–2700)
very broad O–H 3300–2500 **and** C=O about 1710	**carboxylic acid**
C=C 1669–1645 and =C–H above 3000	**alkene**
N–H 3500–3300	**amine**
C–X below 800, nothing else special	**halogenoalkane**

WORKED EXAMPLE: FOLLOWING AN OXIDATION

Propan-1-ol is heated under reflux with acidified $K_2Cr_2O_7$. How does the IR spectrum show that propanoic acid has formed?

Before: broad **O–H (alcohol)** absorption at **3750–3200 cm^{-1}**; no C=O.

After: a strong **C=O** absorption appears at **1725–1700 cm^{-1}**, and the O–H absorption moves to a **very broad band at 3300–2500 cm^{-1}** (carboxylic acid O–H).

If propanal formed instead (distillation), there would be **C=O at 1740–1720 cm^{-1}** and **no O–H** at all.

WORKED EXAMPLE: AN ELIMINATION

Butan-2-ol is dehydrated to butenes. IR evidence: the **O–H** absorption at **3750–3200 cm^{-1}** disappears; a **C=C** absorption appears at **1669–1645 cm^{-1}** (and =C–H at 3095–3010 cm^{-1}).

MARK-SCHEME LANGUAGE

- Give **the bond and its wavenumber range**: "Both wavenumber ranges without bonds scores (1)"; a single specific wavenumber inside a range is accepted, but you still need the bond (WCH12 Jun 2023).
- Product v reactant: "O–H absent / C=O present". "One mark for two correct bonds identified." C–H peaks and the fingerprint region are ignored in these questions (WCH12 Jun 2025).
- Elimination: "O–H present in the reactant / absent in the product" and "C=C in the product" (WCH12 Jun 2019).
- Ranges are those in the **Data Booklet**; any value within the booklet range is accepted.

Often asked as: give one IR difference between a reactant and product; assign labelled peaks; choose the compound that matches a spectrum (6-mark with MS, WCH12 Jan 2025).

BEYOND THE SPEC, BUT EXAMINED

- The **fingerprint region** changes when a reaction happens, which can be used to show it has occurred (WCH12 Oct 2025, Oct 2020).
- The C=O in an **acyl chloride** ($COCl_2$) absorbs at a higher wavenumber (about 1795 cm^{-1}) than in ketones; mark schemes accept a wide range for unfamiliar C=O compounds (WCH12 Jun 2021).

COMMON MISTAKE

Quoting a bond with no wavenumber, or a wavenumber with no bond.
Not using the Data Booklet ranges (WCH12 Jan 2025 report).
Confusing alcohol O–H (3750–3200) with carboxylic acid O–H (3300–2500, much broader).
Using C–H absorptions to tell compounds apart: nearly every organic compound has them.

SPEC 10.23

Core Practical 8: analysis of some inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Use chemical tests to identify unknown inorganic and organic substances.

10.23 Inorganic tests

Testing for	Method	Positive result
Group 1 / 2 cations	**flame test**: clean nichrome wire dipped in conc. HCl, then the solid, into a hot (blue) flame	Li^+ red; Na^+ yellow; K^+ lilac; Ca^{2+} orange-red; Sr^{2+} red (crimson); Ba^{2+} pale green
NH_4^+	warm with **NaOH(aq)**	gas (NH_3) turns **damp red litmus blue**
CO_3^{2-} / HCO_3^-	add **dilute acid**	effervescence; CO_2 turns **limewater milky**
SO_4^{2-}	add **dilute HCl**, then **$BaCl_2$(aq)**	**white precipitate** ($BaSO_4$)
Cl^-, Br^-, I^-	add **dilute HNO_3**, then **$AgNO_3$(aq)**; then **NH_3(aq)**	AgCl **white**, dissolves in dilute NH_3; AgBr **cream**, dissolves in conc. NH_3; AgI **yellow**, insoluble in conc. NH_3

- The HCl (for sulfate) and HNO_3 (for halides) remove **carbonate** ions first, which would also give a precipitate.
- Use HNO_3, not HCl, before $AgNO_3$: HCl would add Cl^- ions.

10.23 Organic tests

Testing for	Reagent	Positive result
C=C (alkene)	**bromine water** (shake)	**orange → colourless**
–OH (alcohol, acid, water)	**PCl_5** (dry sample)	**steamy fumes** of HCl; white smoke with NH_3
1° or 2° alcohol, or aldehyde	warm with **acidified $K_2Cr_2O_7$**	**orange → green** (tertiary alcohols and ketones: stays orange)
Aldehyde	warm with **Benedict's or Fehling's**	**blue → red precipitate** (ketones: no change)
Carboxylic acid	**$NaHCO_3$** or Na_2CO_3	**effervescence**; CO_2 turns limewater milky
Halogenoalkane	warm with **NaOH(aq)**, add **dilute HNO_3**, then **$AgNO_3$(aq)**	precipitate: white (Cl), cream (Br), yellow (I)

CORE PRACTICAL 8: APPROACH

1. Record the **appearance** of each unknown (state, colour, smell only by wafting).
2. Plan a **sequence** of tests that narrows the options: start with tests that split the groups (e.g. flame test, then anion tests; for organics: bromine water, then PCl_5, then dichromate, then Benedict's/Fehling's or $NaHCO_3$).
3. Use **small amounts** (a few drops or a spatula tip) in test tubes; add reagents dropwise.
4. Record **observations** (what you see), not just conclusions. "Effervescence" not "CO_2 produced".
5. Combine results to identify each unknown. Use IR or mass spectra if they are given.

MARK-SCHEME LANGUAGE

- Test for –OH (6-mark, WCH12 Jun 2024): **PCl_5** gives misty fumes with all alcohols; acidified dichromate fails for a tertiary alcohol; acidified $KMnO_4$ also reacts with C=C.
- Acid test: "effervescence" with carbonate (WCH12 Oct 2020). Aldehyde test: Benedict's or Fehling's.
- Halide: $AgNO_3$ with dilute nitric acid, then ammonia solution (WCH12 Jan 2021).
- Describe **observations**: "a white precipitate", "effervescence", "orange to green", not just the product's name.

Often asked as: choose a test to tell two compounds apart and give the result for each; 6-mark test choice; combine tests with spectra.

COMMON MISTAKE

Writing the name of a product ("silver chloride forms") when an **observation** is asked.
Giving the result for only **one** of the two compounds in a "distinguish" question: say what happens with **each**.
Testing a halogenoalkane directly with $AgNO_3$(aq) with no hydrolysis step, or skipping the HNO_3 after NaOH.